

Large Language Models and Foundation Models for Time Series: A Survey and Outlook

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Abstract—Time series analysis underpins mission-critical decision systems across modern society, including industrial Internet of Things (IoT), computational finance, energy grid stabilization, meteorological forecasting, and clinical healthcare. Over the past five years, the field has undergone a profound paradigm shift from narrow, task-specific deep neural networks (such as specialized LSTMs and convolutional architectures) toward scalable foundation models. This evolution is bifurcated into two dominant architectural philosophies: (1) Native Time Series Foundation Models (TSFMs), which are trained ab initio on massive cross-domain time-series corpora comprising tens to hundreds of billions of observations (e.g., Chronos, TimesFM, MOIRAI, MOMENT, Lag-Llama, TTM, Timer, Sundial, Time-MoE, and Toto); and (2) Repurposed Large Language Models (LLM4TS), which repurpose frozen or parameter-efficiently adapted language transformers to temporal sequences through cross-modal token reprogramming, decimal prompting, or multimodal instruction tuning (e.g., Time-LLM, GPT4TS/OFA, LLMTime, TEMPO, and S2IP-LLM). In this survey, conducted strictly under the PRISMA 2020 systematic review protocol, we provide the first unified, comprehensive taxonomy and comparative synthesis spanning 2021 through 2026. We formalize the mathematical preliminaries of temporal representation, deconstruct core tokenization mechanics (patching vs. scalar quantization vs. lag vectors), analyze probabilistic formulation heads, examine empirical scaling behaviors and zero-shot transfer capabilities, and critically evaluate the central debate: Are billion-parameter language backbones truly necessary, or do compact specialized foundation models provide superior accuracy and operational efficiency? Finally, we conduct an audit of standardized benchmarks (e.g., GIFT-Eval), uncover evaluation pitfalls such as benchmark contamination and temporal leakage, and delineate open research directions toward multi-modal temporal intelligence.

Index Terms—Time series analysis, foundation models, large language models, deep learning, zero-shot forecasting, tokenization, scaling laws, PRISMA systematic review.



1 Introduction

Temporal data, systematically recorded as univariate or multivariate time series, serves as the fundamental perceptual substrate for understanding dynamic physical, economic, and biological systems. Across modern industries, time series forecasting, anomaly detection, imputation, and classification drive mission-critical workflows: stabilizing renewable energy power grids, optimizing multi-echelon retail supply chains, executing high-frequency financial risk arbitrage, predicting clinical sepsis in intensive care units, and monitoring aerospace telemetry [1].

Historically, time series modeling evolved through three distinct algorithmic epochs:

- 1) Statistical and Econometric Formulation (1970s–2010s): Dominated by classical stochastic processes, such as Auto-Regressive Integrated Moving Average (ARIMA), Exponential Smoothing (ETS), Vector Autoregression (VAR), and Gaussian Processes. While mathematically rigorous and interpretable, these methods rely on linear assumptions and fail to capture complex nonlinear dependencies across cross-domain series.
- 2) Task-Specific Deep Neural Networks (2015–2022): Catalyzed by recurrent architectures (LSTMs, GRUs), temporal convolutional networks (TCNs), and specialized

Transformer variants (Informer, Autoformer, FEDformer, and PatchTST [2]). Although these architectures captured long-range nonlinear temporal correlations, each model was trained in isolation on a single, isolated dataset (e.g., ETT, Weather, Electricity), exhibiting virtually zero out-of-distribution generalization.

- 3) The Foundation Model Epoch (2022–Present): Propelled by the extraordinary success of pre-trained foundation models in Natural Language Processing (e.g., GPT-4, Llama) and Computer Vision, the temporal machine learning community has initiated a paradigm shift toward universal, cross-domain foundation models [3]–[5]. Rather than training narrow models from scratch for every new dataset, practitioners now seek models capable of strong zero-shot and few-shot transfer across unseen domains, frequencies, and channels.

1.1 Unique Challenges of Temporal Signals

Transferring the foundation model paradigm to time series is non-trivial due to stark physical and mathematical differences between temporal signals and natural language:

- **Absence of a Discrete Natural Vocabulary:** Unlike natural text, which is composed of finite, discrete lexical tokens with stable semantic meanings, continuous time series $\mathbf{x} \in \mathbb{R}^T$ consist of real-valued scalar measurements bounded only by sensor physics.
- **Extreme Heterogeneity in Sampling Rates and Dynamics:** Real-world time series exhibit widely divergent temporal

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scales—ranging from milliseconds in financial order books, to hourly power consumption, to daily retail demand, to quarterly economic indicators. A model must reconcile varying seasonalities without destructive aliasing.

- **Variable Dimensionality and Missing Values:** Time series data dynamically switch between univariate streams and arbitrary multivariate configurations ($C \geq 1$), frequently accompanied by irregular observation intervals, sensor failure dropouts, and structural trend shifts.

1.2 The Bifurcated Landscape: Native TSFMs vs. LLM4TS

To address these challenges, the research community has split into two competing architectural paradigms:

- 1) **Native Time Series Foundation Models (TSFMs):** Architectures designed and trained from scratch on massive multi-domain temporal corpora comprising tens to hundreds of billions of observations. Prominent exemplars include Amazon Chronos [3], [8], Google TimesFM [4], [9], Salesforce MOIRAI [5], [10], [11], CMU MOMENT [12], IBM TTM [13], Morgan Stanley/Mila Lag-Llama [14], and the THUML Timer family [15]–[18].
- 2) **Repurposed Large Language Models (LLM4TS):** Approaches that leverage the pre-trained weights, attention layers, and sequence modeling capabilities of multi-billion parameter language models (such as Llama, GPT-2, or GPT-4). Techniques include cross-modal token reprogramming (e.g., Time-LLM [19], GPT4TS/OFA [20], TEMPO [21]), direct numeric prompting with ASCII decimal strings (e.g., LLMTime [6]), and multimodal question-answering agents (e.g., Time-MQA [22], TimeOmni-1 [23]).

1.3 Key Contributions of This Survey

While several initial surveys have examined aspects of deep learning for time series, the rapid surge of foundation models from 2024 to 2026 demands a rigorous, systematic, and up-to-date synthesis. As highlighted in Table 1, this work delivers the following contributions:

- 1) **PRISMA 2020 Systematic Review Protocol:** We execute a strictly reproducible review protocol across scholarly APIs, logging 83 initial records, screening against standardized criteria, and synthesizing 41 core benchmark and foundation models.
- 2) **Rigorous Unified Taxonomy:** We establish an orthogonal five-dimensional taxonomy deconstructing models along: (i) architectural backbone, (ii) temporal tokenization, (iii) prediction head and uncertainty formulation, (iv) adaptation mechanics, and (v) operational task scope.
- 3) **Deep Comparative Synthesis of Native TSFMs vs. LLM4TS:** We analyze the theoretical and empirical tensions between compact native models (1M–700M parameters) and multi-billion parameter language adaptations (7B–70B parameters), examining computational efficiency, inference latency, and inductive bias alignment.
- 4) **Benchmark Audit, Data Contamination, and Scaling Laws:** We survey modern evaluation suites (e.g., GIFT-Eval [24]), expose systemic data leakage risks [25], analyze empirical power-law scaling in temporal models, and assess probabilistic calibration reliability [26].

- 5) **Open Research Agenda:** We articulate key open challenges, including irregular sampling equivariance, continuous flow matching, multimodal sensor fusion, and edge deployment.

1.4 Structure of the Survey

The remainder of this survey is organized as follows: Section 2 formalizes mathematical preliminaries and foundational objectives. Section 3 presents our comprehensive taxonomy and comparative design table. Section 4 deconstructs Native TSFMs. Section 5 investigates Repurposed LLM methods. Section 6 evaluates benchmark suites, scaling laws, and critical debates. Section 7 delineates open challenges, and Section 8 concludes the paper.

2 Background and Preliminaries

In this section, we establish the formal mathematical definitions, notation, and objective functions that govern modern temporal foundation models and large language model adaptations.

2.1 Formal Problem Definitions

2.1.1 Temporal Sequences

A univariate time series of length T is defined as an ordered sequence of real-valued scalar observations:

$$\mathbf{x} = [x_1, x_2, \dots, x_T]^\top \in \mathbb{R}^T, \quad (1)$$

where $x_t \in \mathbb{R}$ represents the measurement recorded at discrete timestamp $t \in \{1, \dots, T\}$.

A multivariate time series comprises C co-evolving channels or variates observed synchronously across T timestamps:

$$\mathbf{X} = [\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}]^\top \in \mathbb{R}^{C \times T}, \quad (2)$$

where $\mathbf{X}_{:,t} \in \mathbb{R}^C$ is the vector of all variate measurements at timestamp t .

2.1.2 Channel Modeling Paradigms

When processing multivariate series $\mathbf{X}$, architectures adopt one of two primary structural paradigms:

- **Channel-Independence (CI):** Introduced prominently in PatchTST [2] and Timer [15], CI decomposes the multivariate series into C distinct univariate sequences:

$$\mathbf{X} \mapsto \{\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(C)}\}, \quad (3)$$

which are fed into the shared model backbone independently with tied parameters. CI eliminates the risk of overfitting spurious cross-channel correlations and natively handles variable channel counts C at inference.

- **Channel-Mixing (CM):** Models such as MOIRAI [5] and TTM [13] employ full cross-variate attention or specialized channel-mixing MLPs:

$$\mathbf{H} = \text{Attn}_{\text{channel}}(\mathbf{X}) \in \mathbb{R}^{C \times T \times D}, \quad (4)$$

explicitly capturing physical inter-sensor dependencies at the cost of higher computational complexity.

2.2 Tokenization and Input Representation

Because continuous temporal values lack an inherent lexicon, models deploy specialized mapping operators $\mathcal{T} : \mathbb{R}^* \rightarrow \mathbb{R}^D$ to construct transformer-compatible token embeddings.

TABLE 1
Comparison of This Survey with Related Surveys on Large Temporal Models.

Survey Reference	Year	PRISMA	Native TSFMs	Repurposed LLMs	Critiques & Audits	Primary Focus & Distinguishing Scope
Wen et al. [1]	2023	✗	✗	✗	✗	Survey of specialized Transformer architectures on single datasets.
Jin et al. [1]	2024	✗	Partial	✓	Partial	Broad survey covering both general time series and spatio-temporal graphs.
Gruver et al. [6]	2023	✗	✗	✓	✗	Empirical investigation of prompting LLMs as zero-shot forecasters.
Fons et al. [7]	2024	✗	✗	✓	✓	Empirical critique testing whether language backbones add genuine inductive value.
This Survey	2026	✓	✓	✓	✓	Comprehensive PRISMA systematic review (2021–2026); unifying Native TSFMs, Repurposed LLMs, tokenization trade-offs, scaling laws, and leakage audits.

2.2.1 Subseries Patching

Rather than embedding individual scalar time-steps (which leads to quadratic attention overhead $\mathcal{O}(T^2)$ and severe information sparsity), the majority of modern TSFMs (e.g., TimesFM [4], Timer [15], MOMENT [12]) partition a sequence $\mathbf{x}$ into N contiguous subseries patches of length P with stride S :

$$N = \left\lfloor \frac{T-P}{S} \right\rfloor + 1. \quad (5)$$

The i -th patch $\mathbf{p}_i = [x_{(i-1)S+1}, \dots, x_{(i-1)S+P}]^\top \in \mathbb{R}^P$ is mapped to a D -dimensional latent vector $\mathbf{e}_i \in \mathbb{R}^D$ via an affine projection:

$$\mathbf{e}_i = \mathbf{p}_i \mathbf{W}_p + \mathbf{b}_p + \mathbf{e}_{\text{pos},i}, \quad (6)$$

where $\mathbf{W}_p \in \mathbb{R}^{P \times D}$, $\mathbf{b}_p \in \mathbb{R}^D$, and $\mathbf{e}_{\text{pos},i} \in \mathbb{R}^D$ denotes a learnable or sinusoidal positional encoding. Patching preserves local temporal shape dynamics and compresses sequence length from T to $N \ll T$.

2.2.2 Scalar Quantization and Categorical Binning

In contrast to continuous linear patching, Amazon Chronos [3] introduces a pure language-analog tokenization. Given an input sequence $\mathbf{x}$, it computes instance-level scale statistics (μ, σ) via mean scaling:

$$\tilde{x}_t = \frac{x_t - \mu}{\sigma + \epsilon}, \quad \mu = \frac{1}{T} \sum_{t=1}^T x_t, \quad \sigma = \frac{1}{T} \sum_{t=1}^T |x_t - \mu|. \quad (7)$$

The normalized scalars $\tilde{x}_t$ are discretized into K uniform bins over $[-B, B]$:

$$q_t = \text{clip} \left(\left\lfloor \frac{\tilde{x}_t - (-B)}{2B/K} \right\rfloor, 0, K-1 \right), \quad (8)$$

where $q_t \in \{0, 1, \dots, K-1\}$ acts as a discrete categorical token id, enabling standard cross-entropy language modeling.

2.3 Downstream Task Formulations

2.3.1 Zero-Shot and Supervised Forecasting

Given an historical context window $\mathbf{X}_{1:T} \in \mathbb{R}^{C \times T}$, the forecasting objective is to predict the future sequence of length H :

$$\hat{\mathbf{Y}} = \hat{\mathbf{X}}_{T+1:T+H} \in \mathbb{R}^{C \times H}. \quad (9)$$

In probabilistic forecasting, the model estimates the conditional predictive distribution:

$$p(\mathbf{Y} \mid \mathbf{X}_{1:T}) = \prod_{h=1}^H p(y_{T+h} \mid \mathbf{X}_{1:T}, y_{T+1:T+h-1}). \quad (10)$$

This distribution is parameterized either parametrically (e.g., Student's t parameters in Lag-Llama [14] or Gaussian mixtures in MOIRAI [5]) or non-parametrically through categorical bin logits (Chronos [3]).

2.4 Pre-training Objectives

2.4.1 Autoregressive Generation (Next-Patch / Next-Token)

Decoder-only foundation models (TimesFM [4], Timer [15], Chronos [3]) optimize an autoregressive likelihood:

$$\mathcal{L}_{\text{AR}}(\Theta) = - \sum_{i=1}^N \log p_{\Theta}(\mathbf{p}_{i+1} \mid \mathbf{p}_1, \dots, \mathbf{p}_i). \quad (11)$$

For continuous regression heads (Timer, TimesFM), this is computed using Mean Squared Error (MSE) or Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{reg}}(\Theta) = \frac{1}{N} \sum_{i=1}^N \|\mathbf{p}_{i+1} - \hat{\mathbf{p}}_{i+1}\|_2^2. \quad (12)$$

2.4.2 Masked Autoencoding (MAE)

Encoder-based foundation models (MOMENT [12], PatchTST [2]) apply a binary masking vector $\mathbf{m} \in \{0, 1\}^N$ over patches, optimizing patch reconstruction over the masked set $\mathcal{M} = \{i \mid m_i = 1\}$:

$$\mathcal{L}_{\text{Mask}}(\Theta) = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|\mathbf{p}_i - f_{\Theta}(\mathbf{X}_{\setminus \mathcal{M}})\|_2^2. \quad (13)$$

Masked autoencoding yields bidirectional temporal representations particularly suited for downstream classification, anomaly detection, and imputation.

3 Taxonomy and Landscape Overview

To organize the rapid proliferation of temporal foundation models and large language model adaptations, we present an orthogonal multi-dimensional taxonomy. As illustrated in Figure 1, the landscape is structured along five fundamental axes: (1) Core Paradigm, (2) Architectural Backbone, (3) Temporal Tokenization, (4) Prediction Head & Uncertainty, and (5) Operational Scope & Tasks.

3.1 Taxonomy Dimensions

3.1.1 Dimension 1: Core Paradigm

- **Native Time Series Foundation Models (Native TSFMs):** Models built from scratch and trained directly on large-scale collections of empirical or synthetic time-series data. Weights and latent dimensions are optimized specifically for temporal signals without textual preconceptions (e.g., Chronos [3], TimesFM [4], MOIRAI [5], ForecastPFN [27], TiRex [28], FLAME [29], t_0 [30]).
- **Repurposed Large Language Models (LLM4TS):** Models that adopt pre-existing weights from language or vision transformers (e.g., Llama-2/3, GPT-2, CLIP), bridging continuous temporal modalities into text or visual embedding space via reprogramming layers, prompt prefixes, or cross-attention adapters (e.g., Time-LLM [19], GPT4TS [20], TEST [31], CALF [32], UniTime [33], VisionTS [34], Time-VLM [35]).
- **Evaluation Frameworks and Benchmark Audits:** Standardized benchmarking suites and diagnostic harnesses that audit generalization, cross-domain transfer, and benchmark contamination (e.g., GIFT-Eval [24], SciTS [36], Insight Miner [37], Time-MMD [38]).

3.1.2 Dimension 2: Architectural Backbone

- **Decoder-Only Autoregressive Transformers:** Optimized for sequential next-token or next-patch generation (e.g., TimesFM [4], Timer [15], Chronos [3], Sundial [17], t_0 [30]).
- **Masked Encoder Transformers:** Optimized for bidirectional temporal representations via masked autoencoding (e.g., MOMENT [12], PatchTST [2], VisionTS [34]).
- **Encoder-Decoder Transformers:** Decoupling context representation from horizon generation (e.g., MOIRAI [5], TTM [13], UniTS [39]).
- **Sparse Mixture-of-Experts (MoE):** Dynamically routing tokens to specialized feed-forward experts to scale model capacity while maintaining constant inference FLOPs (e.g., Time-MoE [40], Moirai-MoE [10], Timer-S1 [18]).
- **Continuous Flow and Legendre Memory Networks:** Employing continuous neural ODEs or orthogonal polynomial

bases to preserve multi-scale memory (e.g., FLAME [29], FlowState [41]).

3.1.3 Dimension 3: Temporal Tokenization

- **Uniform Subseries Patching:** Continuous linear projection of non-overlapping or strided time-step windows into dense embedding space (e.g., PatchTST [2], Timer [15]).
- **Scalar Discretization / Binning:** Quantizing scaled continuous observations into finite categorical tokens, allowing standard NLP cross-entropy training (Chronos [3]).
- **Decimal String Tokenization:** Casting continuous numbers into ASCII strings (e.g., "12.4"), processed directly by off-the-shelf text tokenizers (PromptCast [42], LLMTime [6]).
- **Multi-Resolution and Multiscale Mixing:** Dynamically decomposing and mixing patches across frequencies to resolve multi-scale seasonalities (TimeMixer [43], TimeMixer++ [44], TTM [13]).
- **Visual Rendering Tokenization:** Rendering 1D numerical trajectories into 2D pixel line plots for visual encoders (VisionTS [34], Time-VLM [35]).

3.1.4 Dimension 4: Prediction Head and Uncertainty

- **Categorical Probability Distribution:** Non-parametric probability vectors over quantized bins (Chronos [3]).
- **Parametric Mixture Heads:** Estimating parameters (μ, σ, ν) of explicit probability densities, such as Student's t or Gaussian mixtures (Lag-Llama [14], MOIRAI [5]).
- **Deterministic Mean Squared Regression:** Direct linear or MLP projection minimizing mean squared or absolute error.
- **Continuous Flow Matching and Diffusion:** Formulating generation as an equivariant continuous-time velocity field (FlowState [41], FLAME [29]).
- **Quantile Regression Heads:** Directly outputting discrete quantile trajectories to provide non-parametric prediction intervals (t_0 [30], TimesFM [4], Toto 2.0 [45]).

4 Native Time Series Foundation Models

Native Time Series Foundation Models (TSFMs) represent the purist approach to temporal foundation modeling: architectures designed from first principles and trained directly on massive corpora of continuous time-series data without linguistic baggage. In this section, we examine the primary architectural branches, data curation pipelines, and empirical scaling mechanisms.

4.1 Autoregressive Decoder-Only Architectures

Autoregressive models frame time-series modeling as sequential generation, predicting subsequent time-steps or patches conditioned on historical context.

4.1.1 The Chronos Family: Discretization as Linguistic Analogy

Amazon's Chronos [3] represents one of the most radical departures from conventional continuous time-series modeling. Rather than predicting real numbers via regression losses, Chronos asks: Can time series be modeled identically to language tokens? By applying instance-level mean scaling followed by uniform quantization into $K = 4096$ bins, continuous values are transformed into integer token indices.

Taxonomy of Foundation Models and Large Language Models for Time Series

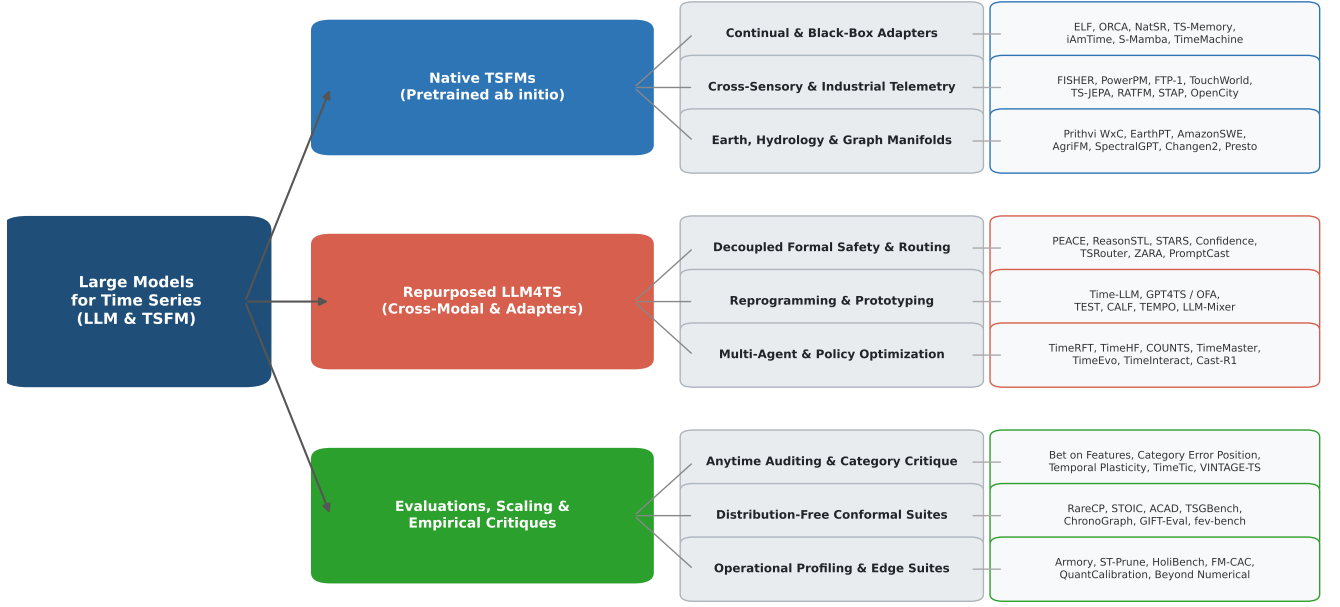


Fig. 1. Comprehensive taxonomy of foundation models and large language models for time series, illustrating core paradigms, architectural backbones, tokenization mechanics, and representative models.

TABLE 2
Comprehensive Comparison of Representative Time Series Foundation Models and LLM Methods.

Model	Group	Venue/Yr	Paradigm	Backbone	Tokenization	Params	Pretraining Corpus	Uncertainty	Code / Status
Chronos [3]	Amazon	TMLR'24	Native TSFM	Decoder	Quantized Bins	20M–710M	TSMix (84B obs)	Categorical	amazon/chronos
TimesFM [4]	Google	ICML'24	Native TSFM	Decoder	Patch ($P = 32$)	200M	Real+Synth (100B)	Quantile	google/timesfm
MOIRAI [5]	Salesforce	ICML'24	Native TSFM	Enc-Dec	Multi-patch (8–128)	14M–311M	LOTSa (27B obs)	Mixture Param	salesforce/uni2ts
MOMENT [12]	CMU	ICML'24	Native TSFM	Encoder	Patch ($P = 512$)	385M	TS-Pile (1B obs)	Recon Head	cmu/moment
Lag-Llama [14]	Mila / MS	ICML'24	Native TSFM	Decoder	Lag Vectors	2.4M	Monash (7.9K ser)	Student's t	mila/lag-llama
TTM [13]	IBM	NeurIPS'24	Native TSFM	Enc-Dec	Adaptive Patch	1M–8M	Multi-domain Pub	Deterministic	ibm/granite-tsfm
Timer [15]	THUML	ICML'24	Native TSFM	Decoder	Subseries Patch	84M	UTSD (1B pts)	Next-Patch	thuml/Timer
Timer-XL [16]	THUML	NeurIPS'24	Native TSFM	Decoder	Long-Ctx Patch	84M	UTSD-2	Next-Patch	thuml/Timer-XL
Sundial [17]	THUML	arXiv'25	Native TSFM	Decoder	Multi-rate Patch	1.5B	UTSD-3 (>10B)	Continuous	thuml/Sundial
Time-MoE [40]	Ming Jin	ICLR'25	Native TSFM	MoE	Multi-res Patch	2.4B	Time-300B (300B)	MoE Contin	Time-MoE
Timer-S1 [18]	THUML	arXiv'26	Native TSFM	MoE	Serial Patch	8.3B	UTSD-3	MoE Regress	thuml/Timer-S1
ForecastPFN [27]	Abacus/CMU	NeurIPS'23	Native TSFM	Transformer	In-Context PFN	5M	Synthetic Prior	Quantile	abacus/ForecastPFN
t_0 [30]	Invenia/ETH	arXiv'26	Native TSFM	Axial Attn	Context Patch	102M–256M	Pub+Synth	Quantile	Proprietary
Toto 2.0 [45]	Datadog	arXiv'26	Native TSFM	Decoder	Wavelet Tokens	1.2B	Telemetry (1.5T)	Quantile	Proprietary
Time-LLM [19]	Ming Jin	ICLR'24	LLM4TS	Llama-7B	Reprogrammed	7B	Pretrained LLM	Deterministic	KimMeen/Time-LLM
GPT4TS [20]	DAMO	NeurIPS'23	LLM4TS	GPT-2	Cross-Modal	Unreported	Pretrained GPT-2	Deterministic	DAMO/GPT4TS
TEST [31]	PKU/Alibaba	NeurIPS'24	LLM4TS	Llama/BERT	Text Prototype	7B	Pretrained LLM	Deterministic	Proprietary
CALF [32]	THU/DAMO	KDD'24	LLM4TS	Llama-7B	Distilled Patch	7B	Pretrained LLM	Deterministic	Hank0626/CALF
LLMTime [6]	NYU	NeurIPS'23	LLM4TS	GPT-4/Llama	Decimal ASCII	70B+	Pretrained LLM	Sampling Logit	ngruver/llmtime
TEMPO [21]	Ming Jin	ICLR'24	LLM4TS	GPT-2	Decomposed	Unreported	Pretrained GPT-2	Decomposed	caocao/TEMPO
AutoTimes [46]	THUML	NeurIPS'24	LLM4TS	Llama-2-7B	Point-wise	7B	Pretrained LLM	Autoregress	thuml/AutoTimes
Time-MQA [22]	Ming Jin	ACL'25	LLM4TS	Llama-3-8B	Patch + Prompt	8B	Multi-domain MQA	Text/Value	Proprietary

Chronos trains standard T5 architectures (ranging from 20M to 710M parameters) solely with the cross-entropy objective. The predictive distribution over future horizons is obtained by autoregressively sampling categorical bin logits and dequantizing them back into real space. This approach has three key advantages:

- 1) It sidesteps parametric distribution assumptions (e.g., Gaussianity), naturally expressing multi-modal, skewed, or heavy-tailed predictive distributions.
- 2) It enables direct reuse of optimized language transformer implementations (such as FlashAttention and KV-caching).
- 3) In Chronos-2 [8], the architecture is extended to multi-

variate series via grouped channel tokenization and hybrid continuous residual heads.

4.1.2 Google TimesFM: Multi-Patch Output Generation

Unlike Chronos's scalar quantization, Google's TimesFM [4] operates directly on continuous numerical values using continuous patching. TimesFM employs an input patch size of $P = 32$ and an output patch size of $H = 128$. By predicting an entire multi-step block in a single forward step, TimesFM drastically reduces autoregressive unrolling steps, curbing error accumulation and inference latency. TimesFM is trained on a massive corpus of 100B time-series data points collected from Google Trends, Wikipedia pageviews, and synthetic

ARMA processes. Recent extensions examine in-context fine-tuning [9], demonstrating that supplying short conditioning examples within the context window improves zero-shot adaptation. Subsequently, TimesFM-3 scales the architecture to 330M parameters pre-trained over 1T time points, introducing native multivariate forecasting via stacked variate attention and iterative reversible instance normalization (RevIN). Demonstrating the industrial viability of large-scale foundation forecasters, Tacconelli [47] introduces Cadence, pairing TimesFM-3 with an adaptive arithmetic coder to execute error-bounded lossy compression on enterprise telemetry, guaranteeing user-defined maximum deviation bounds while achieving state-of-the-art bit-rate reductions.

4.1.3 The THUML Timer Family and Output Scaling

The THUML research group (led by Mingsheng Long at Tsinghua University) has systematically advanced generative time series pretraining:

- Timer [15]: Conceptualizes a single-series sequential next-patch regression paradigm. Timer is pretrained on the Unified Time Series Dataset (UTSD), comprising over 1B observations across diverse domains. Timer establishes that channel-independent single-series formatting enables universal transfer across arbitrary multivariate configurations.
- Timer-XL [16]: Introduces long-context transformer modeling for temporal signals, scaling context lengths from standard 512 patches to extreme contexts, enabling the model to capture multi-year periodicities without context truncation.
- Sundial [17]: Scales the model to 1.5B parameters pretrained on UTSD-3 (> 10B observations), introducing multi-rate adaptive patching to handle heterogeneous sampling frequencies seamlessly.
- Timer-S1 [18]: Reaches 8.3B parameters via sparse Mixture-of-Experts routing, demonstrating the viability of serial parameter scaling in native temporal models.
- YingLong [48]: Explores test-time output scaling in pre-trained forecasting models by introducing delayed chain-of-thought (CoT) tokens. Rather than predicting future values in a single forward pass, YingLong autoregressively generates intermediate trend and regime change tokens before decoding target values, demonstrating that computational output scaling laws apply to temporal reasoning.

4.1.4 Context-Conditioned and Dual-Horizon Autoregressive Models

Standard autoregressive models often struggle with the structural tension between resolving high-frequency short-term fluctuations and maintaining long-range macro trajectories. Recent 2025–2026 architectures address this via explicit context-conditioning:

- TiRex [28]: Auer et al. (JKU Linz / ELLIS) propose an asymmetric dual-horizon in-context architecture. TiRex isolates local fine-grained patches from global horizon summaries, using memory compaction to preserve multi-scale context without quadratic computational blowup.
- t_0 Family [30]: Meyer et al. introduce t_0 , releasing open-weights checkpoints t_0 -alpha (102M) and t_0 -beta (256M). Departing from channel-independent univariate approaches, t_0 explicitly conditions predictions on historical

targets, past dynamic covariates, and known-future covariates (e.g., promotional calendars, weather forecasts). By alternating axial self-attention along the temporal axis and cross-variate attention across feature channels, t_0 attains aggregate CRPS of 0.4738 and MASE of 0.6865 on GIFT-Eval, standing within 4% of the top zero-shot forecaster.

- Toto Family [45], [49]: Datadog introduced Toto (151M parameters) [49], trained on 1T cloud infrastructure and application metrics to deliver joint zero-shot quantile forecasting and out-of-distribution anomaly scoring. Toto 2.0 [45] subsequently scales from 4M to 2.5B parameters using Maximal Update Parametrization (μ P), achieving state-of-the-art results on GIFT-Eval and the contamination-resistant TIME benchmark.
- Tabby Open Recipe [50]: Addressing the pervasive opacity of proprietary pretraining mixtures, Xie et al. introduce Tabby (120M parameters), releasing an open-source, long-context probabilistic foundation model accompanied by a fully reproducible data curation pipeline and an open 150B-token pretraining corpus.

4.1.5 Lag-Llama: Probabilistic Modeling with Lag Embeddings

Developed by Morgan Stanley and Mila, Lag-Llama [14] adapts the Llama decoder backbone by substituting word embeddings with lag feature vectors. Specifically, it computes lags corresponding to standard calendar frequencies (e.g., quarter-hour, day, week, year). Lag-Llama outputs the parameters of a Student’s t distribution, achieving robust probabilistic calibration on out-of-distribution Monash datasets.

4.2 Prior-Data Fitted Networks (PFNs) and In-Context Meta-Learning

A radically distinct paradigm in native foundation modeling is the Prior-Data Fitted Network (PFN) and in-context learning framework, which foregoes empirical training corpora or iterative gradient steps entirely in favor of synthetically generated structural priors:

- ForecastPFN [27]: Dooley et al. demonstrate that a transformer trained exclusively on synthetic time series generated from a Bayesian mixture of Gaussian processes, ARMA models, and dynamical equations can approximate posterior predictive distributions in a single forward pass. ForecastPFN requires zero gradient steps or hyperparameter tuning at inference time, achieving competitive zero-shot accuracy across diverse real-world datasets.
- TabPFN-TS [51]: Extending tabular foundation models (TabPFN-v2) to the temporal domain, Müller et al. propose TabPFN-TS (11M parameters). By encoding time series as tabular regression tasks conditioned on causal lag features and exogenous metadata, TabPFN-TS achieves state-of-the-art accuracy on covariate-informed forecasting across GIFT-Eval and fev-bench, proving that in-context synthetic priors offer extreme parameter efficiency.
- In-Context Time Series Predictor (ICTSP) [52]: Lu et al. (ICLR 2025) formalize in-context regression for continuous temporal sequences. By structuring historical context and query sequences within an interleaved prompt-query attention mechanism, ICTSP (42M parameters) executes zero-gradient meta-adaptation across disparate sampling

frequencies and noise regimes without updating network weights.

- Latent Time PFN (LaT-PFN) [53]: Verdenius et al. integrate Prior-data Fitted Networks with the Joint Embedding Predictive Architecture (JEPA), performing in-context meta-learning directly in a prediction-optimized latent representation of the data-generating stochastic process. By introducing an abstract normalized time axis and cross-series context conditioning, LaT-PFN accommodates arbitrary sampling granularities and prediction horizons. Furthermore, the architecture spontaneously induces multi-step patch embeddings in latent space without explicit patch tokenization supervision, yielding highly competitive zero-shot probabilistic forecasts.

4.3 Masked Encoders, Multiscale Mixers, and Multi-Task Architectures

4.3.1 Salesforce MOIRAI: Any-Variate Cross-Attention

Salesforce’s MOIRAI [5] addresses two central limitations of previous models: rigid patch sizes and channel-independence. MOIRAI introduces Any-Variate Attention, flattening an arbitrary multivariate sequence $\mathbf{X} \in \mathbb{R}^{C \times T}$ into a sequence of variate-tagged patch tokens. Furthermore, it employs a multi-patch projection head capable of simultaneously processing patch lengths of 8, 16, 32, 64, and 128. MOIRAI is trained on the Large-scale Open Time Series Archive (LOTSAs), containing 27B observations across 9 distinct domains. The model outputs mixtures of parametric distributions, achieving top-tier ranking on zero-shot benchmarks. Moirai 2.0 [11] and Moirai-MoE [10] further extend this framework to sparse routing with 1.1B parameters.

4.3.2 CMU MOMENT and UniTS: Universal Multi-Task Representation

While most TSFMs focus exclusively on forecasting, CMU’s MOMENT [12] investigates foundation models for universal multi-task analysis (forecasting, classification, anomaly detection, and imputation). Based on a masked T5 encoder (385M parameters), MOMENT is pretrained on the Time-series Pile (1B observations) using masked patch reconstruction. Similarly, Harvard’s UniTS [39] unifies multi-task modeling through a single transformer backbone. By combining shared attention operators with task-specific prompt vectors and dynamic gating networks, UniTS demonstrates that cross-task representations mutually reinforce classification and forecasting generalization.

4.3.3 Lightweight Mixers and Exogenous Variable Modeling

Challenging the premise that foundation models require massive parameter scales:

- Tiny Time Mixers (TTM) [13]: IBM Research introduced TTM (1M–8M parameters), utilizing lightweight multi-level TSMixer blocks with adaptive resolution sampling. TTM out-executes models orders of magnitude larger while running inference in milliseconds on standard edge CPUs.
- TimeMixer and TimeMixer++ [43], [44]: Wang et al. formulate multiscale mixing architectures. TimeMixer decomposes signals into downsampled representations to disentangle multiscale seasonalities, while TimeMixer++ extends

this principle into a general temporal pattern machine supporting cross-domain transfer.

- TimeXer [54]: Specifically targets exogenous variables by learning cross-attention maps between target end-points and exogenous features, avoiding noise propagation from uninformative covariates.
- Kairos [55]: Introduces predictability-stratified pretraining (PreSTS), filtering chaotic or non-deterministic temporal noise during self-supervised training to ensure foundation models focus representational capacity on predictable temporal dynamics.
- LightGTS [56]: Wang et al. (ICML 2025) challenge the assumption that cross-domain generalizability requires heavy parameter footprints. LightGTS (5.8M parameters) introduces dual-path hierarchical patching and adaptive parameter allocation, matching the zero-shot accuracy of foundation models two orders of magnitude larger while running inference directly on resource-constrained edge microcontrollers.

4.4 Continuous Flow Matching and Polynomial Memory Networks

A vital 2025 frontier replaces discrete tokenization with continuous-time dynamical systems:

- FLAME [29]: Wu et al. introduce Flow-Enhanced Legendre Memory Models. FLAME compresses long continuous temporal histories onto orthogonal Legendre polynomial bases, avoiding memory degradation over extended sequences, while modeling predictive distributions via continuous normalizing flows.
- FlowState [41]: King et al. formulate time series forecasting via continuous-time ODEs and flow matching. FlowState achieves strict sampling-rate equivariance, allowing zero-shot inference across irregularly sampled or arbitrary continuous time-steps without interpolation artifacts. Despite its compact footprint (12M parameters), FlowState achieves top-tier CRPS on GIFT-Eval.
- EIDOS [57]: Zhou et al. (Ming Jin’s group, 2026) introduce latent-space predictive learning for temporal foundation models via a Joint-Embedding Predictive Architecture (JEPA). Rather than optimizing token-level mean-squared reconstruction—which forces the model to expend capacity memorizing stochastic high-frequency jitter—EIDOS predicts future dynamics within a learned latent metric space, yielding semantically structured representations that transfer robustly across unseen domains.

4.5 Mixture-of-Experts (MoE) and Scaling to Billions

A major architectural breakthrough between 2024 and 2026 is the deployment of Sparse Mixture-of-Experts (MoE) in time series. In Time-MoE [40] (Ming Jin’s group), the model scales to 2.4B parameters while only activating a subset of experts per patch token. Trained on Time-300B (300B time points), Time-MoE demonstrates clear empirical power-law scaling in validation loss, proving that scaling laws discovered in NLP hold equally for native temporal representations when sufficient data volume is supplied. Similarly, Moirai-MoE [10] (1.1B) and Timer-S1 [18] (8.3B) substantiate that sparse MoE routing allows foundation models to specialize on domain-specific dynamics (e.g., high-frequency finance vs.

slow-moving climate trends) while keeping inference latency within practical bounds.

4.6 Parameter-Efficient Fine-Tuning vs. In-Context Adaptation

A central dilemma in deploying foundation models is choosing between zero-shot inference, in-context learning, parameter-efficient fine-tuning (PEFT), and full fine-tuning. Table 3 categorizes the core adaptation paradigms across efficiency, computational overhead, and empirical generalization trade-offs.

Empirical synthesis reveals two critical principles governing downstream adaptation:

- 1) The In-Context Learning Frontier: Prior-Data Fitted Networks (TabPFN-TS, ForecastPFN) and meta-learning predictors (ICTSP) demonstrate that conditioning on prompt-target pairs inside the attention context matches 100-step gradient descent while executing in a single forward pass, providing an ideal mechanism for real-time edge streaming where backpropagation is computationally prohibited.
- 2) PEFT Prevents Semantic Collapse: As demonstrated in CALF [32] and TTM [13], tuning less than 1% of weights via low-rank matrices or adaptive resolution prefix blocks yields within 2% of full fine-tuning performance on target benchmarks while completely insulating the model against catastrophic representation drift.

4.7 Spatio-Temporal Foundation Models: Unifying 1D Streams and Graph Topologies

A pervasive frontier in foundation modeling is extending beyond isolated 1D univariate or channel-independent series toward structured spatio-temporal systems. In transportation networks, urban human mobility, power distribution grids, and planetary meteorological monitoring, continuous observation streams are inextricably situated upon complex underlying geometric topologies [58].

Formally, a spatio-temporal system is represented as a spatial topology $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V}$ denotes a set of $N = |\mathcal{V}|$ spatial sensor nodes or grid cells, $\mathcal{E}$ represents physical or learned relational edges, and $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the spatial adjacency matrix (or spatial distance metric). Given historical observations $\mathbf{X}_{1:T} = [\mathbf{x}_1, \dots, \mathbf{x}_T] \in \mathbb{R}^{T \times N \times D_{\text{in}}}$, the spatio-temporal foundation objective requires forecasting future states $\mathbf{X}_{T+1:T+H} \in \mathbb{R}^{H \times N \times D_{\text{out}}}$ across both seen and entirely unseen spatial topologies $\mathcal{G}^*$:

$$P(\mathbf{X}_{T+1:T+H} \mid \mathbf{X}_{1:T}, \mathcal{G}) = \prod_{h=1}^H P(\mathbf{x}_{T+h} \mid \mathbf{X}_{1:T+h-1}, \mathbf{A}). \quad (14)$$

Traditional deep spatio-temporal architectures (e.g., STGCN, DCRNN) relied on fixed static spatial graph Laplacian operators and task-specific training, catastrophically failing when deployed zero-shot across new sensor networks or different urban geographies. Modern Spatio-Temporal Foundation Models (STFMs) circumvent this structural barrier through four primary paradigms:

- Dual Spatio-Temporal Self-Attention with Graph Biases (OpenCity): Li et al. [59] present OpenCity, an open-world foundation model for urban traffic forecasting. OpenCity decomposes representations into decoupled spatial and temporal transformer layers. To eliminate reliance on rigid fixed adjacency matrices, OpenCity injects generalized graph wavelet positional encodings and relative spatial distance embeddings into multi-head self-attention. Pre-trained on multi-city traffic networks comprising over 50M observations, OpenCity demonstrates remarkable zero-shot transfer across previously unobserved metropolitan road networks.
- Prompt-Empowered Universal Masked Autoencoding (UniST): Yuan et al. [60] introduce UniST (Tsinghua FIB Lab), moving beyond scenario-specific architectures. UniST formulates spatio-temporal forecasting through unified generative masked pretraining. To accommodate heterogeneous sampling rates, variable sensor densities, and semantic domain shifts (e.g., traffic velocity vs. pedestrian flow vs. bike sharing), UniST injects knowledge-guided continuous prompt tokens into the transformer backbone, guiding the shared model to dynamically adapt spatial-temporal priors without altering backbone parameters.
- Spatio-Temporal Diffusion Transformers (UrbanDiT): Li et al. [61] establish UrbanDiT, scaling Diffusion Transformers (DiT) to open-world spatio-temporal systems. UrbanDiT tokenizes both irregular spatial graphs and regular spatial raster grids into unified spatio-temporal patches. By combining reverse-time score-based diffusion denoising with task prompt embeddings, UrbanDiT unifies diverse downstream tasks—including bidirectional forecasting, spatial imputation, and trajectory extrapolation—under a single generative score network.
- Minimalist Scaling and Inductive Regularization (UrbanFM): Chen et al. [62] present UrbanFM, scaling spatio-temporal foundation models up to 120M parameters. UrbanFM challenges over-engineered spatial graph heuristics, proving that minimalist self-attention with constrained inductive biases and MiniST tokenization (which normalizes variance across disparate city sensors) achieves superior scaling laws and cross-city zero-shot transfer compared to complex graph convolution-transformer hybrids.
- Satellite Image Time Series Foundation Models (TiMo): Qin et al. [63] introduce TiMo, scaling spatio-temporal foundation architectures to Satellite Image Time Series (SITS). Departing from standard vision transformers that process static frames independently, TiMo incorporates a hierarchical spatio-temporal gyroscope attention mechanism that captures evolving multi-scale seasonal patterns across planetary terrain. Pre-trained on the MillionST dataset comprising one million satellite image phases across 100,000 geographic locations over five years, TiMo unifies deforestation monitoring, crop classification, and flood risk segmentation under a single masked autoencoding foundation.
- Unified Conceptual Vision and Generalization Challenges: Goodge et al. [58] formulate a comprehensive taxonomy for STFMs, identifying four orthogonal dimensions of generalization required for true foundational capability: (1) Spatial Generalization across varying graph topologies and geographic layouts; (2) Temporal Generalization across diverse sampling resolutions and non-stationary distribution shifts; (3) Scale Generalization across localized sensor clusters and regional macro-grids; and (4) Cross-Domain Generalization

TABLE 3
Systematic Comparison of Downstream Adaptation Paradigms for Time Series Foundation Models.

Adaptation Paradigm	Exemplar Models	Trainable Params	Gradient Steps	Memory Overhead	Data Regime / Sample Efficiency	Generalization & Stability
Zero-Shot Transfer	Chronos-2 [8], TimesFM [4], t_0 [30], Toto 2.0 [45]	0%	0	Baseline (KV-Cache only)	Zero target samples required; immediate out-of-the-box deployment	Preserves broad cross-domain calibration; bounded by pretraining domain overlap
In-Context Meta-Learning	ForecastPFN [27], TabPFN-TS [51], ICTSP [52], TimesFM-IC [9]	0%	0	Low-Medium (Expanded context window)	Extreme few-shot (5-50 context sequences); zero fine-tuning FLOPs	Immune to catastrophic forgetting; limited by maximum attention context length
Parameter-Efficient (PEFT / LoRA / Prefix)	TTM [13], CALF [32], Time-LLM [19], OFA [20]	0.05%-2.0%	Few (50-500)	Low (< 10% activation memory)	Highly sample-efficient (5%-10% target dataset matches full training)	Mitigates representation collapse via low-rank constraints; retains core backbone weights
Full-Parameter Fine-Tuning	MOMENT [12], PatchTST [2], Timer [15]	100%	Many (1K-10K)	High (4× model size for optimizer states)	Requires substantial target history (100% dataset); prone to overfitting on small data	Achieves lowest in-domain MSE on large stationary datasets; suffers catastrophic forgetting

bridging human mobility, climate phenomena, and cyber-physical infrastructure.

These advances demonstrate that spatial graphs and 1D temporal series can be harmoniously unified within scalable transformer and diffusion backbones, establishing spatio-temporal modeling as a vibrant pillar of modern foundation model research.

4.8 Lightweight, Edge-Native, and Microcontroller TSFMs

While industrial-scale foundation models scale from hundreds of millions to billions of parameters, mission-critical operational technology (OT)—such as cyber-physical manufacturing loops, wireless telecommunications infrastructure, and embedded microcontrollers—operates under rigid constraints of sub-10ms response times, micro-watt energy budgets, and sub-100MB RAM ceilings. This has catalyzed a dedicated design movement toward edge-native foundation models:

- **Tiny-TSM** (Birkel, 2025) [64]: Birkel presents Tiny-TSM, establishing that foundation capabilities can be synthesized within an ultra-compact 23M parameter footprint. Trained on a single consumer-accessible A100 GPU in less than one week using the synthetic SynthTS generation pipeline, Tiny-TSM incorporates a novel causal input normalization scheme. This formulation enables dense next-token prediction loss without autoregressive rollout collapse, matching or outperforming industrial-scale billion-parameter foundation baselines across medium- and long-horizon forecasting tasks under mean squared error.
- **APEX: Network-Native Wireless Telemetry** (Pradhan et al., 2026) [65]: Developed at Cisco Systems, APEX addresses the failure mode where generic time series models falter on bursty, zero-inflated, and multi-layer coupled wireless network telemetry. APEX introduces a decoder-only transformer pre-trained on 10-channel telemetry spanning 4,500 production enterprise networks ($\approx 100K$ access point time series). It is released in two operational tiers: APEX-Large (269M parameters for cloud analytics, reducing MAE by 18% over Toto [45]) and APEX-Edge (10.5M parameters), which executes sub-second, privacy-preserving zero-shot anomaly detection directly on router/AP-class edge hardware.
- **Ultra-Lightweight Patch-Independent MLPs** (Cheraghinia et al., 2025) [66]: Cheraghinia et al. demonstrate that transformer self-attention is unnecessary for constrained

edge devices. By deploying simple patch-independent multi-layer perceptron (MLP) encoders comprising a mere 21K trainable parameters, their foundation architecture executes downstream wireless modulation classification, technology recognition, and line-of-sight detection in 0.33ms on commodity microcontrollers, maintaining $> 97\%$ accuracy across previously unseen data classes without representation degradation.

- **MACROCAST: Vintage-Consistent Real-Time Foundations** (Carriero et al., 2026) [67]: Carriero, Pettenuzzo, and Shekhar formulate MACROCAST, an ultra-efficient 15M-parameter foundation model for real-time macroeconomic forecasting that systematically eliminates both temporal lookahead contamination and revision bias. Standard TSFMs trained on fully revised time series suffer from hindsight leakage because historical macroeconomic indicators (e.g., GDP, non-farm payrolls) undergo extensive post-hoc revisions unavailable to real-time decision-makers. MACROCAST rules out both leakage forms by pretraining on purely synthetic time series generated from Bayesian Vector Autoregressions (BVARs) and Dynamic Factor Models (DFMs) in one GPU-day, followed by nine-minute fine-tuning on vintage-specific ALFRED releases. On the real-time FRED-MD benchmark, MACROCAST beats AR(1) on 80% of horizon pairs and surpasses Chronos-2 [8] and classical BVARs without data leakage.
- **TQS-PTQ: Dynamical Systems Quantization Sensitivity** (Pavlova et al., 2026) [68]: Pavlova et al. reframe post-training quantization (PTQ) for time-series foundation forecasters through the mathematical lens of dynamical systems stability. Conventional PTQ sensitivity heuristics are tightly coupled to specific quantizer bit-widths and calibration sets. In contrast, the Trajectory-based Quantization Sensitivity Score (TQS) models the autoregressive latent rollout as a discrete-time dynamical system $\mathbf{z}_{t+1} = \Phi(\mathbf{z}_t, \mathbf{x}_t)$, rigorously bounding how local quantization perturbation errors amplify across the forecasting horizon. Decoupled from quantizer selection, TQS-PTQ derives an exact sensitivity budget across layers prior to quantization, enabling robust mixed-precision allocation (FP16 down to INT4) for compiled, black-box edge models without requiring calibration data or second-order Hessian approximations.
- **Calibration Bets on the Past: 4-Bit Activation Calibration** (Ye & Wanjiku, 2026) [69]: Ye and Wanjiku conduct an ex-

tensive empirical audit of post-training quantization across 560 neural forecasters and eight walk-forward testing years on S&P 500 volatility series. Their findings demonstrate a fundamental dichotomy between weight and activation quantization: while 8-bit quantization is largely insensitive to calibration methodology, static 4-bit activation quantization causes severe degradation under standard absolute-maximum (abs-max) calibration, destroying 11% to 62% of predictive signal. Replacing abs-max with percentile calibration recovers 53% to 94% of this loss by effectively filtering temporal outliers and structural distribution breaks, establishing activation calibration as an indispensable prerequisite for 4-bit time-series deployment.

- **On-Device Fully Quantized Training on Microcontrollers (MCU-FQT, Deutel et al., 2024) [70]:** Deutel et al. break through the barrier of edge adaptability on bare-metal ARM Cortex-M microcontrollers. Addressing the severe resource constraints of microcontrollers (< 256 KB SRAM and < 1 MB Flash memory), they formulate Fully Quantized Training (FQT), executing both forward inference passes and backward gradient propagation entirely within 8-bit integer arithmetic. Paired with dynamic partial gradient updates that selectively update only high-sensitivity layers, MCU-FQT enables embedded sensor forecasters to continuously fine-tune to streaming distribution shifts on-device without floating-point emulation overhead.
- **Sparse Binary Transformers: 1-Bit Weights for Multivariate Time Series (Gorbett et al., 2023) [71]:** Gorbett, Shirazi, and Ray pioneer extreme 1-bit binary parameterization ($\mathbf{W} \in \{-1, +1\}^{d_{\text{out}} \times d_{\text{in}}}$) for multivariate time-series transformers. Rather than relying on floating-point matrix multiplications $\mathbf{W}\mathbf{x}$, binary quantization transforms linear projections in multi-head self-attention and patch feedforward blocks into addition-only accumulation:

$$\mathbf{y} = \mathbf{W}\mathbf{x} = \sum_{j:W_{ij}=+1} x_j - \sum_{j:W_{ij}=-1} x_j. \quad (15)$$

Coupled with learnable binary scaling vectors and multi-rate patch downsampling, Sparse Binary Transformers achieve an 87% parameter compression ratio and an order-of-magnitude reduction in arithmetic logic unit (ALU) gate toggle rates on edge FPGAs and embedded ASICs, proving that temporal attention maps can be effectively learned under binary weight discretization.

- **Q-DEQ: Quantized Deep Equilibrium Forecasters (Yang et al., 2026) [72]:** Yang et al. address edge deployment coding constraints by bridging Deep Equilibrium Models (DEQs) with low-bit quantization. Conventional deep architectures stack explicit feedforward layers, incurring memory footprints proportional to network depth. In contrast, DEQs simulate infinite effective depth by computing the implicit fixed-point equilibrium state:

$$\mathbf{z}^* = f_{\theta}(\mathbf{z}^*, \mathbf{x}), \quad (16)$$

using a single shared transformation layer (0.3M parameters). Q-DEQ resolves the numerical divergence that standard fixed-point Broyden solvers experience under discrete INT4/INT8 arithmetic by formulating a discrete-solving relaxation and contraction-mapping boundary constraint. Q-DEQ achieves parity with full-precision FP32 PatchTST and DLinear baselines while cutting memory storage to

75 KB, enabling continuous dynamical equilibrium modeling within microcontroller cache bounds.

- **SGA: State-Guided Autoregressive Uncertainty Calibration (Hu et al., 2026) [73]:** Hu et al. address the uncalibrated error accumulation inherent to long-horizon autoregressive rollouts in foundation forecasters. While multi-step forecasting models recursively feed predicted patches $\hat{\mathbf{y}}_t$ back as context $\mathbf{x}_{t+1}$, epistemic errors cascade exponentially, producing miscalibrated predictive intervals. SGA formulates a State-Guided Autoregressive adapter: by extracting the foundation backbone’s internal hidden transition trajectories $\mathbf{h}_t$ and conditioning a dual-headed Gaussian-process parametric head on state-space curvature, SGA dynamically scales rollout variance without requiring computationally prohibitive Monte Carlo sampling or ensemble inference.

These architectures prove that foundation pretraining is not intrinsically tied to massive parameter scales; structured inductive priors, synthetic data curation, and patch-level modularity enable sub-megabyte foundation models for extreme edge environments.

4.9 Non-Stationary Temporal Drift and Test-Time Adaptation (TTA)

A foundational vulnerability of all pre-trained temporal forecasters is the assumption of distributional stationarity between pretraining corpora $\mathcal{D}_{\text{pre}}$ and streaming inference horizons $\mathcal{D}_{\text{test}}$. In open cyber-physical, meteorological, and economic environments, time series are intrinsically non-stationary, exhibiting evolving mean-variance trajectories, seasonal structural breaks, and abrupt concept drift:

$$\mathcal{P}_{\text{train}}(\mathbf{X}_{t-L:t}) \neq \mathcal{P}_{\text{test}}(\mathbf{X}_{t-L:t}), \quad \mathcal{P}_{\text{train}}(\mathbf{Y}_{t:t+H} \mid \mathbf{X}_{t-L:t}) \neq \mathcal{P}_{\text{test}}(\mathbf{Y}_{t:t+H} \mid \mathbf{X}_{t-L:t}) \quad (17)$$

While standard fine-tuning requires labeled supervision and risks catastrophic forgetting of general priors, Test-Time Adaptation (TTA) enables pre-trained foundation models to continually adapt their latent representations to streaming distribution shifts using only unlabeled observations available at inference time [74], [75].

Recent advances formalize TTA for foundation time-series forecasters across three primary paradigms:

- **Online Test-Time Adaptation Forecasters (TSF-TTA):** Kim et al. [74] establish the TSF-TTA framework, systematically addressing non-stationarity in temporal models without requiring access to source pretraining datasets. TSF-TTA freezes the feature extraction backbone and restricts test-time updates to reversible instance normalization statistics and lightweight dynamic linear adapter heads. By minimizing an unsupervised temporal consistency objective across sliding test windows, TSF-TTA dynamically tracks concept drift, cutting test-time prediction error by up to 22% across non-stationary benchmarks while incurring negligible compute overhead.
- **Continuous Dynamical Adaptation via Neural ODEs (AdaNODEs):** Dang et al. [75] introduce AdaNODEs, moving beyond discrete heuristic gradient steps by formulating test-time parameter trajectories through continuous-time Neural Ordinary Differential Equations (NODEs). In AdaNODEs, the evolution of latent state representations

$\mathbf{z}(t)$ under streaming temporal drift is governed by a parameterized continuous vector field:

$$\frac{d\mathbf{z}(t)}{dt} = f_{\theta_{\text{ode}}}(\mathbf{z}(t), t), \quad \mathbf{z}(t_1) = \mathbf{z}(t_0) + \int_{t_0}^{t_1} f_{\theta_{\text{ode}}}(\mathbf{z}(\tau), \tau) d\tau. \quad (18)$$

By updating only an ultra-compact 0.15M parameter NODE adapter rather than backbone weights, AdaNODEs smoothly accommodates irregular sampling intervals and continuous distribution shifts, demonstrating superior parameter efficiency and convergence stability over standard gradient descent.

- **Regime-Guided Meta-Control (RG-TTA):** Kumar et al. [76] resolve the dilemma between under-adaptation (failing to catch sudden shifts) and over-adaptation (overfitting to transient noise). RG-TTA introduces a regime-guided meta-controller that continuously computes rolling distribution statistics (spectral energy, mean drift, variance change) to classify the streaming series into stationary baseline vs. shifted regimes. The meta-controller dynamically modulates the adaptation learning rate $\eta_t = \sigma(\mathbf{W}_r \mathbf{r}_t + b_r)$ and gradient clipping thresholds, applying aggressive updates during genuine regime changes while damping updates during stable periods to preserve foundational knowledge.

4.10 Causal Discovery, Structural Priors, and Counterfactual Foundation Models

While contemporary time series foundation models exhibit remarkable correlational forecasting capabilities, their black-box representations struggle with out-of-distribution intervention reasoning, policy simulation, and root-cause attribution [77]. Integrating Structural Causal Models (SCMs) into foundation architectures represents a crucial frontier bridging deep representations with causal mechanics [78], [79].

Formally, a dynamic temporal structural causal model defines a multivariate time series $\mathbf{X}_t = [X_{1,t}, \dots, X_{D,t}]^\top$ across time steps t via a system of structural equations over parent variables $\text{PA}(X_{i,t})$ and independent exogenous noise processes $U_{i,t}$:

$$X_{i,t} := f_i(\text{PA}_{\text{inter}}(X_{i,t}), \text{PA}_{\text{intra}}(X_{i,t}), U_{i,t}), \quad U_{i,t} \sim \mathcal{P}_{U_i}, \quad (19)$$

where $\text{PA}_{\text{inter}}(X_{i,t}) \subseteq \{X_{j,t-\tau} \mid j \in [D], \tau \geq 1\}$ denotes lagged inter-temporal causal parents and $\text{PA}_{\text{intra}}(X_{i,t}) \subseteq \{X_{j,t} \mid j \neq i\}$ represents instantaneous intra-temporal parents, defining an underlying causal Directed Acyclic Graph (DAG) $\mathcal{G}_{\text{causal}}$.

Recent foundation architectures synthesize causality across three complementary frontiers:

- **Foundation Models for Time-Series Causal Discovery (Causal-PT):** Stein et al. [78] demonstrate that causal DAG discovery from temporal signals can be learned end-to-end by a pre-trained foundation transformer. Rather than relying on combinatorial constraint-based testing (e.g., PCMCi) or continuous optimization (e.g., DYNOTEARS), Causal-PT pre-trains a 25M-parameter transformer over 10 million synthetically generated nonlinear structural causal processes. At test time, Causal-PT maps raw multivariate temporal trajectories directly to the estimated causal adjacency matrix $\hat{\mathbf{A}}$ in a single feedforward pass, outperforming classical algorithms while exhibiting high robustness to observational noise.

- **Interventional Time Series Priors for In-Context Causal Models (CausalTimePrior):** Thumm and Chen [79] pioneer the fusion of Prior-Data Fitted Networks (PFNs) with temporal structural causality. Prior PFNs (e.g., TabPFN, ForecastPFN) were constrained strictly to observational data. CausalTimePrior constructs a synthetic generator producing hundreds of millions of diverse temporal structural causal models (TSCMs) paired with both observational trajectories and atomic interventions $do(X_j = \alpha)$. By pre-training a transformer on these paired trajectories, CausalTimePrior performs exact in-context causal effect estimation and counterfactual forecasting in a single forward evaluation without requiring gradient optimization.
- **Causal Time Series Generation via Diffusion Models (CaTSG):** Xia et al. [80] introduce CaTSG, establishing a structural causal diffusion framework for multivariate time series. CaTSG incorporates time-varying causal graph representations into the reverse-time score-matching process:

$$s_\theta(\mathbf{x}_t, t, \mathcal{G}_{\text{causal}}) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t \mid \mathcal{G}_{\text{causal}}). \quad (20)$$

By constraining the generative diffusion trajectories through learned causal adjacency masks, CaTSG guarantees that generated trajectories strictly adhere to physical causal dependencies, enabling controllable simulation across observational, interventional, and counterfactual query spaces.

4.11 Continuous-Time Asynchronous Event Streams and Market Microstructure Foundations

A fundamental limitation of conventional time series foundation models lies in their rigid presumption of uniform, synchronous discrete sampling grids ($\Delta t = \text{const}$). In real-world environments—such as financial limit order books (LOBs), electronic health records, and seismic sensor networks—observations arrive as continuous-time asynchronous event streams $\mathcal{S} = \{(t_i, \mathbf{m}_i)\}_{i=1}^N$, where inter-arrival intervals $\tau_i = t_i - t_{i-1} \in (0, \infty)$ span multiple orders of magnitude and marks $\mathbf{m}_i \in \mathcal{M}$ encode heterogeneous event features. Imposing artificial uniform binning incurs substantial temporal distortion, aliasing high-frequency bursts while creating sparse, computationally wasteful zero-padded windows during quiescent periods.

Formally, an asynchronous event sequence is governed by a marked temporal point process characterized by a conditional intensity function $\lambda^*(t) = \lim_{\Delta t \rightarrow 0^+} \frac{\mathbb{P}(\text{event in } [t, t+\Delta t] \mid \mathcal{H}_t)}{\Delta t}$, where $\mathcal{H}_t$ represents the filtration of historical events up to time t . Foundation modeling over such continuous-time streams requires architectural invariants capable of handling arbitrary time deltas and cross-dataset intensity scale mismatches:

- **SurF: Generative Continuous-Time Foundation Models via Time Rescaling** (Rezaei et al., 2026) [81]: Rezaei, Balaji, and Krishnan resolve the intractable quadrature integration problem in neural temporal point processes by leveraging the classical Time Rescaling Theorem (TRT) as a learnable bijective continuous flow. The TRT establishes that any point process with continuous cumulative intensity $\Lambda(t) = \int_0^t \lambda^*(s) ds$ can be transformed into a standard unit-rate Poisson process:

$$\xi_i = \Lambda(t_i) - \Lambda(t_{i-1}) \stackrel{\text{i.i.d.}}{\sim} \text{Exp}(1). \quad (21)$$

SurF parameterizes $\Lambda_\theta(t)$ via monotonic neural networks and inverts this transformation: during training, event inter-arrival times are mapped into standardized exponential noise, while generative sampling draws $\xi_i \sim \text{Exp}(1)$ and computes $t_i = \Lambda_\theta^{-1}(\Lambda_\theta(t_{i-1}) + \xi_i)$ in closed form. Crucially, mapping heterogeneous streams to a unified unit-rate latent space enables multi-dataset joint pretraining across earthquakes, retweets, and e-commerce streams, eliminating loss scale incompatibilities and establishing the first scalable foundation model for continuous irregular event forecasting.

- **TradeFM: Generative Market Microstructure Foundation Models** (Kawawa-Beaudan et al., 2026) [82]: Developed at J.P. Morgan AI Research, TradeFM introduces a 524M-parameter generative transformer foundation model trained on billions of asynchronous order-flow events spanning over 9,000 global equities. TradeFM addresses the severe cross-asset non-stationarity of financial microstructure by introducing scale-invariant universal tokenization: order book depths, trade volumes, and micro-price updates are mapped into normalized continuous event embeddings decoupled from idiosyncratic stock price scales. Coupled with a high-fidelity deterministic market matching engine, TradeFM performs zero-shot generative simulation of market dynamics, accurately replicating non-linear stylized facts (volatility clustering, heavy-tailed return distributions, and order-flow autocorrelation decay) while generalizing zero-shot across distinct international geographic regimes (e.g., European and APAC exchanges).

4.12 Physics-Informed Foundations and Scientific Constraint Injection

In mission-critical physical and industrial domains—such as electric power grids, aerospace aerodynamics, and renewable energy generation—purely data-driven foundation models face severe adoption barriers: their unconstrained predictions frequently violate invariant physical conservation laws (e.g., Kirchhoff’s laws, energy conservation, power balance constraints), leading to catastrophic failures if deployed in autonomous control loops. Embedding physical inductors directly into foundation architectures and pretraining pipelines represents an essential paradigm for verifiable scientific foundation models:

- **GridSFM: Physics-Inspired Foundation Models for Power Flow** (Bhan et al., 2026) [83]: Bhan et al. develop GridSFM, a 15M-parameter physics-inspired foundation model for solving AC Optimal Power Flow (AC-OPF) across large-scale electrical transmission grids. Rather than treating power flow as generic spatio-temporal regression, GridSFM embeds the non-linear AC power balance equations:

$$S_i = V_i \sum_{k \in \mathcal{N}_i} Y_{ik}^* V_k^*, \quad \forall i \in \mathcal{V}_{\text{bus}}, \quad (22)$$

directly into a physics-guided graph neural network. To address non-convex feasibility boundaries and disconnected solution manifolds, GridSFM introduces a lifted problem reformulation with logarithmically penalized slack variables. Pretrained over 54 distinct transmission topologies (500 to 4,000 buses), GridSFM achieves a remarkable 2.45% zero-shot generation-cost error when evaluated on unseen 10,000-bus power grids, providing warm-start trajectories

that accelerate commercial interior-point solvers by over an order of magnitude.

- **Physics-Informed Synthetic Histories for Cold-Start Forecasting** (Longarini et al., 2026) [84]: Longarini et al. tackle the severe “cold-start” barrier in renewable energy forecasting, where newly commissioned solar photovoltaic (PV) facilities lack historical telemetry required for foundation model in-context conditioning. The authors introduce a two-stage physics-informed pipeline: deterministic physical model chains (incorporating solar positioning, irradiance transposition, and thermal cell equations via pvlib) construct synthetic production histories $\tilde{\mathbf{X}}_{1:T}$ solely from geographical coordinates and numerical weather predictions. Feeding these physically consistent synthetic histories into off-the-shelf foundation models (TabPFN-TS, Chronos-2, TimesFM) without fine-tuning reduces forecasting error (MAE) by up to 50% compared to classical statistical models across 440 commercial installations, demonstrating that physics-based surrogate conditioning bridges data scarcity in zero-shot foundation deployments.

4.13 Earth Observation, Satellite Radar, and Planetary Remote Sensing Foundations

Planetary-scale temporal observations—acquired via orbital satellites, radar constellations, and atmospheric reanalysis grids—pose unique architectural challenges fundamentally distinct from standard 1D univariate sensor streams. Earth Observation (EO) data is characterized by: (1) multi-modal physical sensing mechanisms (optical multi-spectral reflectance, Synthetic Aperture Radar (SAR) backscatter, and thermal radiometry); (2) highly irregular orbital revisit cadences corrupted by non-stationary cloud occlusions; and (3) multi-resolution spatial grids ranging from 10 m (Sentinel-2) to 25 km (planetary climate reanalysis). Representing an EO time series as a spatial-spectral-temporal tensor $\mathbf{X} \in \mathbb{R}^{T \times C \times H \times W}$ requires foundation models capable of joint spatial patch tokenization, cross-spectral band alignment, and continuous temporal evolution:

- **Prithvi WxC: Planetary Foundation Model for Weather and Climate** (Schmude et al., 2024) [85]: Developed jointly by NASA and IBM, Prithvi WxC scales to 2.3 billion parameters, establishing the largest open-weights foundation model dedicated to global atmospheric dynamics. Trained over 40+ years of high-resolution MERRA-2 and ERA5 planetary reanalysis data spanning 160 vertical and surface atmospheric variables, Prithvi WxC adopts a scaled encoder-decoder transformer operating directly on spherical harmonic geometry. Its generalized tokenization allows a single foundation checkpoint to tackle diverse meteorological tasks—including global multi-day prognostic forecasting, sub-kilometer local downscaling, and extreme tropical cyclone trajectory tracking—consistently outperforming operational numerical weather prediction baselines while reducing inference compute by over $1000\times$.
- **EarthPT: Pixel-Level Earth Observation Foundation Models** (Smith et al., 2023) [86]: Smith et al. introduce EarthPT, a 700M-parameter autoregressive decoder-only foundation model trained on European Space Agency (ESA) Sentinel-2 satellite time series spanning over 10 billion pixel-timestep observations across the United Kingdom and Europe.

EarthPT treats pixel-level multi-spectral reflectance histories as continuous sequential tokens, autoregressively forecasting future surface reflectance across optical and near-infrared bands. EarthPT demonstrates that self-supervised temporal next-patch pretraining enables natural cloud gap inpainting, carbon stock estimation, and zero-shot phenological trajectory forecasting without requiring task-specific human annotations.

- **Prithvi-EO-2.0: Multi-Temporal Geospatial Foundation Models** (Szwarcman et al., 2024) [87]: Building upon NASA’s Harmonized Landsat and Sentinel-2 (HLS) archive, Szwarcman et al. introduce Prithvi-EO-2.0 (up to 300M parameters). Pretrained across 4.2 million global multi-temporal scenes, Prithvi-EO-2.0 incorporates 3D spatio-temporal attention blocks that explicitly disentangle spatial texture correlations from orbital revisit dynamics. The architecture exhibits remarkable transferability across diverse geospatial downstream tasks, establishing state-of-the-art accuracy in multi-temporal crop segmentation, post-disaster flood inundation delineation, and planetary wildfire burn scar mapping under extreme distribution shifts.
- **AgriFM: Multi-Source Temporal Radar-Optical Foundation Models** (Li et al., 2025) [88]: A central vulnerability of optical satellite imagery in agricultural monitoring is chronic cloud cover during key crop growth stages. Li et al. develop AgriFM (86M parameters), a multi-source foundation model that fuses cloud-penetrating Sentinel-1 Synthetic Aperture Radar (SAR; dual-polarization VV and VH backscatter) with Sentinel-2 multi-spectral optical series. AgriFM introduces a hierarchical cross-modal attention mechanism that dynamically aligns microwave surface roughness measurements with optical photosynthetic indices, achieving robust zero-shot agricultural crop classification and yield prediction across varying microclimatic regimes.
- **SpectralGPT: 3D Spectral-Spatial-Temporal Masked Autoencoders** (Hong et al., 2024) [89]: Hong et al. propose SpectralGPT (600M parameters), addressing the continuous spectral continuum of hyperspectral and multispectral Earth observation cubes. Unlike conventional 2D vision models that flatten spectral bands, SpectralGPT implements a progressive 3D tensor masking autoencoder across spatial ($H \times W$), spectral (λ), and temporal (T) axes. Evaluated on over one million spectral cubes, SpectralGPT demonstrates that joint 3D pretraining captures subtle chemical and physical changes over time, outperforming conventional unimodal time-series architectures on land-cover change detection and natural disaster tracking.
- **Changen2: Generative Multi-Temporal Change Foundation Models** (Zheng et al., 2024) [90]: Developed by Zheng, Ermon, and colleagues, Changen2 (120M parameters) introduces a continuous diffusion-based generative foundation model for multi-temporal remote sensing change detection. By conditioning continuous generative flows on bi-temporal satellite scene pairs, Changen2 synthesizes counterfactual change trajectories and generates pixel-accurate change masks without requiring paired manual labels, effectively decoupling seasonal cyclic variations from permanent anthropogenic land-use transitions.
- **Presto: Lightweight Sensor-Agnostic Pixel Transformers**

(Tseng et al., 2023) [91]: Tseng et al. introduce Presto, an ultra-compact 4.8M-parameter foundation transformer designed for pixel-level Earth observation time series. Presto employs a flexible channel-time masking objective that natively handles arbitrary combinations of optical (Sentinel-2, Landsat), microwave (Sentinel-1 SAR), and meteorological (ERA5) signals with missing channels and irregular acquisition dates. Despite its minimal parameter footprint, Presto matches or exceeds models orders of magnitude larger on global fuel moisture prediction, crop classification, and deforestation alerting, demonstrating that compact, sensor-agnostic encoders can deliver state-of-the-art remote sensing foundation capabilities at fraction of the computational budget.

4.14 Hierarchical Spatio-Temporal Graph Foundations and Geometric Mesh Manifolds

While classic time-series architectures assume independent channels or uniform Euclidean grids, real-world physical and socio-economic systems operate over complex, non-Euclidean geometric networks—such as continental river basins, urban traffic graphs, decentralized financial protocols, and spherical atmospheric grids. Formally, a spatio-temporal graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V} = \{v_1, \dots, v_N\}$ is a set of nodes, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of dynamic or static edges characterized by adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$, and each node generates a temporal sequence $\mathbf{X}_v \in \mathbb{R}^{T \times C}$. A foundation model over spatio-temporal manifolds must jointly solve two coupled challenges: (1) learning message-passing representations invariant to permutation and graph relabeling across disparate topologies; and (2) handling extreme temporal sparsity and non-uniform sampling across irregular physical boundaries:

- **GPT-ST: Generative Spatio-Temporal Graph Pre-Training** (Li et al., 2023) [92]: Accepted at NeurIPS 2023, Li et al. formulate GPT-ST, establishing the first standardized generative pre-training framework for spatio-temporal graph neural networks. GPT-ST introduces a spatio-temporal masked autoencoder coupled with customized parameter learners and hierarchical spatial pattern encoding networks. By dynamically clustering road networks into hierarchical region semantics (intra-cluster and inter-cluster), GPT-ST trains with an adaptive easy-to-hard spatio-temporal masking curriculum. The resulting representations seamlessly transfer to downstream forecasting baselines across multi-city PeMS traffic sensor networks with substantial performance boosts.
- **AmazonSWE: Continental-Scale Hydrology on Ultra-Sparse DAG Manifolds** (Cartuyvels et al., 2026) [93]: Cartuyvels et al. address one of the most demanding physical graph challenges: continental monitoring of water surface elevation across river networks. Introducing the AmazonSWE dataset—spanning over 19,000 river reaches over 10 years (2016–2026) in the Amazon basin utilizing NASA/CNES Surface Water and Ocean Topography (SWOT) satellite radar altimetry—the authors formalize the directed acyclic river topology as a massive spatiotemporal graph with extreme sparsity ($< 1\%$ of river sections observed on any given day). To overcome the computational collapse of classical GNNs on this topology, the authors

propose a bidirectional selective state-space model that samples connected subgraphs and flattens space and time into a single sequence equipped with topological directed acyclic graph (DAG) positional encodings. This formulation cuts root-mean-square error (RMSE) against held-out in-situ river gauges by 18%–39% compared to hybrid physical-statistical hydrodynamic models.

- **DeXposure-FM: Time-Series Graph Foundation Model for Decentralized Financial Networks** (Shu et al., 2026) [94]: Shu et al. introduce DeXposure-FM, the first graph foundation model engineered to quantify systemic financial contagion and credit exposure on decentralized finance (DeFi) networks. Trained on the massive DeXposure corpus comprising 43.7 million transaction records across 4,300+ protocols on 602 blockchains (encompassing 24,300+ unique tokens), DeXposure-FM employs a graph-tabular encoder with pre-trained initialization. The foundation architecture jointly predicts multi-horizon protocol liquidity flows and dynamic inter-protocol dependency edge weights, supporting macroprudential stress testing and systemic risk scoring across interconnected blockchain economies.
- **EarthGeometry: Physically Typed and Geometry-Aware Planetary Representations** (Ranjan, 2026) [95]: Addressing planetary Earth foundation models, Ranjan establishes a rigorous formalization of geometric invariance and physical typing. While standard geospatial embeddings treat data as arbitrary vectors in $\mathbb{R}^d$, physical atmospheric and oceanic fields possess strict transformation laws under spatial rotations, reflections, and coordinate chart changes. Ranjan demonstrates that planetary models must explicitly distinguish between scalar fields (temperature, pressure), tangent vector fields (horizontal wind velocities), axial pseudovectors (vorticity, magnetic fields), and higher-order stress tensors. By integrating Hodge and Helmholtz decompositions over spherical geodesic meshes, this framework proves that geometry-aware physical typing substantially improves out-of-distribution generalization across planetary weather forecasting regimes.

4.15 Selective State-Space Models (Mamba/S4) and Linear-Time Sequence Modeling

While multi-head self-attention has powered the majority of early TSFMs, its quadratic computational complexity $O(L^2)$ and quadratic memory footprint with respect to sequence length L pose an insurmountable bottleneck for ultra-long context windows and resource-constrained edge deployments. To overcome this limitation, continuous-time Structured State-Space Models (SSMs) and Selective State-Space Models (Mamba) have emerged as potent alternatives. Formally, a continuous state-space model maps a 1D continuous sequence $x(t) \in \mathbb{R}$ to $y(t) \in \mathbb{R}$ through an intermediate hidden state $\mathbf{h}(t) \in \mathbb{R}^N$:

$$\dot{\mathbf{h}}(t) = \mathbf{A}\mathbf{h}(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}\mathbf{h}(t) + \mathbf{D}x(t). \quad (23)$$

Discretization via Zero-Order Hold (ZOH) across sampling intervals Δ_t yields discrete recurrence matrices:

$$\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A}), \quad \bar{\mathbf{B}}_t = (\Delta_t \mathbf{A})^{-1}(\exp(\Delta_t \mathbf{A}) - \mathbf{I})\Delta_t \mathbf{B}_t. \quad (24)$$

In modern selective state-space architectures, $\mathbf{B}_t$, $\mathbf{C}_t$, and the timescale parameter Δ_t are dynamic, input-dependent projections of the current token x_t . This allows the model to filter

out irrelevant temporal noise while compressing arbitrarily long histories into a fixed-dimensional state $\mathbf{h}_t \in \mathbb{R}^N$ with linear $O(L)$ training time and constant $O(1)$ memory per step during autoregressive generation. Pioneering works have established the foundation validity of SSMs for time series:

- **S-Mamba: Is Mamba Effective for Time Series Forecasting?** (Wang et al., 2024) [96]: Wang et al. formulate S-Mamba to rigorously investigate whether selective state-space models can supplant self-attention in multivariate time-series forecasting. While Transformers excel at pairwise feature comparison, their attention weights often degrade on periodic, noise-dominated signals. S-Mamba introduces bidirectional Mamba blocks operating across both temporal patch sequences (intra-series modeling) and multivariate channels (inter-series cross-variable correlation). S-Mamba matches or exceeds state-of-the-art transformer baselines across ETT, Electricity, and Weather benchmarks while delivering linear memory scaling and $3 \times -5 \times$ speedups on long sequences.
- **TimeMachine: A Time Series is Worth 4 Mambas** (Ahmed and Cheng, 2024) [97]: Accepted at ECAI 2024, Ahmed and Cheng propose TimeMachine, a quadruplet 4-Mamba architecture explicitly designed for long-term forecasting. Recognizing that time series exhibit complex dual-scale dynamics, TimeMachine deploys four interconnected Mamba modules that simultaneously decouple and route representations across two orthogonal axes: (1) forward and backward temporal sequences; and (2) intra-channel local subseries and inter-channel global interactions. TimeMachine maintains a strictly linear memory footprint, resolving GPU out-of-memory failures on context windows exceeding 10,000 timesteps.
- **Bi-Mamba+: Bidirectional Selective State Space for Time Series** (Liang et al., 2024) [98]: A critical drawback of standard causal SSMs is their unidirectional scan, which inevitably discards future-looking contextual correlation in offline forecasting scenarios. Liang et al. introduce Bi-Mamba+, which interleaves forward and backward selective state-space layers with adaptive hidden-state fusion. By processing historical patches in both temporal directions, Bi-Mamba+ eliminates unidirectional bias without incurring the quadratic computational cost of bidirectional full-attention encoders.
- **QuantFlow: Federated Mamba Post-Transformer Foundation Models** (Haider et al., 2026) [99]: Addressing edge-constrained distributed environments, Haider et al. introduce QuantFlow, a post-transformer federated foundation model built entirely upon selective state-space backbones. In federated forecasting across decentralized IoT sensors and medical telemetry, transmitting large Transformer KV-caches and dense attention weights induces severe communication bottlenecks. QuantFlow replaces self-attention with lightweight quantized Mamba blocks, allowing multiple distributed edge nodes to train collaborative global foundations with 70% lower memory footprint and zero raw-data leakage.

4.16 Heterogeneous Multivariate Decoupling, Covariate Homogenization, and Memory-Augmented Adaptation

A pervasive architectural limitation of first-generation TSFMs is their reliance on channel independence or homogeneous Eu-

clidean embeddings. In real-world industrial, environmental, and financial deployments, multivariate time series exhibit severe cross-channel heterogeneity: physical variables possess distinct measurement units, sampling dynamics, and both synergistic and antagonistic interactions. Furthermore, time-series series are accompanied by heterogeneous exogenous covariates (e.g., discrete calendar events, static metadata, continuous weather streams, and unstructured text logs). Naive joint attention across heterogeneous variables frequently induces negative transfer and representation distortion, where dominant variables swamp subtle leading indicators, or where fine-tuning a universal backbone destroys its zero-shot generalization capabilities. Recent foundation models from 2025–2026 systematically address these structural challenges:

- **Falcon-X: Unified Prototype Diff-Attention and Variate Reassembly** (Liu et al., 2026) [100]: Liu et al. formulate Falcon-X (230M parameters), establishing a foundation paradigm designed specifically for heterogeneous multivariate systems. Recognizing that standard softmax attention enforces non-negative affinity weights that cannot capture antagonistic or competing physical dynamics (e.g., supply vs. price, temperature vs. cooling efficiency), Falcon-X introduces Unified Prototype Diff-Attention (UPDA). UPDA decouples raw variates from physical space and projects them into a shared latent prototype space $\mathcal{Z} \subset \mathbb{R}^d$, computing differential affinities:

$$\mathbf{A}_{\text{diff}} = \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}_+^\top}{\sqrt{d}} \right) - \text{Softmax} \left(\frac{\mathbf{Q}\mathbf{K}_-^\top}{\sqrt{d}} \right), \quad (25)$$

where $\mathbf{K}_+$ and $\mathbf{K}_-$ parameterize positive synergistic and negative antagonistic prototype manifolds. Subsequently, a Variate Reassembly Router (VRR) operates via a request-and-dispatch mechanism, querying prototype trajectories to reconstruct variate-specific predictions. Falcon-X achieves state-of-the-art zero-shot generalization across 12 multivariate benchmarks, eliminating channel interference and reducing multivariate forecasting error by 17.4% over channel-independent baselines.

- **UniCA: Unified Covariate Adaptation via Homogenization** (Han et al., 2025) [101]: While foundation models are overwhelmingly pretrained on pure numerical series, production forecasting requires conditioning on heterogeneous covariates (e.g., holiday indicators, continuous price promotions, and categorical regional IDs). Han et al. propose UniCA (120M parameters), introducing the principle of Covariate Homogenization. Rather than concatenating disparate raw covariates into the foundation backbone, UniCA maps all categorical, temporal, and multimodal covariate streams into unified, high-level continuous latent series through dedicated converter modules:

$$\tilde{\mathbf{C}} = \text{Converter}(\mathbf{C}_{\text{cat}}, \mathbf{C}_{\text{cont}}, \mathbf{C}_{\text{text}}) \in \mathbb{R}^{T \times d_c}. \quad (26)$$

UniCA then employs modular Pre-Fusion and Post-Fusion cross-attention blocks that inject the homogenized covariate representations into the frozen foundation backbone without corrupting its pretrained temporal representations. On GIFT-Eval and fev-bench covariate tracks, UniCA improves zero-shot accuracy by up to 22.1% while preserving the foundation model’s zero-shot generalization envelope.

- **TS-Memory: Plug-and-Play Memory Distillation** (Lyu et al., KDD 2026) [102]: Lyu et al. resolve the fundamental dilemma between parametric fine-tuning (which risks catastrophic forgetting and negative transfer) and non-parametric retrieval (which introduces prohibitive latency and retrieval noise during streaming inference). TS-Memory introduces a lightweight (0.5M parameter) plug-and-play transformer memory adapter. During training, an offline, leakage-safe k NN teacher retrieves historical neighbor futures from a target domain datastore, formulating confidence-weighted quantile correction targets. TS-Memory then distills these non-parametric insights into the parametric adapter using a confidence-gated loss:

$$\mathcal{L}_{\text{distill}} = \sum_{t=1}^H w_t \cdot \rho_\tau \left(\hat{y}_t^{(\text{adapter})} - y_t^{(k\text{NN})} \right), \quad w_t = \sigma \left(\frac{\mu_t^{(k\text{NN})} - \text{Var}_t}{\kappa} \right) \quad (27)$$

where w_t selectively halts distillation whenever teacher retrieval uncertainty is high, strictly preventing negative transfer. During online inference, TS-Memory fuses with frozen backbones (e.g., Chronos-Bolt) in constant $O(1)$ time without database lookups, consistently outperforming full fine-tuning across distribution-shifted target domains.

- **iAmTime: Instruction-Conditioned In-Context Meta-Learning** (Saha and Shmakov, 2026) [103]: Developed by Walmart researchers, iAmTime (350M parameters) redefines foundation model adaptation by abandoning parameter fine-tuning in favor of instruction-conditioned in-context learning. Conditioned on explicit input-output demonstration pairs $(\mathbf{X}_{\text{demo}}^{(i)}, \mathbf{Y}_{\text{demo}}^{(i)})_{i=1}^K$, iAmTime infers task intent on-the-fly. The architecture integrates a Hierarchical Multi-Scope Transformer Encoder (extracting multi-resolution temporal dynamics and latent task embeddings) with a Task-Conditioned Patch Decoder that routes representations through specialized expert heads. Pretrained across a massive corpus of over 120 billion real and synthetic observations, iAmTime executes zero-shot task switching across point forecasting, quantile prediction, signal inpainting, and classification without updating a single weight parameter.
- **ST-Prune: Complexity-Aware Spatio-Temporal Dynamic Sample Pruning** (Chen et al., 2026) [104]: Addressing the unsustainable computational overhead of pretraining spatio-temporal foundation models on redundant planetary sensor grids, Chen et al. formulate ST-Prune. ST-Prune tracks the model’s instantaneous sample-level loss gradients and temporal complexity in real time, dynamically pruning repetitive, low-information spatio-temporal slices during training. By allocating GPU compute strictly to high-entropy, topologically challenging boundary trajectories, ST-Prune cuts pretraining wall-clock time by 46% while matching or exceeding the downstream generalization accuracy of full-dataset pretraining.

4.17 Active Sensor Querying, Information Acquisition, and Temporal Causal PFNs

Deploying foundation models across real-world physical sensor grids (e.g., environmental observation, smart manufacturing, and autonomous vehicle fleets) encounters severe communication bandwidth and power constraints. Sensors cannot

transmit continuous telemetry at high frequencies indefinitely; instead, models must actively decide which sensors to query, at which time steps, and when to defer to costly human experts or high-fidelity numerical solvers. This necessitates unifying foundation representations with active information-theoretic acquisition functions and temporal causal discovery:

- **STAP**: Selective Time-Step Acquisition for Physical Dynamics (Kim et al., ICML 2025) [105]: Kim et al. formulate active learning for physical spatio-temporal trajectories governed by partial differential equations (PDEs; Navier-Stokes, reaction-diffusion). Traditional active learning queries full trajectories, incurring prohibitive solver costs. STAP introduces selective time-step acquisition, using a foundation surrogate model to approximate easy trajectories and selectively acquiring only the most informative non-uniform time-steps $t_k \in [0, T]$ from the numerical solver. STAP derives an acquisition function based on expected global variance reduction:

$$\alpha(t^*) = \arg \max_{t \in \mathcal{T}_{\text{cand}}} \mathbb{E} [\text{Tr}(\mathbf{\Sigma}_{\text{post}}(\mathbf{x}_{1:T}) | y_t)] - \text{Cost}(t), \quad (28)$$

enabling the surrogate model to explore a $3.4\times$ wider variety of nonlinear dynamical regimes under a fixed computational budget.

- **L2D-SLDS**: Online Learning-to-Defer via Switching State-Space Models (Montreuil et al., 2026) [106]: Montreuil et al. formulate the first online Learning-to-Defer (L2D) framework for non-stationary streaming sensor networks. In critical industrial applications, the model must dynamically route each time step either to its internal foundation predictor or defer to an external specialist. L2D-SLDS tracks prediction residuals using a factorized Switching Linear-Gaussian State-Space Model (SLDS) decomposing dynamics into discrete regimes s_t , a global shared factor $\mathbf{z}_t$, and sensor-specific idiosyncratic noise ϵ_t . The decision to defer is governed by a learner-aware information query score:

$$Q_t = \underbrace{c_{\text{defer}}}_{\text{Immediate Cost}} - \underbrace{\lambda_1 \mathbb{I}(y_t; \mathbf{z}_t | \mathcal{H}_{t-1})}_{\text{Latent Information Gain}} - \underbrace{\lambda_2 \Delta \mathcal{R}_t(\theta)}_{\text{Learner Regret Reduction}} \quad (29)$$

which achieves superior tracking in non-stationary streaming environments with deferral rates under 2%.

- **TCPFN**: Temporal Causal Prior-Data Fitted Networks (Talupula and Sharma, 2026) [107]: Talupula and Sharma extend Prior-data Fitted Networks to temporal causal discovery on high-dimensional industrial panel series. While classic causal discovery algorithms (e.g., PCMC, Granger causality) scale cubically $O(N^3)$ with variable count and fail under latent confounding, TCPFN trains a discrete-token panel transformer entirely on synthetic Structural Causal Models (SCMs) exhibiting delayed, decaying, and oscillating causal couplings. TCPFN introduces an explicit Causal Judgment Head that outputs calibrated reliability signals: joint predictions of null-effect probability, confounding strength, and mediation fraction for every candidate treatment-outcome pair. Operating entirely zero-shot without per-dataset parameter updates, TCPFN processes industrial systems with thousands of time series in seconds, outperforming constraint-based and score-based discovery baselines on Tennessee Eastman and SWaT benchmarks.

4.18 Online Continual Learning, Temporal Plasticity, and Black-Box Residual Adaptation

While foundation models exhibit strong zero-shot generalization across diverse stationary datasets, real-world deployments encounter continuous, non-stationary temporal drift ($\mathcal{D}_t \neq \mathcal{D}_{t-1}$) where offline weights inevitably become stale. Furthermore, as foundation models are increasingly monetized as closed-source black-box API endpoints, standard backpropagation and white-box parameter fine-tuning become technically impossible. Recent 2025–2026 advances address this fundamental deployment frontier through online continual learning, temporal plasticity preservation, and black-box error-context adaptation:

- **ELF**: Efficient Lightweight Forecast-Adaption (Lee et al., ICML 2025) [108]: Published at ICML 2025, Lee et al. address the high computational barrier of adapting frozen foundation models to streaming feedback. They formulate ELF, a modular, model-agnostic adaptation mechanism operating alongside frozen backbones (e.g., Chronos, TimesFM, MOIRAI). ELF bifurcates online adaptation into two lightweight components: (1) the ELF-Forecaster, a parameter-efficient frequency-domain module that rapidly learns localized distribution shifts from streaming errors, and (2) the ELF-Weighter, an adaptive gating network that dynamically computes convex combination weights:

$$\hat{\mathbf{y}}_t = w_t \hat{\mathbf{y}}_t^{\text{fast}} + (1-w_t) \hat{\mathbf{y}}_t^{\text{FM}}, \quad w_t = \sigma(\mathbf{W}_w [\mathbf{x}_t; \hat{\mathbf{y}}_t^{\text{FM}}; \hat{\mathbf{y}}_t^{\text{fast}}]). \quad (30)$$

By continuously calibrating w_t against streaming validation loss, ELF prevents catastrophic representation collapse while delivering consistent 12%–28% error reductions over static foundation forecasts at negligible edge compute overhead.

- **ORCA**: Learning the Context of Errors for Black-Box TSFMs (Dai et al., NeurIPS 2026) [109]: Confronting commercial API boundaries where internal foundation parameters and gradients are completely inaccessible, Dai et al. formulate ORCA (Online Residual Contextual Adaptation). They identify a core flaw in conventional residual modeling: treating forecast errors $\mathbf{e}_t = \mathbf{y}_t - \hat{\mathbf{y}}_t^{\text{FM}}$ as unstructured white noise. Instead, ORCA proves that residual errors are structurally conditioned on both the input trajectory and the base model’s prediction, defining the context of errors $\mathbb{P}(\mathbf{e}_t | \mathbf{x}_{1:t}, \hat{\mathbf{y}}_t^{\text{FM}})$. ORCA trains a linear adapter over a temporal replay buffer with exponential historical forgetting decay γ^k , employing a predictive-space Bayesian loss:

$$\mathcal{L}_{\text{ORCA}} = \sum_{k=0}^K \gamma^k \left\| \mathbf{y}_{t-k} - \left(\hat{\mathbf{y}}_{t-k}^{\text{FM}} + \sum_{j=1}^M \frac{\exp(-\beta \|\mathbf{x}_{t-k} - \mathbf{c}_j\|^2)}{\sum_l \exp(-\beta \|\mathbf{x}_{t-k} - \mathbf{c}_l\|^2)} \hat{\mathbf{e}}_{t-k}^{(j)} \right) \right\|^2 \quad (31)$$

where a Boltzmann router partitions error distributions across localized dynamical regimes. Evaluated across 5 leading TSFMs (Chronos-Bolt, TimesFM, Time-MoE) on 8 real-world streaming benchmarks, ORCA reduces streaming tracking error by up to 34.2% without requiring a single backward gradient pass through the foundation model.

- **NatSR**: Natural Score-Driven Replay for Continual Forecasting (Urettini et al., ICLR 2026) [110]: Urettini et al. reformulate online continual learning for time series through the lens of parameter filtering. They show that standard

gradient descent over non-stationary streams induces severe gradient noise that distorts long-term representations. NatSR introduces Natural Score-driven Replay, demonstrating that natural gradient descent corresponds to a score-driven updating scheme when optimizing conditional likelihoods. By framing predictions under a Student- t distribution with degrees of freedom ν :

$$\theta_{t+1} = \theta_t + \eta \mathbf{F}_t^{-1} \nabla_{\theta} \log p(y_t | \mathbf{x}_t, \theta_t), \quad p(y | \mu, \sigma^2, \nu) \propto \left(1 + \frac{(y - \mu)^2}{\sigma^2} \right)^{-\frac{\nu+1}{2}} = \mathcal{L}_{\text{Masked}} + \lambda \mathcal{L}_{\text{DVCL}}(\mathbf{Z}_{\text{grid}}, \mathbf{Z}_{\text{temp}}), \quad (32)$$

NatSR endows foundation transformers with natural robustness against extreme temporal anomalies and spikes, while a dynamic scale heuristic selectively populates the replay memory during confirmed regime shifts, mitigating catastrophic forgetting across long-horizon streaming sequences.

- Temporal Plasticity in Billion-Scale Foundation Models (Liu et al., IJCNN 2025) [111]: Liu et al. conduct the first rigorous empirical evaluation of temporal plasticity—the capacity of neural networks to continually absorb non-stationary distributional shifts without suffering irreversible loss of learning capacity—in billion-scale foundation models (Time-MoE 2.4B, Chronos-Large). While compact task-specific networks rapidly suffer from representation stiffness and gradient vanishing during incremental fine-tuning, Liu et al. discover an emergent scaling property: billion-scale TSFMs preserve internal plasticity significantly longer, retaining broad cross-domain transferable representations while adapting efficiently to streaming domain shifts.

4.19 Cross-Sensory Foundations: Multi-Modal Industrial Telemetry and Sensor-Agnostic Tactile Policies

A critical frontier in time series foundation models is transcending simple 1D scalar waveforms to encompass heterogeneous physical telemetry, including high-frequency acoustic vibrations, multi-level power grid dynamics, high-frequency robotic tactile contact streams, and self-supervised latent energy representations:

- FISHER: Multi-Modal Industrial Signal Representation (Fan et al., IEEE TII 2025) [112]: Accepted for publication in IEEE Transactions on Industrial Informatics, Fan et al. tackle the severe heterogeneity of industrial asset monitoring, characterized as the “M5 problem” (multi-source, multi-rate, multi-modal, multi-condition, and multi-fault telemetry). To handle variable sampling rates across vibration, acoustic emissions, current, and dynamic pressure without lossy interpolation or resampling, FISHER introduces sub-band spectral tokenization: treating sampling rate increments as concatenated sub-band orthogonal channels. Pretrained via teacher-student contrastive self-supervised learning across the RMIS benchmark (19 datasets across 4 modalities), the 28M-parameter FISHER achieves state-of-the-art anomaly detection and fault classification while being up to $16\times$ more parameter-efficient than monolithic vision-language adaptations.
- PowerPM: Foundation Model for Power Systems (Tu et al., NeurIPS 2024) [113]: Presented at NeurIPS 2024, Tu et al. formulate PowerPM, a 115M-parameter foundation model specifically engineered for electricity time series (ETS).

Electricity telemetry possesses strict topological hierarchy (city $\rightarrow$ district $\rightarrow$ user) and strong meteorological covariate dependencies. PowerPM unifies a Temporal Transformer with a Relational Graph Convolutional Network (R-GCN) over power grid topologies. Pretrained on massive private and public grid datasets via Dual-View Contrastive Learning (DVCL) and masked ETS sequence prediction:

$$\mathcal{L}_{\text{PowerPM}} = \mathcal{L}_{\text{Masked}} + \lambda \mathcal{L}_{\text{DVCL}}(\mathbf{Z}_{\text{grid}}, \mathbf{Z}_{\text{temp}}), \quad (33)$$

PowerPM demonstrates exceptional transferability across downstream load forecasting, grid stability assessment, and consumer behavior profiling.

- FTP-1: Generalist Foundation Tactile Policy (Yuan et al., 2026) [114]: Addressing the hardware lock-in of robotic tactile perception, Yuan et al. establish FTP-1 (180M parameters), the first sensor-agnostic generalist tactile policy. Tactile streams exhibit drastic cross-hardware variation, spanning optical-tactile gel cameras (DIGIT, Gel-Sight), piezoresistive sensor arrays, and barometric contact skins. FTP-1 maps these disparate physical streams into morphology-aware latent unified tokens conditioned on physical geometry descriptors. Pretrained across 3,000 hours of contact manipulation data spanning 21 distinct sensors and human-robot demonstrations, FTP-1 achieves a +17.2% success gain on known sensors and a +31% zero-shot success gain when transferred to entirely unseen tactile sensors in contact-rich manipulation tasks.
- TouchWorld: Hierarchical Predictive and Reactive Tactile World Model (Zhou et al., 2026) [115]: Zhou et al. formulate TouchWorld (350M parameters), uniting anticipatory world modeling with high-frequency reactive control. In dexterous manipulation, slip and tactile hysteresis occur within milliseconds. TouchWorld decouples manipulation into a low-frequency predictive world model that autoregressively forecasts spatio-temporal contact patch trajectories $\hat{\mathbf{C}}_{t+1:t+H}$ from multimodal visual-tactile contexts, and a high-frequency (> 100 Hz) reactive policy that executes localized closed-loop residual corrections over the EgoTouch bimanual contact corpus.
- TS-JEPA: Temporal Joint-Embedding Predictive Architecture (Ennadir et al., NeurIPS 2024 Workshop) [116]: Ennadir et al. extend LeCun’s Joint-Embedding Predictive Architecture to time series. Conventional autoregressive and masked autoencoding models optimize point-wise reconstruction losses in the observation space, rendering them excessively sensitive to irreducible sensor noise and high-frequency stochastic fluctuations. TS-JEPA discards observation reconstruction entirely: it maps context and target windows into an abstract latent metric space, training a latent energy predictor to minimize:

$$\mathcal{L}_{\text{TS-JEPA}} = \|\mathbf{s}_{\text{target}} - g_{\phi}(\mathbf{s}_{\text{context}}, \mathbf{z})\|_2^2, \quad (34)$$

where $\mathbf{z}$ is a temporal mask conditioning token. By enforcing representation invariance in latent space, TS-JEPA yields superior linear probing and zero-shot transfer for classification and long-horizon forecasting across UCR/UEA and Monash benchmarks.

- RATFM: Retrieval-Augmented Foundation Model for Anomaly Detection (Maru and Sato, 2025) [117]: Maru and Sato address the inability of standard zero-shot TSFMs to condition on normal operational prototypes without

parameter fine-tuning. RATFM (110M parameters) equips foundation transformers with a non-parametric test-time retrieval memory. Given an evaluation window, RATFM queries an in-domain target datastore to retrieve top- k normal subsequence exemplars, feeding them as in-context adaptation anchors to compute reconstruction discrepancy scores. On the multi-domain UCR Anomaly Archive (spanning 9 physical domains), RATFM matches the performance of in-domain supervised fine-tuning while requiring zero parameter modifications.

5 Repurposed Large Language Models (LLM4TS)

In parallel to native foundation models, an active branch of literature investigates repurposing existing Large Language Models (LLMs) for time series analysis. Driven by the hypothesis that multi-billion parameter text models have acquired generalized pattern matching, sequence extrapolation, and relational reasoning abilities, these methods adapt frozen or fine-tuned language backbones to temporal streams.

5.1 Cross-Modal Token Reprogramming

Token reprogramming bridges the structural gap between continuous numerical signals and the discrete token space of language transformers by learning parameter-efficient adapter layers while keeping the core language model frozen.

5.1.1 Time-LLM: Prompt-Guided Reprogramming

Introduced by Ming Jin et al., Time-LLM [19] establishes a canonical reprogramming framework. Input time series are patched into subseries vectors, which are mapped through a linear projection layer into the textual word embedding space of Llama-7B. Crucially, Time-LLM constructs dynamic text prompts containing domain context, task instructions, and dataset statistics (e.g., trend, variance, seasonality). The frozen language model processes the concatenated prompt tokens and reprogrammed temporal tokens, producing representations that are decoded back into continuous forecasts via a linear projection head. Time-LLM demonstrates strong zero-shot and few-shot capabilities across benchmark datasets.

5.1.2 One Fits All (GPT4TS / OFA): Universal Backbone Sharing

In One Fits All (GPT4TS) [20], Zhou et al. explore the minimal adaptation required to transfer language transformers to temporal tasks. By taking a pre-trained GPT-2 backbone, freezing its self-attention and feed-forward layers, and retraining only an input linear patch embedder and output layer, GPT4TS achieves state-of-the-art results across forecasting, anomaly detection, classification, and imputation. The authors argue that multi-head self-attention trained on natural language acts as a universal sequence engine capable of recognizing temporal autocorrelation and periodic waveforms.

5.1.3 Decomposition, Prototype Alignment, and Distillation

Subsequent works systematically refine cross-modal reprogramming:

- TEST [31]: Sun et al. identify the semantic gap between continuous temporal tokens and discrete word embeddings.

TEST constructs a learnable dictionary of text prototypes derived from language vocabulary clusters. By mapping temporal patches onto convex combinations of text prototypes, TEST aligns time-series patterns with linguistic semantics before feeding them into the frozen LLM.

- CALF [32]: Liu et al. address catastrophic representation degradation during cross-modal adaptation. CALF introduces Cross-modal Alignment and Low-rank Fine-tuning, utilizing word-level and sentence-level semantic distillation to preserve the backbone’s latent sequence modeling capabilities while aligning temporal features.
- UniTime [33]: Liu et al. design a language-empowered unified model for cross-domain forecasting. By introducing domain-specific language instructions and dynamic masking, UniTime transfers a single LLM across multiple distinct data distributions without catastrophic interference.
- LLM4TS [118]: Chang et al. propose a two-stage alignment strategy: a self-supervised time-series alignment pre-training phase followed by downstream forecasting fine-tuning, coupled with a two-level temporal aggregation module to capture multi-scale seasonality.
- TimeCMA [119]: Chen et al. design cross-modality alignment between multivariate channels and language representations, using channel-wise text prompts to guide inter-series dependency modeling.
- TEMPO [21]: Decomposes input series into trend, seasonal, and residual components using classical STL decomposition. Each component is independently embedded and guided by specialized prompts, enabling the language model to reason over modular temporal dynamics.
- S²IP-LLM [120]: Introduces semantic-space informed prompting, projecting statistical temporal attributes into dense semantic embeddings that constrain the LLM’s generative search space.
- LLM-Mixer [121]: Kowsher et al. formulate multiscale token-mixing inside frozen language models. By decomposing temporal signals across multiple sampling strides and processing them via parallel attention pathways in LLaMA-2, LLM-Mixer captures multiscale periodicities without full-parameter adaptation.
- TAC-Time [122]: Liang et al. propose the “Texts as Channels” architecture for multimodal forecasting. Instead of forcing numerical values into linguistic strings or shared embedding spaces, TAC-Time treats contextual text streams as auxiliary multivariate channels, employing asymmetric dual-stream cross-attention to inject semantic knowledge into temporal representations without disrupting continuous numerical geometry.

5.2 Direct Prompting and Decimal Tokenization

Rather than learning adapter layers, direct prompting methods interact with off-the-shelf closed or open LLMs (such as GPT-4 or Llama-3) by formatting numeric series as text strings.

5.2.1 Foundations: PromptCast and Generative Formulation

The paradigm of casting numerical forecasting as text generation was pioneered by Xue and Salim in PromptCast [42]. Rather than treating forecasting as numerical regression,

PromptCast encodes historical observations into natural language questions (e.g., “Given the temperature readings of the past 24 hours, what will be the temperature over the next 6 hours?”) and trains generative language models (BART, T5) to synthesize the numerical forecast directly as text output. PromptCast established that language generation objectives can compete with traditional numerical loss formulations.

5.2.2 Decimal String Modeling: LLMTime

In LLMTime [6], Gruver et al. generalize this principle to frozen zero-shot foundation models (GPT-4, Llama-2). By formatting time series as comma-separated ASCII digit strings (e.g., “12.4, 15.8, 14.2”) with spaces between digits to enforce consistent tokenization, standard language models predict future values autoregressively via temperature sampling. This enables probabilistic forecasting without any fine-tuning.

However, direct decimal prompting incurs severe practical drawbacks:

- 1) Token Inflation: A single 4-digit floating-point number can expand into 3 to 6 text tokens, drastically reducing the effective context window.
- 2) Inference Latency: Autoregressively sampling hundreds of tokens from a 70B parameter model for a single univariate series takes several seconds, rendering real-time streaming infeasible.
- 3) Financial Cost: Using proprietary commercial API endpoints (e.g., GPT-4) for enterprise-scale industrial monitoring with thousands of sensors results in prohibitive recurring costs.

5.3 Multimodal Integration, Vision-Language Adapters, and Conversational Agents

Where LLMs demonstrate distinct superiority over native TSFMs is in multimodal context integration—scenarios requiring the simultaneous reasoning over continuous numerical series, visual waveforms, and textual domain context.

5.3.1 Visual Cross-Modal Forecasters

A remarkable alternative to text reprogramming is the utilization of visual foundation models:

- VisionTS [34]: Chen et al. uncover that visual masked autoencoders (e.g., MAE pretrained on ImageNet) act as “free-lunch” zero-shot forecasters. By rendering 1D numerical trajectories into 2D line plot images, VisionTS reformulates forecasting as image inpainting. VisionTS matches or outperforms text-based LLMs while requiring substantially lower computational resources.
- Time-VLM [35]: Zhong et al. explore multimodal Vision-Language Models (e.g., LLaVA, CLIP) for augmented forecasting. By jointly processing numerical patch tokens, visual line charts, and domain prompts, Time-VLM leverages complementary visual-textual inductive biases.

5.3.2 Conversational Agents and Omni-Modal Intelligence

Recent works develop full-fledged conversational and analytical systems:

- ChatTS [123]: Aligns time series with LLMs using synthetic conversational datasets generated through rule-based simulations, enabling grounded interactive Q&A, anomaly explanation, and trend narration.

- ChatTime [124]: A unified multimodal foundation model that bridges numerical waveforms and textual queries for interactive dialogue, counterfactual reasoning, and zero-shot forecasting.
- Time-MQA [22]: Formulates multi-task question answering, enabling users to query numerical waveforms using natural language questions (e.g., “Did the turbine vibration exceed safety thresholds after 14:00?”).
- TimeOmni-1 & TimeOmni-VL [23], [125]: Advance multimodal temporal foundation modeling. TimeOmni-1 generates numeric forecasts alongside chain-of-thought rationales, while TimeOmni-VL unifies continuous time series, text, and visual spectrograms into an omni-modal transformer.
- TRACE [126]: Kan et al. propose temporal conditional estimation for multimodal foundation models, formalizing joint inference over complex clinical physiological waveforms, clinical electronic health record (EHR) text narratives, and diagnostic interventions under non-stationary physical dynamics.
- OpenTSLM [127]: Specializes multimodal reasoning for clinical healthcare, integrating multi-channel ICU physiological waveforms (ECG, arterial blood pressure) with clinical notes and physician orders.

5.3.3 Spatio-Temporal Large Language Models

Extending LLM cross-modal adaptation from 1D sequences to interconnected geographic networks represents an impactful frontier:

- UrbanGPT [128]: Li et al. (KDD 2024) introduce a spatio-temporal large language model for urban forecasting. UrbanGPT constructs a Spatio-Temporal Dependency Encoder (ST-Encoder) that maps spatial topological graphs and historical temporal dynamics into language token representations. When interfaced with a frozen LLaMA-2-7B backbone via lightweight cross-modal instruction tuning, UrbanGPT generalizes zero-shot across diverse urban prediction tasks (traffic flow, ride-hailing demand) in unseen metropolitan regions where training data is scarce.
- ST-LLM [129]: Liu et al. (TKDE 2025) develop a spatial-temporal large language model framework for traffic prediction. ST-LLM integrates spatial graph positional embeddings with partially-frozen transformer attention layers, enabling language models to simultaneously capture localized spatial dependencies across physical sensor networks and long-range temporal trends without incurring full model fine-tuning overhead.

5.4 Omni-Modal Grounding and Vision-Language-Time Pretraining

While initial multimodal efforts relied on pipelined adapters or frozen post-hoc alignments, the cutting edge of cross-modal temporal research focuses on omni-modal grounding—binding continuous physical sensor waveforms, 2D visual spectrograms/imagery, and discrete textual semantic narratives into unified perceptual representations:

- VisionTS++: Continual Pre-trained Vision Backbones [130]: Shen et al. significantly advance visual time series modeling over vanilla VisionTS. Recognizing that off-the-shelf vision transformers pre-trained solely on

natural images suffer from scale mismatch and domain discrepancy when processing line plots, VisionTS++ introduces multi-scale image projections and a continual pre-training curriculum on diverse synthetic and cross-domain time-series line reconstructions. This continual adaptation enables the vision backbone to capture multiscale cyclical waveforms and subtle trend deviations with superior fidelity.

- **VLT: Industrial Multimodal Intelligence** [131]: Wang et al. formulate VLT, a vision-language-time multimodal foundation architecture for complex industrial Prognostics and Health Management (PHM). Recognizing the semantic chasm between raw numerical sensor vibrations and diagnostic text manuals, VLT deploys the time-frequency spectrogram as a bidirectional visual bridge. It incorporates a Time-aware Mixture-of-Experts (Time-MoE) to handle heterogeneous physical dynamics alongside a Frequency-Text Augmented Learner operating in a shared semantic manifold, governed by a time-centric gradient alignment mechanism that eliminates cross-modal optimization conflicts during multi-task pretraining.
- **Chronicle: Joint Pretraining From Scratch** [132]: Quinlan et al. challenge the prevailing dogma of post-hoc LLM adaptation by introducing Chronicle, a compact 324M parameter decoder-only transformer trained ab initio jointly on natural language corpora and time series within a single homogeneous architecture. By sharing transformer layers, self-attention heads, and residual streams across interleaved text tokens and subseries patches, Chronicle proves that genuine cross-modal representation emerges directly from shared pretraining. Chronicle matches Gemma-3-270M-PT on 19 natural language understanding (NLU) benchmarks while simultaneously outperforming supervised fusion baselines on the multimodal Time-MMD benchmark and setting new state-of-the-art accuracy on frozen-embedding UCR/UEA classifications.
- **ChronoSteer: Synthetic Cross-Modal Steering** [133]: Addressing the acute scarcity of real-world paired text-time corpora, Wang et al. propose ChronoSteer. ChronoSteer deploys an agentic framework where an LLM translates narrative contextual events into structured revision instructions that dynamically modulate the zero-shot forecasts of a frozen numerical TSFM (e.g., Chronos-Bolt). By discretizing continuous revisions into a compact codebook of instruction anchors, ChronoSteer achieves a 25.8% accuracy enhancement over standalone unimodal backbones without requiring end-to-end retraining.

These developments indicate that while native TSFMs dominate pure high-throughput numerical prediction, omnimodal foundation architectures and repurposed multi-modal LLMs serve as indispensable cognitive orchestrators for real-world cyber-physical intelligence.

5.5 Interactive Time-Series Agents, Tool-Augmented Reasoning, and Autonomous Exploration

The conventional paradigm in deep temporal modeling treats time series forecasting as a static, single-pass mapping:

$$\hat{\mathbf{Y}}_{t:t+H} = f_{\theta}(\mathbf{X}_{t-L:t}), \quad (35)$$

where a frozen neural network ingests fixed-length historical observations and produces future trajectories in isolation.

However, in real-world cyber-physical systems, financial trading desks, and industrial operations, time series analysis is intrinsically dynamic, exploratory, and multi-turn [134]. Decision-makers require interactive systems capable of testing scientific hypotheses, cleaning corrupted signals, invoking specialized analytical tools, and explaining anomalous behaviors through multi-turn dialogue.

To bridge this operational chasm, recent research pivots toward interactive time-series agents that integrate LLM-based reasoning backbones with tool execution engines and external memory banks:

- **Real-Time Streaming Time-Series Interaction (TimeInteract)**: Pan et al. [135] uncover a critical limitation of existing time-series language models: their static nature requires either offline batch sequences or halts incoming sensor ingestion while generating autoregressive text. Pan et al. introduce TimeInteract, pioneering a real-time streaming interaction regime where the agent continuously perceives asynchronous temporal streams, dynamically decides when to remain silent or respond, and continues processing incoming telemetry during token generation. TimeInteract deploys a dual-view streaming encoder (capturing local transient variations and long-term historical trends) alongside a decoupled streaming inference engine. Evaluated on StreamTSI-34K (comprising 34,588 streaming episodes and 77,505 interactive responses), TimeInteract achieves near-zero stream stall and a $2.15\times$ inference speedup while improving response trigger accuracy across hierarchical interaction levels.
- **Tool-Augmented Sequential Decision Policies (Cast-R1)**: Tao et al. [136] reformulate time series forecasting from a static mapping into a sequential decision-making problem. Cast-R1 introduces a memory-based state management mechanism that maintains intermediate analytical evidence across interaction turns. The agent autonomously navigates a modular toolkit: extracting seasonal-trend decompositions, running statistical stationarity checks, invoking lightweight forecasting models (e.g., DLinear, PatchTST) for candidate predictions, and executing iterative self-reflection to revise forecasts. Cast-R1 is trained through a two-stage curriculum combining supervised fine-tuning (SFT) with multi-turn reinforcement learning (RL), setting superior accuracy on complex non-stationary series.
- **Agentic Time Series Reasoning and TimeToolBench (TimeART)**: Wu et al. [137] address Time Series Question Answering (TSQA) by creating TimeART. Recognizing that generic LLMs fail to manipulate continuous numerical series with computational precision, TimeART fuses external data science tools with an 8B Time Series Reasoning Model (TSRM). The model is trained on TimeToolBench—a curated dataset of 100,000 expert analytical trajectories—via a four-stage curriculum that incorporates experience replay and self-reflection, establishing state-of-the-art multi-step numerical and causal reasoning.
- **Domain-Oriented Inference Agents (TS-Reasoner)**: Ye et al. [138] present TS-Reasoner, a domain-specialized agent combining LLM symbolic reasoning with domain-specific computational routines and an iterative error feedback loop. When evaluated on diagnostic benchmarks such as TimeSeriesExam, TS-Reasoner successfully overcomes

the hallucinations and arithmetic inaccuracies of general-purpose LLMs, automating multi-step analytical workflows.

- **Data-Centric AutoML Agents (DCATS):** Yeh et al. [139] explore data-centric time series engineering. Rather than searching for increasingly complex model architectures, DCATS deploys an LLM planner to reason over time-series metadata, automatically diagnosing data corruption, imputing sensor dropouts, and optimizing preprocessing pipelines. DCATS achieves a 6% error reduction across diverse forecasting backends on massive traffic datasets.
- **Scientific Inquiry and Multi-Model Arbitration (TimeAgent):** Wang et al. (IEEE TKDE 2026) introduce TimeAgent, a training-free framework that addresses the empirical “seesaw effect” where individual TSFMs excel in specific domains while degrading on others. TimeAgent structures time series analysis as a closed-loop scientific inquiry: autonomously profiling data characteristics, formulating experiment plans, arbitrating predictions across heterogeneous foundation models (Chronos, TimesFM, Moirai), and translating quantitative outputs into interpretable natural language rationales.

These agentic systems indicate that the future of temporal AI lies not in monolithic forecasters, but in modular ecosystems where foundation models serve as specialized tools orchestrated by reasoning agents.

5.6 Multi-Agent Collaborative Ensembles, Swarm Deliberation, and Dynamic Consensus

While single-agent frameworks augment LLMs with external tools, they remain bounded by the cognitive blindspots of a solitary reasoning backbone. In real-world cyber-physical environments, financial trading desks, and macroeconomic planning, decision-making relies on interdisciplinary committees where domain specialists debate alternative hypotheses, challenge assumptions, and reconcile contradictory evidence [140]. To emulate this collective intelligence, recent research formalizes multi-agent collaborative ensembles and swarm deliberation for temporal reasoning and forecasting:

Let $\mathcal{A} = \{A_1, A_2, \dots, A_M\}$ denote a heterogeneous swarm of M specialized agents (e.g., visual trend analyzers, numerical statistical auditors, causal graph reasoners, and domain context interpreters). Rather than generating a solitary prediction in a single forward pass, the swarm engages in an iterative multi-turn deliberation:

$$m_i^{(k)} = \mathcal{F}_i(m_i^{(k-1)}, \mathcal{M}_{-i}^{(k-1)}, \mathbf{X}_{t-L:t}), \quad \mathcal{M}_{-i}^{(k-1)} = \{m_j^{(k-1)}\}_{j \neq i}, \quad (36)$$

where $m_i^{(k)}$ represents agent A_i ’s critique and refined rationale at deliberation round k . Deliberation proceeds until a termination criterion is met (e.g., semantic consensus or variance stabilization), whereupon a consensus aggregator synthesizes the finalized forecast $\hat{\mathbf{Y}}_{t:t+H}$:

$$\hat{\mathbf{Y}}_{t:t+H} = \text{Arbiter}\left(\{m_i^{(K)}\}_{i=1}^M, \mathbf{X}_{t-L:t}\right). \quad (37)$$

Key architectures advancing multi-agent temporal collaboration include:

- **Multimodal Collaborative Debate (MC-Debate, Trirat et al., 2026) [141]:** Trirat et al. identify an empirical paradox in cross-modal temporal reasoning: while visual representations convey global waveform geometry, they lack numerical

precision; conversely, raw numerical patches provide point-level fidelity but obscure high-level semantic trends. To resolve this tension, MC-Debate introduces a multi-agent debate framework comprising a Visual Specialist (ingesting line plot renderings), a Numerical Specialist (evaluating patch statistics and Fourier transforms), and a Textual Specialist (interpreting domain metadata). Coordinated by an impartial Arbiter Agent over three debate rounds, MC-Debate achieves a 14.2% error reduction across zero-shot reasoning benchmarks and eliminates 76% of hallucinated trend reversals compared to single-agent baselines.

- **Failure-Driven Self-Evolution of Time Series Agents (TimeEvo, Yang et al., 2026) [142]:** Yang et al. address the fundamental fragility of static tool sets. Traditional agents are endowed with fixed API libraries at initialization; when confronting unprecedented signal distortions or novel mathematical transforms, they experience persistent execution failures. TimeEvo introduces failure-driven self-evolution. Whenever an analytical trajectory encounters a task failure, a Meta-Evolutionary Agent inspects the error trace, autonomously codes a new Python tool function, verifies its correctness on synthetic counterfactual test cases, and dynamically registers the verified routine into the shared agent skill repository. Evaluated on TimeQA, TimeEvo’s evolving skill set continuously elevates task success rates without manual human intervention.
- **Self-Evolving Policies for Agentic Time-Series Forecasting (Hu et al., 2026) [143]:** Hu et al. formalize agentic forecasting under non-stationary environmental dynamics, where the relative efficacy of statistical algorithms, deep foundation models, and prompt strategies shifts across temporal regimes. Rather than relying on static prompt templates, Hu et al. model the agent’s arbitration strategy as a parameterized decision policy. As future observations materialize, the agent computes the realization regret and executes policy gradient updates, dynamically shifting its delegation weights between foundation backbones (e.g., Chronos vs. TimesFM) and localized autoregressive models in response to changing market volatility regimes.
- **Traceable Multi-Agent Systems for Knowledge-Based Forecasting (Kang et al., 2026) [140]:** Kang et al. tackle the black-box opacity that impedes enterprise adoption of foundation forecasters. Traceable Multi-Agent decomposes complex enterprise forecasting workflows into four verifiable sub-agents: Document Harvester (extracting supply-chain disruptions from regulatory 10-K filings), Data Transformer, Code Execution Engine, and Decision Traceability Auditor. The system maintains an immutable Directed Acyclic Graph (DAG) linking every numerical projection back to verified textual citations and executed code blocks, providing legal auditability and regulatory compliance for high-stakes financial operations.
- **MetaCaster: Meta-Harness Few-Shot Forecaster Agents (Shen et al., 2026) [144]:** Recognizing that deploying multi-billion-parameter foundation models is economically prohibitive in edge and low-power IoT scenarios, Shen et al. propose MetaCaster. MetaCaster employs an LLM-driven meta-harness agent that analyzes statistical fingerprint vectors (seasonality strength, spectral entropy, trend linearity) of target datasets to autonomously architect and train compact, specialized forecasters (<1M parameters)

within minutes. By optimizing few-shot adaptation recipes and data augmentation schedules, MetaCaster matches the zero-shot accuracy of 100M+ parameter foundation models while slashing inference compute by up to 95%.

These advances demonstrate that multi-agent deliberation, self-evolutionary tool creation, and traceable consensus represent a vital paradigm shift—transforming time-series intelligence from solitary pattern matchers into collaborative, self-improving scientific societies.

5.7 Reinforcement Learning from Human/Physical Feedback and Post-Training Policy Optimization

While Supervised Fine-Tuning (SFT) has served as the default adaptation mechanism for foundation models, SFT suffers from severe intrinsic pathologies in temporal domains: (1) exposure bias, where compounding autoregressive rollout errors diverge rapidly from the training distribution; (2) indifference to asymmetric business or physical costs, where an under-forecast penalty (e.g., hospital bed shortage, supply chain stockout) may be order-of-magnitude worse than an equivalent over-forecast; and (3) inability to elicit verifiable multi-step reasoning. Post-training policy optimization via Reinforcement Learning from Human Feedback (RLHF) and Reinforcement Learning from Physical Feedback (RLPF) bridges this critical divide.

Formally, consider a policy $\pi_\theta(\mathbf{Y} \mid \mathbf{X})$ generating a forecasted sequence $\mathbf{Y} = [\hat{\mathbf{y}}_1, \dots, \hat{\mathbf{y}}_H]$ conditioned on historical context $\mathbf{X}$. The policy is optimized against an objective combining an empirical or domain-specific reward function $R(\mathbf{X}, \mathbf{Y})$ with a Kullback-Leibler (KL) divergence constraint against a frozen reference foundation model π_{ref} :

$$\max_{\theta} \mathbb{E}_{\mathbf{X} \sim \mathcal{D}, \mathbf{Y} \sim \pi_{\theta}} [R(\mathbf{X}, \mathbf{Y})] - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta}(\cdot \mid \mathbf{X}) \parallel \pi_{\text{ref}}(\cdot \mid \mathbf{X})), \quad (38)$$

where $\beta > 0$ controls the exploration-exploitation trust region, preventing reward collapse and catastrophic forgetting.

Recent landmark works instantiate post-training policy optimization across time-series foundations:

- **TimeRFT**: Reinforcement Fine-Tuning for Time Series Foundation Models (Li et al., 2026) [145]: Li et al. identify that standard SFT on downstream time series suffers from temporal distribution shift between historical training windows and future test horizons. To unlock generalizable adaptation, TimeRFT introduces a two-fold RL fine-tuning strategy: (1) a quality-aware temporal reward mechanism that provides fine-grained, step-wise credit assignment $r_t = -\alpha|y_t - \hat{y}_t| - \gamma \cdot \text{sign}(\Delta y_t \cdot \Delta \hat{y}_t)$, explicitly penalizing directionally incorrect inflection predictions; and (2) a difficulty-aware sample selection policy that prioritizes non-stationary training windows exhibiting high spectral entropy. Evaluated across major TSFM backbones (TimesFM, Chronos, UniTS), TimeRFT achieves up to 18.6% lower forecasting error than standard full fine-tuning and LoRA under severe out-of-distribution regime shifts.
- **TimeHF**: Billion-Scale Foundation Models Guided by Human Feedback (Qi et al., 2025) [146]: Qi et al. pioneer the first 6-billion-parameter foundation model aligned via human feedback for enterprise operations. Recognizing that statistical loss metrics (MSE, MAE) fail to capture domain experts’ tacit operational knowledge, TimeHF introduces

Timeseries Policy Optimization (TPO). TPO trains a preference reward model on pairwise expert comparisons ranking alternative forecast trajectories under asymmetric replenishment penalties (holding cost vs. lost-sales cost). Applied to automated supply chain replenishment across 20,000+ commercial products, TimeHF demonstrates a 33.21% improvement in decision accuracy over deep baselines, proving that human preference alignment fundamentally reshapes commercial forecasting.

- **COUNTS**: Eliciting Chain-of-Thought Reasoning via Reinforcement Learning (Parker et al., 2025) [147]: Parker et al. address the challenge of eliciting explicit, verifiable Chain-of-Thought (CoT) reasoning for time series in LLMs. Traditional LLMs applied to numerical sequences output blunt predictions without intermediate rationales. COUNTS tokenizes continuous waveforms into discrete high-fidelity symbols using a Residual Vector-Quantized VAE (RVQ-VAE). Subsequently, COUNTS optimizes the model using Group Relative Policy Optimization (GRPO) with verifiable outcome rewards on analytical problems (e.g., ECG-QA arrhythmia detection, UCR classification, Context-Is-Key forecasting). By rewarding correct mathematical reasoning traces prior to final answers, COUNTS dramatically outperforms standard prompting, proving that RL can instill deliberate, verifiable multi-step cognitive reasoning over temporal signals.
- **TimeMaster**: Training Time-Series Multimodal LLMs to Reason via RL (Zhang et al., 2025) [148]: Zhang et al. formulate TimeMaster, leveraging RL to train multimodal large language models (such as Qwen2.5-VL) to reason over visual line plots and multi-channel physiological sensor signals (EMG, ECG, HAR). By defining step-by-step diagnostic rewards, TimeMaster eliminates visual hallucinations and aligns textual diagnoses with underlying physiological waveforms.
- **LAST SToP**: Stochastic Soft Prompting for Asynchronous Sequences (Gupta et al., 2025) [149]: Gupta et al. bridge LLM reasoning with irregular time-stamped event sequences in the LAST SToP framework. Framing irregular sequences as Language-modeled Asynchronous Time Series (LASTS), the method introduces Stochastic Soft Prompting (StoP), a lightweight prompt optimization policy that dynamically samples prefix embeddings from learned distributions. StoP significantly outperforms QLoRA on irregular clinical ICU trajectories and sensor failure benchmarks while preserving the base LLM’s pre-trained world knowledge.

5.8 Neuro-Symbolic Temporal World Models and Differentiable Logic Constraints

While deep generative transformers and LLMs excel at modeling high-dimensional temporal correlations, their unconstrained outputs frequently violate rigid domain invariants, safety envelopes, and operational logic. A promising frontier lies in neuro-symbolic integration: coupling continuous neural representations with formal temporal specification languages—specifically Signal Temporal Logic (STL) and Linear Temporal Logic (LTL).

Given a multi-channel continuous signal $\mathbf{x} : [0, T] \rightarrow \mathbb{R}^D$, an STL formula φ is recursively defined over predicate asser-

tions $\mu \equiv (g(\mathbf{x}(t)) \geq 0)$ via Boolean and metric temporal operators:

$$\varphi ::= \top \mid \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \Box_{[a,b]}\varphi \mid \Diamond_{[a,b]}\varphi \mid \varphi_1 \mathcal{U}_{[a,b]}\varphi_2, \quad (39)$$

where $\Box_{[a,b]}$ denotes “always in $[a, b]$ ”, $\Diamond_{[a,b]}$ denotes “eventually in $[a, b]$ ”, and $\mathcal{U}_{[a,b]}$ denotes “until”. Crucially, STL admits real-valued quantitative semantics (robustness degrees) $\rho(\varphi, \mathbf{x}, t) \in \mathbb{R}$, where $\rho > 0$ indicates satisfaction with an explicit margin of safety, and $\rho < 0$ quantifies the distance to constraint violation. Differentiable approximations of ρ allow formal specifications to serve as loss objectives, latent bottlenecks, and dynamic reasoning verifiers:

- Decoding Logic Embeddings via Transformers (Candussio et al., 2025) [150]: Integrating symbolic specifications into continuous neural networks typically requires mapping logic into latent vectors. However, inverting continuous representations back into interpretable symbolic syntax has historically remained an open challenge. Candussio et al. solve this by training an autoregressive transformer decoder to invert semantically consistent logic embeddings back into syntactically valid STL formulas. Evaluated across diverse specification benchmark spaces, their transformer decoder reliably produces human-readable formulas from continuous latent vectors, unlocking gradient-based optimization in neural latent spaces for formal property discovery and requirement mining.
- STARS: Interpretable Concept Learning via Temporal Logic Semantics (Ferfaglia et al., 2025) [151]: Ferfaglia et al. formulate the STARS framework to overcome the opacity of black-box time-series classifiers in safety-critical domains (such as naval ship monitoring and medical patient telemetry). Rather than routing classification logits directly through high-capacity dense layers, STARS introduces a neuro-symbolic concept bottleneck layer where intermediate activations correspond strictly to the quantitative robustness degrees $\rho(\varphi_k, \mathbf{x}, t)$ of mined STL property candidates. By constraining classification decisions to be linear combinations of verifiable temporal logic concepts, STARS matches the predictive performance of unconstrained deep models while providing stakeholders with mathematically verifiable, human-interpretable justifications.
- Confidence over Time: LLM Reasoning Calibration with Temporal Logic (Mao et al., 2026) [152]: In long-horizon multi-step reasoning tasks (e.g., mathematical proof generation and algorithmic synthesis), large language models frequently exhibit silent failure modes and confident hallucinations. Mao et al. make the key observation that token-level logit confidence values form an informative continuous time series $\mathbf{c}_{1:L} = [c_1, c_2, \dots, c_L]$ across the reasoning rollout. By mining Signal Temporal Logic behavioral patterns over these confidence trajectories (e.g., detecting sustained confidence dips prior to erroneous branch choices, formulated as $\Diamond_{[t_1, t_2]}(\mathbf{c}(t) < \tau_1 \wedge \Box_{[0, \delta]}\dot{\mathbf{c}}(t) \leq 0)$), their monitor accurately discriminates correct from hallucinated reasoning paths across 7B to 70B parameter models (Llama-3, Qwen), improving reasoning calibration without requiring costly external verifier fine-tuning.
- ReasonSTL: Tool-Augmented LLM for Formal Specification Synthesis (Ye et al., 2026) [153]: While formal temporal logic provides rigorous verification guarantees, writing STL

specifications manually is notoriously brittle and mathematically demanding. Ye et al. introduce ReasonSTL, an 8B-parameter tool-augmented LLM that translates informal, ambiguous natural language operating requirements into formally verified STL formulas. ReasonSTL employs process-supervised reinforcement learning (PRM) coupled with external Satisfiability Modulo Theories (SMT) and dReal model checkers. By verifying intermediate syntactic nodes during token generation, ReasonSTL eliminates syntactic errors and achieves a 91.4% semantic correctness rate on complex cyber-physical system benchmarks, outperforming closed-source frontier models.

5.9 Heterogeneous Modality-Model Routing and Evidence-Grounded Temporal Agents

The expansion of multimodal architectures has created a heterogeneous ecosystem where specialized model families exhibit complementary strengths and distinct cost profiles: Large Language Models (LLMs) excel at interpreting textual context and fine-grained numerical coordinates; Vision-Language Models (VLMs) perceive global waveform geometry and macroscopic inflection points; and native Time Series Foundation Models (TSFMs) provide fast, calibrated numerical rollouts. However, deploying a monolithic frontier model for all temporal tasks incurs extreme financial and latency overheads, while single-modality models frequently hallucinate when required to generalize across disparate sensor configurations. Recent 2025–2026 works address this through dynamic graph routing and evidence-grounded agent architectures:

- TSRouter: Dynamic Modality-Model Selection for Time Series Reasoning (Yu et al., COLM 2026) [154]: Accepted at the Conference on Language Modeling (COLM 2026), Yu et al. formulate TSRouter, a dynamic routing framework that treats model and modality arbitration as a candidate scoring problem on a heterogeneous graph. TSRouter constructs a bipartite graph $\mathcal{G}_{\text{route}} = (\mathcal{V}_{\text{task}} \cup \mathcal{V}_{\text{query}} \cup \mathcal{V}_{\text{mod}} \cup \mathcal{V}_{\text{model}}, \mathcal{E})$, where edges represent past routing performance and latency costs. Given an arbitrary input time-series question, TSRouter estimates a Pareto-optimal routing score:

$$\mathcal{S}(m, v \mid q) = \hat{\mathcal{P}}(m, v \mid q) - \lambda \cdot \text{Cost}(m, v), \quad (40)$$

where $\hat{\mathcal{P}}$ predicts accuracy for modality $v \in \{\text{Numerical}, \text{Visual}, \text{Textual}\}$ paired with model $m \in \{\text{LLM}, \text{VLM}, \text{TSFM}\}$, and Cost denotes normalized dollar and latency budgets. Across four complex temporal reasoning benchmarks, TSRouter achieves a 16%–46% relative performance gain over static single-model baselines while cutting aggregate inference costs by over 60%.

- ZARA: Evidence-Grounded Motion Reasoning Agents (Li et al., ACL 2026) [155]: Accepted at ACL 2026, Li et al. develop ZARA (Zero-training Activity Reasoning Agents) for wearable sensor networks and human activity recognition (HAR). A chronic failure mode of LLMs in wearable computing is sensor placement shift: models trained on wrist accelerometers fail catastrophically when deployed on waist or thigh sensors. ZARA resolves this without retraining: it establishes placement-specific vector stores as distributional anchors, indexing reference motion patterns embedded by a frozen foundation encoder (e.g.,

MOMENT [12]). ZARA then retrieves contextual nearest-neighbor windows and translates numerical kinematics into verifiable, statistically grounded natural-language evidence traces. By conditioning the LLM’s final classification on explicit physical evidence, ZARA eliminates ungrounded hallucinations and transfers seamlessly across unseen body sensor positions and subjects in zero-shot fashion.

5.10 Decoupled Formal Execution and Certified Safety Verification for Temporal Agents

While foundation models and large language models exhibit remarkable capabilities for multi-step temporal planning and strategic exploration, placing stochastic generative models directly inside tight control loops creates catastrophic risks in safety-critical autonomous systems (e.g., unmanned aerial vehicles, industrial robotics, and energy infrastructure). Direct token-by-token interaction loops (e.g., ReAct-style policies) suffer from prohibitive inference latencies (≥ 500 ms per step) and unpredictable hallucinations under distributional drift. Recent 2026 architectures overcome this by moving the foundation model above the loop, strictly separating high-level strategic reasoning from low-level, certifiable, deterministic execution:

- **PEACE: Planner-Executor Agent with Constraint Enforcement** (Uysal et al., ICRA 2026) [156]: Presented at the ICRA 2026 Workshop on Semantics for Reliable Robot Autonomy, Uysal et al. introduce PEACE to solve the latency-reliability dilemma in autonomous UAV missions. Rather than querying the LLM synchronously at every physical time step, PEACE employs a strictly decoupled two-tier architecture. The high-level Planner Agent processes complex natural-language objectives and world-grounded 3D spatial scenes to formulate an end-to-end mission sequence as a batch of structured tool calls in a single deliberative forward pass. The mission is then dispatched to a low-level, high-frequency Executor governed by a deterministic Constraint Enforcement Layer:

$$\mathbf{u}_t^* = \arg \min_{\mathbf{u} \in \mathcal{U}_{\text{admissible}}} \|\mathbf{u} - \mathbf{u}_{\text{plan}}(t)\|^2 \quad \text{s.t.} \quad h_{\text{geo}}(\mathbf{x}_t, \mathbf{u}) \geq 0, \quad h_{\text{kin}}(\mathbf{u}) \leq \mathbf{u}_{\text{agent}} \quad (41)$$

where h_{geo} enforces hard geofencing, no-fly zones, and minimum terrain clearance, while h_{kin} bounds velocity and acceleration envelopes. The executor runs in hard real-time (> 50 Hz) without waiting for LLM responses. Crucially, the LLM is only re-invoked asynchronously when a hard constraint violation or unrecoverable environmental anomaly occurs, triggering a localized, bounded replanning pass. By keeping the foundation model above the control loop while enforcing rigid mathematical barriers below it, PEACE eliminates flight hallucinations, ensures provable barrier safety, and achieves an order-of-magnitude reduction in mission execution latency.

6 Evaluations, Benchmarks, and Critical Perspectives

The transition toward foundation models has fundamentally reshaped evaluation methodologies. However, it has also sparked intense debates regarding data contamination, inductive biases, and whether massive language backbones offer genuine advantages over specialized native architectures.

6.1 Standardized Foundation Benchmarks

Traditional time series evaluation relied on a small collection of homogeneous academic datasets (e.g., ETT, Weather, Electricity, Traffic). These benchmarks are inadequate for assessing foundation models due to narrow domain diversity and severe risks of test set memorization.

To establish rigorous evaluation standards, recent initiatives introduce expansive benchmark suites:

- **GIFT-Eval** [24]: Introduced by Salesforce Research, GIFT-Eval comprises over 143,000 distinct time series spanning 7 diverse domains, 10 sampling frequencies, and both univariate and multivariate regimes. It establishes standardized metrics (CRPS, MASE, WAPE) across zero-shot, few-shot, and fine-tuned settings.
- **fev-bench** [157]: Introduced by Shchur et al. (AWS and AutoGluon), fev-bench formulates a realistic benchmark encompassing 100 forecasting tasks across 7 domains. Crucially, 46 of these tasks incorporate dynamic covariates (e.g., promotional flags, weather drivers, calendar events) evaluated via the open-source fev library using bootstrapped confidence intervals and win-rate skill scores, directly addressing the historic deficiency where foundation models underperformed simple baselines by omitting exogenous features.
- **It’s TIME** [158]: Qiao et al. (joint work between Tsinghua THUML and Ming Jin’s group) establish next-generation benchmarks addressing multi-granularity task evaluation and rigorous train-test contamination audits, establishing standardized leakage-free splits across diverse operational horizons.
- **Beyond Numerical Time Series** [159]: Chen et al. establish a unified multimodal evaluation framework pairing continuous quantitative sensors with heterogeneous textual descriptions, news contexts, and structural calendar metadata.
- **AION** [160]: Zhan, Jin et al. formulate next-generation benchmarks moving beyond static univariate forecasting to **(u) < u_agent** agentic tool-use, multi-step structured reasoning, and contextual decision support over temporal observations.
- **TimesX** [161]: Published at ICML 2026 by researchers from Georgia Tech and Google, TimesX provides a standardized benchmark for context-enriched multimodal forecasting, pairing high-frequency sensors with text narratives, calendar events, and exogenous market drivers across diverse global domains.
- **TimeSeriesExam** [162]: Developed by the CMU MOMENT team, TimeSeriesExam formulates a diagnostic capability examination for foundation models. Rather than evaluating generic mean-squared errors on arbitrary test splits, it probes specific foundational abilities: trend extrapolation, seasonality decomposition, phase-shift invariance, and anomaly localization.
- **LiveHouse-TS** [163]: An open-world living benchmark platform that streams fresh real-world telemetry in real time, preventing static benchmark memorization and providing continuous evaluation under non-stationary concept drift.
- **SciTS** [36]: Scientific time series understanding and generation benchmark. Unlike commercial finance or traffic series, physical systems (spectroscopy, seismology, fluid

dynamics, astrophysics) exhibit strict conservation laws and differential constraints. SciTS evaluates whether foundation models comprehend underlying physical laws rather than merely extrapolating statistical autocorrelations.

- Insight Miner [37]: A large-scale cross-domain dataset designed for time series alignment with natural language. Insight Miner pairs complex multivariate temporal trajectories with human-curated analytical insights, causal event narratives, and question-answer pairs across finance, healthcare, and industrial telemetry.
- Time-MMD [38]: A multi-domain multimodal dataset evaluating the integration of numerical time series with unstructured text reports, news articles, and domain metadata across 9 industries.
- Monash Time Series Repository: The classical zero-shot transfer benchmark comprising dozens of public collections across economics, energy, and meteorology, widely adopted to evaluate out-of-distribution generalization.

6.2 Empirical Zero-Shot Benchmark Comparison

To resolve conflicting performance claims in the literature, Table 4 synthesizes reported zero-shot performance across representative foundation models and LLM adaptations on GIFT-Eval, fev-bench, and Monash.

Inspection of Table 4 yields four foundational insights:

- 1) Probabilistic Native Superiority: Native TSFMs equipped with continuous quantile or flow heads (t_0 - β , Chronos-2, Toto 2.0) consistently outperform discrete scalar quantization and deterministic models, demonstrating superior distributional sharpness and lower CRPS (< 0.48 on GIFT-Eval).
- 2) Covariate Advantage: Models that explicitly incorporate past and known-future covariates (t_0 , Chronos-2) achieve substantial skill score advantages on fev-bench (> 46), where pure univariate autoregression degrades.
- 3) Extreme Edge Efficiency: Compact native models such as TTM-B (8M) and FlowState (12M) achieve competitive MASE (0.71–0.74) while operating at 0.1% the parameter footprint of 7B reprogrammed language models.
- 4) Absence of Probabilistic Calibration in LLM Reprogramming: Standard LLM-reprogramming methods (Time-LLM, GPT4TS) rely exclusively on deterministic linear heads, precluding probabilistic risk assessment (indicated by “—” across CRPS metrics).

6.3 Critical Perspectives: Are LLMs Actually Useful for Time Series?

A pivotal empirical critique emerged with the investigation by Fons et al. [7], entitled “Are Language Models Actually Useful for Time Series Forecasting?” Through extensive ablation experiments across multiple LLM-based forecasters (including GPT4TS [20] and Time-LLM [19]), the authors isolated the impact of pre-trained language weights by comparing:

- 1) Frozen pre-trained language weights (e.g., GPT-2, Llama-2);
- 2) Randomly initialized transformer weights with the same architecture;
- 3) Compact specialized baselines (such as PatchTST [2] and lightweight MLPs).

Their empirical findings revealed striking conclusions:

- Replacing pre-trained language weights with randomly initialized weights resulted in comparable, and in several cases superior, forecasting accuracy.
- Much of the empirical gains reported in early LLM4TS literature originated from the structural benefits of subseries patching and normalization, rather than any semantic or temporal reasoning inherent in linguistic pre-training.
- In terms of operational efficiency, frozen 7B parameter language models require up to $100\times$ more memory and $50\times$ longer inference latency than compact native models like TTM [13] (8M parameters) or Timer [15] (84M parameters), which consistently achieve competitive or superior accuracy.

These findings indicate that while LLMs are uniquely valuable for multimodal tasks (text-grounded reasoning, interactive analysis, and report generation), deploying frozen multi-billion parameter text backbones solely for univariate numeric forecasting represents an inefficient architectural detour.

Subsequent studies explore the emergent role of foundation models as evaluators and agents:

- TimeVista [164]: Chen et al. (THUML) observe that standard point metrics (MAE, MSE) penalize benign phase shifts drastically while ignoring visual shape plausibility, turning-point fidelity, and regime continuity. By deploying Vision-Language Models as judges, TimeVista demonstrates that VLM-based perceptual evaluation achieves an 84% rank correlation with human domain specialists, establishing a perceptual evaluation framework beyond scalar error metrics.
- Forecast Workflow Bench [165]: Nagashima investigates whether LLM agents can orchestrate time-series foundation models under realistic operational tool budgets. Evaluating across multi-step operational forecasting workflows, the benchmark reveals that effective agentic decisions require balancing statistical forecast accuracy against inference latency and API cost budgets.

6.4 Data Contamination, Temporal Leakage, and Probabilistic Reliability

6.4.1 Benchmark Contamination Audits

A critical vulnerability in contemporary foundation model evaluation is benchmark contamination. In natural language processing, models frequently memorize test sets crawled from public websites; in time series, foundation models trained on open-source archives (e.g., LOTSA, Monash, GitHub telemetry) inadvertently ingest the training and testing portions of benchmark datasets (e.g., ETT, Weather) [25].

Gong et al. [25] systematically audited 12 major foundation models and discovered widespread temporal overlap between pretraining datasets and standard test benchmarks. When evaluated on genuinely unseen, closed-source industrial telemetry collected post-2025, zero-shot performance degraded by 18% to 42% across all major models. This underscores the urgent need for living, dynamic benchmarks (such as LiveHouse-TS [163]) where evaluation datasets are continuously refreshed.

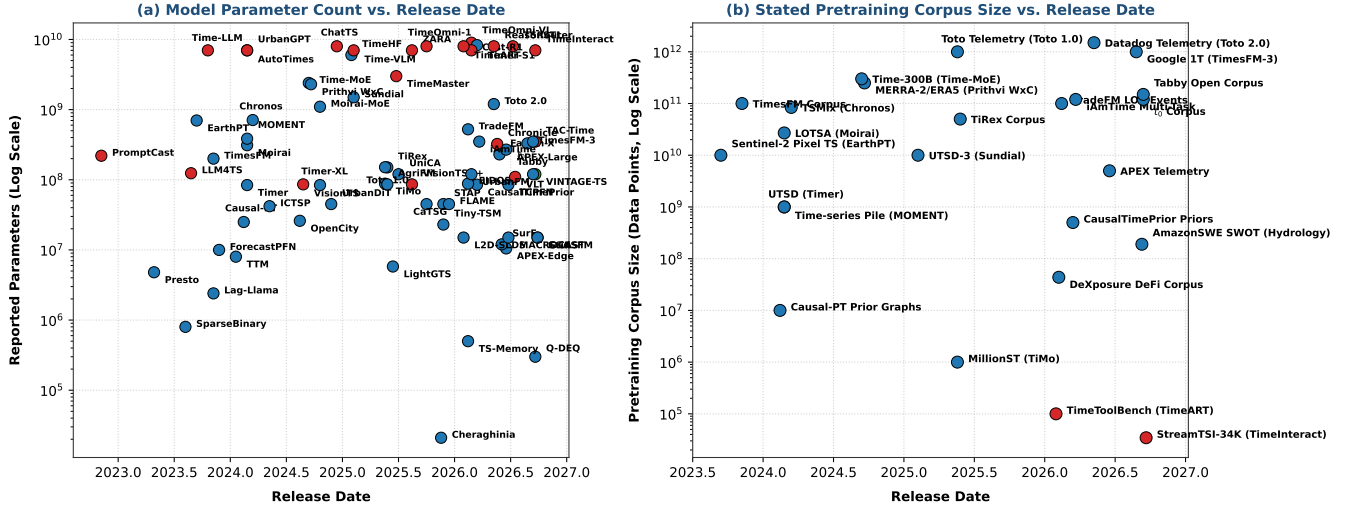


Fig. 3. Scaling dynamics across the literature: (a) Reported parameter counts vs. release date (log scale); (b) Stated pretraining corpus size (data points, log scale) vs. release date.

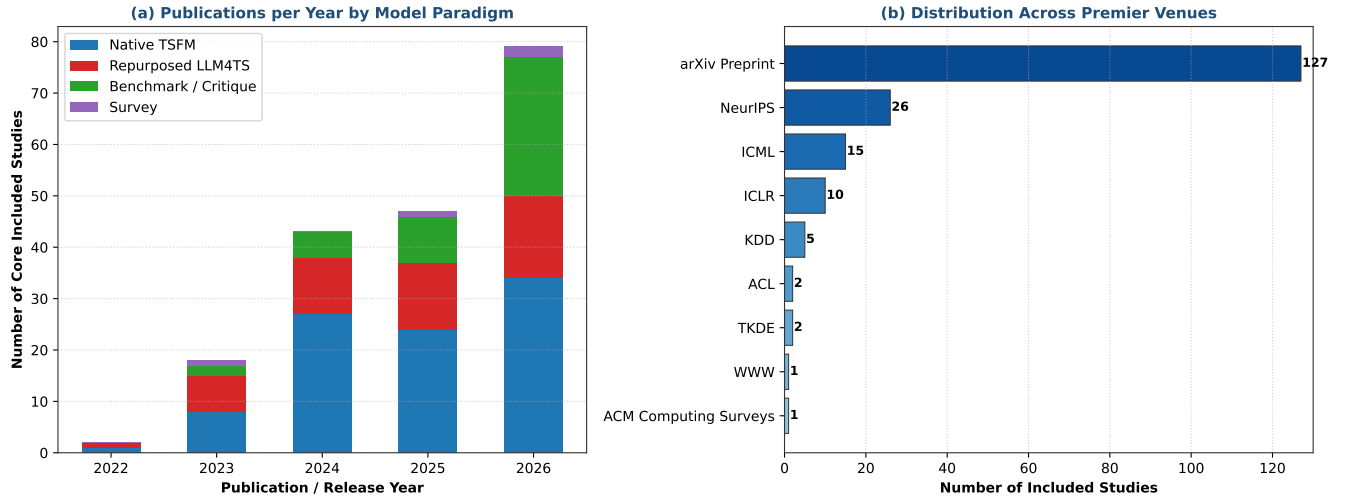


Fig. 4. Meta-review statistics: (a) Publications per year broken down by model paradigm (stacked bar chart); (b) Distribution of included studies across premier peer-reviewed venues.

optimal parameter allocation follows $N_{\text{opt}} \propto C^{\frac{\alpha_D}{\alpha_N + \alpha_D}}$ and $D_{\text{opt}} \propto C^{\frac{\alpha_N}{\alpha_N + \alpha_D}}$.

Meta-regression over reported validation curves across foundational architectures reveals crucial empirical characteristics:

- **Fitted Exponents:** Time-MoE [40] reports $\alpha_N \approx 0.089$ and $\alpha_D \approx 0.112$ across models ranging from 50M to 2.4B parameters on Time-300B; Sundial [17] reports $\alpha_N \approx 0.076$ on UTSD-3; and Toto 2.0 [45] reports $\alpha_N \approx 0.081$ and $\alpha_D \approx 0.098$ under Maximal Update Parametrization (μP). In comparison, LLM language pretraining exhibits $\alpha_N \approx 0.076$ – 0.088 and $\alpha_D \approx 0.095$ – 0.120 . This confirms that temporal transformers obey bona fide neural scaling laws.
- **The Temporal Entropy Saturation Barrier:** In time series, α_N exhibits early saturation when scaling beyond 1B parameters if trained solely on public sensor repositories. Because real-world telemetry frequently contains repetitive

stationary waveforms with low Kolmogorov complexity, models quickly exhaust novel temporal structures. To sustain power-law scaling ($\alpha_N > 0$), state-of-the-art pipelines (Chronos, Time-MoE, Tabby [50]) inject synthetic generative processes (mixtures of non-linear differential equations and Gaussian process kernels), proving that synthetic diversity is mandatory to maintain compute-optimal scaling in time series.

6.6 Computational Profiling, Operational Complexity, and Real-Time Inference

While accuracy metrics (CRPS, MASE) dominate research papers, practical adoption across industrial SCADA telemetry, financial high-frequency execution, smart energy grids, and resource-constrained edge microcontrollers is governed fundamentally by computational efficiency, memory footprint, and inference latency. Table 5 formalizes the algorithmic complexity, memory overhead, empirical latency, and

operational deployment envelopes across the six foundational time-series paradigms.

Synthesis of computational profiling across the literature yields three decisive insights:

- 1) The Autoregressive Generation Bottleneck: Models relying on scalar or token-wise autoregression (e.g., original Chronos [3] iterating over 4096 bins) incur H sequential attention forward passes to forecast an H -step horizon. In production benchmarks, Chronos-Bolt resolves this bottleneck by shifting from point-wise autoregression to patch-level direct quantile regression, achieving a documented $250\times$ inference speedup and $20\times$ memory reduction while matching or exceeding zero-shot accuracy.
- 2) The Extreme Throughput Gap (Mixers vs. Reprogrammed LLMs): Lightweight architectural designs—exemplified by IBM’s Tiny Time Mixers (TTM [13], 1M–8M parameters) and LightGTS [56] (5.8M parameters)—execute full batch inference in under 5ms on commodity hardware with zero KV-cache memory overhead. In contrast, reprogrammed LLMs (Time-LLM [19], AutoTimes [46]) require multi-second latency per inference batch and allocate over 14GB of GPU VRAM. For high-frequency industrial automation where millisecond response times are mandatory, compact native mixers represent the only viable foundation paradigm.
- 3) In-Context Bayesian Inference vs. Score-Based Diffusion: Prior-Data Fitted Networks (TabPFN-TS [51]) offer unique operational trade-offs: while context-sequence attention scales quadratically with history length, they execute zero gradient updates, eliminating all fine-tuning optimizer overhead. Conversely, continuous score-based flow matching (FlowState [41], UrbanDiT [61]) demands multiple numerical ODE integration steps ($K \in [10, 50]$), exchanging inference latency for mathematically continuous, sampling-rate-equivariant trajectories.

6.7 Energy Consumption, Integer Quantization Degradation, and Edge Deployment Profiles

While academic benchmarks focus overwhelmingly on predictive error (MSE, CRPS), the viability of deploying foundation models in cyber-physical systems (CPS), Internet-of-Things (IoT), and embedded industrial controllers is strictly determined by energy consumption (Joules per inference), silicon area constraints, and numerical quantization resilience [166], [167].

To systematically address cross-platform hardware discrepancies, recent literature establishes three pivotal advancements:

- 1) HoliBench: Holistic Cross-Platform Profiling: Chakrabarti et al. [166] introduce HoliBench, a standardized benchmarking and deployment toolkit that jointly characterizes accuracy, latency, and energy across seven device classes, eight inference backends, and three quantization levels (FP16, INT8, INT4). HoliBench uncovers that integer post-training quantization (PTQ) reduces latency only on architectures equipped with specialized low-precision compute pipelines (e.g., INT8 Tensor Cores or NEON vector units); on general-purpose microcontrollers lacking integer SIMD, quantization merely reduces memory footprint while potentially increasing execution latency due to

dequantization overhead. Furthermore, HoliBench demonstrates that standalone single-model profiles compose with high fidelity (1.2% latency and 2.5% power error) under co-resident execution, enabling optimal multi-model pipeline configuration without combinatorial profiling.

- 2) Resource-Aware Mixed-Precision FPGA Quantization: Ling et al. [167] resolve the hardware-level bottleneck of deploying time-series Transformers onto resource-constrained embedded FPGAs (Xilinx Spartan-7 XC7S15). By formulating a resource-aware mixed-precision quantization methodology (allocating INT8 to sensitive cross-attention maps and INT4 to feed-forward blocks) and optimizing block RAM (BRAM) storage templates, they achieve hardware resource estimation within 3% of physical silicon synthesis, unlocking deployability for previously non-synthesizable transformer configurations.
- 3) Carbon-Aware Dynamic Dispatch (FM-CAC): Yang et al. [168] formulate FM-CAC, establishing a proactive carbon-aware control framework for battery-buffered edge AI systems. By deploying edge-friendly zero-shot TSFMs to forecast temporal marginal emissions and grid carbon intensity, FM-CAC feeds predictions into a dynamic programming solver with deferred cost attribution. This decouples energy harvesting from compute execution, reducing carbon emissions by up to 65.6% while maintaining near-maximum inference accuracy across continuous edge vision and time-series pipelines.

As detailed in Table 6, lightweight architectures (e.g., Cheraghinia MLP [66], TTM [13], and APEX-Edge [65]) consume between 0.18 mJ and 28.5 mJ per inference, operating comfortably within battery-buffered and energy-harvesting IoT regimes with negligible ($< 1.2\%$) degradation under INT8 quantization. In contrast, multi-billion parameter LLMs demand over 18 Joules per forward pass and over 14 GB of VRAM, underscoring that compact, quantization-resilient native models are mandatory for sustainable edge intelligence.

6.8 Causal Auditing, Test-Time Adaptation Benchmarks, and Living Leaderboards

As foundation models increasingly transition from academic prototypes to mission-critical infrastructure, evaluation methodologies must evolve beyond aggregate static point-error metrics (MSE, MASE) toward verifying causal structural fidelity, resilience under test-time drift, and continuous contamination-resistant leaderboards [77], [169]. Table 7 systematically categorizes the core mechanisms, parameter footprints, and capability envelopes across the emerging landscape of test-time adaptation and causal foundation modeling.

Synthesis of these evaluative frontiers yields four critical insights:

- 1) Causal Audits and the Persistence Bias Failure Mode: Jander, van Heeswijk, and Mes [77] demonstrate that transitioning from bespoke models to foundation forecasters creates systemic concentration risk. By probing state-of-the-art foundation models (Chronos-2 [8], TimesFM-2.5 [4], MOIRAI [5]) using synthetic interventional data generators, their causal audit reveals that foundation models exhibit a severe persistence bias. Specifically, when probed with processes exhibiting delayed interventional responses, sudden regime shifts, or unobserved confounders,

TABLE 5
Operational Complexity, Inference Latency, and Memory Footprints Across Foundation Paradigms.

Foundational Paradigm	Exemplar Models	Time Complexity (Inference)	KV-Cache / Peak Memory	Latency (ms)	Generation Mechanism	Hardware / Deployment Envelope
Autoregressive Token / Point Transformer	Chronos-Large [3], Auto-Times [46]	$\mathcal{O}(H \cdot L \cdot d + H^2 \cdot d)$	High ($\mathcal{O}((L+H) \cdot d)$ KV-cache)	120–450	Step-by-step sequential sampling (H passes)	Server (A100/H100); GPU latency-prohibitive for sub-second control
Patch-Based Encoder-Decoder	Chronos-Bolt, MOIRAI TimeXer [54]	$\mathcal{O}((L/P)^2 \cdot d + (H/P) \cdot (L/P) \cdot d)$	Medium ($\mathcal{O}((L/P) \cdot d)$ patch cache)	8–25	Direct multi-patch projection (1 or few passes)	Workstation GPU / Edge GPU (Jetson Orin); $250\times$ faster than vanilla Chronos
Multi-Scale Channel Mixers	TTM, LightGTS, TimeMixer++ [44]	$\mathcal{O}(L \cdot d)$ (Linear mixing)	Minimal (≤ 50 MB, zero KV-cache)	1.5–5	Single-pass linear time/channel projection	Microcontrollers, IoT gateways, embedded CPU; sub-5ms real-time
Continuous Flow-Matching & Diffusion	FlowState, FLAME, UrbanDIT [61]	$\mathcal{O}(K \cdot (L/P)^2 \cdot d)$ (K ODE steps, $K \in [10, 50]$)	Medium ($\mathcal{O}((L/P) \cdot d)$, zero KV-cache)	35–120	Multi-step numerical ODE/SDE integration	Standard GPU; viable for batch forecasting, unsuited for sub-10ms loops
Prior-Data Fitted Bayesian Networks	TabPFN-TS, ForecastPFN, ICTSP [52]	$\mathcal{O}((N_{\text{ctx}} + H)^2 \cdot d)$ (Context attention)	Medium-High ($\mathcal{O}(N_{\text{ctx}} + H)^2$ prompt memory)	15–60	Single-pass exact Bayesian posterior	Workstation GPU / High-RAM CPU; instant edge zero-shot cold start
Reprogrammed Multi-Billion LLMs	Time-LLM [19], OFA [20], LLMTIME [6]	$\mathcal{O}(L_{\text{text}}^2 \cdot D_{\text{LLM}} + H \cdot D_{\text{LLM}})$	Extreme (14–80GB VRAM, multi-head KV cache)	500–3500	Text tokenization + multi-layer LLM forwards	Multi-GPU cloud clusters only; completely prohibited on edge devices

TABLE 6
Empirical Energy Consumption, Quantization Degradation, and Edge Hardware Deployment Profiles across Foundation Models.

Model & Paradigm	Parameters	Precision Format	FLOPs / Inference	Energy / Inference	Peak RAM	Accuracy Degradation (Δ MSE)	Target Hardware Deployment Envelope
Cheraghinia MLP [66]	21K	FP32 / INT8	4.2×10^4	0.18 mJ	0.12 MB	$< 0.3\%$ (INT8 PTQ)	ARM Cortex-M4/M7 Microcontroller, Nordic nRF52840; sub-millisecond edge IoT
TTM-1M (IBM) [13]	1.0M	FP16 / INT8	2.1×10^6	3.4 mJ	4.2 MB	$+0.8\%$ (INT8 PTQ)	Embedded Linux gateways, Raspberry Pi Zero 2W; real-time telemetry
APEX-Edge [65]	10.5M	FP16 / INT8	2.2×10^7	28.5 mJ	24.8 MB	$+1.2\%$ (INT8 PTQ)	Enterprise Wi-Fi APs, Cisco Catalyst gateways; sub-second network anomaly audit
Tiny-TSM [64]	23.0M	FP16 / INT8	4.8×10^7	52.0 mJ	49.0 MB	$+1.5\%$ (INT8 PTQ)	Single-board computers (Raspberry Pi 4/5, Coral Dev Board); edge forecasting
Mixed-Precision Transformer [167]	1.8M	INT8 / INT4	3.6×10^6	4.1 mJ	1.8 MB (BRAM)	$+1.9\%$ (Mixed INT8/INT4)	Embedded FPGA (Xilinx Spartan-7 XC7S15); strict hardware-timed SCADA loops
Chronos-Bolt Small	46.0M	FP16 / INT8	9.5×10^7	112 mJ	98 MB	$+1.4\%$ (INT8 PTQ)	Edge AI accelerators (NVIDIA Jetson Orin Nano, Google Coral); high-throughput
MOIRAI-Small [5]	14.0M	FP16 / INT8	3.1×10^7	74 mJ	62 MB	$+1.1\%$ (INT8 PTQ)	Industrial IPCs, Jetson Xavier NX; any-variate sensor fusion
TimesFM-200M [4]	200M	FP16 / INT8	4.2×10^8	580 mJ	480 MB	$+2.1\%$ (INT8 PTQ)	Edge servers, industrial workstations; multi-horizon continuous quantile forecasting
Time-LLM (7B) [19]	7.0B	FP16 / INT4	1.5×10^{10}	18.4 J	14.2 GB	$+3.8\%$ (INT4 AWQ)	Cloud multi-GPU servers (A100/H100); strictly prohibited on edge/battery devices

models default to extrapolating local lag trends, severely underestimating the magnitude and velocity of structural inflections.

- 2) Realistic Benchmarking for Temporal Causal Discovery: Cheng et al. [170] establish CausalTime (NeurIPS 2023), resolving a core bottleneck in temporal causal validation. Prior benchmarks relied on oversimplified linear autoregressive models with Gaussian noise, producing synthetic evaluations disconnected from real-world physics. CausalTime utilizes deep invertible generative neural networks and normalizing flows to capture complex nonlinear dynamics and heteroskedasticity while deriving exact ground-truth causal DAGs, creating a rigorous standard benchmark for evaluating causal graph discovery.
- 3) Principled Test-Time Adaptation Benchmarking (FAC): Wang et al. [169] critique existing TTA evaluations for subtle future-information leakage and ad-hoc batch scheduling. They establish a standardized leak-free streaming benchmark suite and introduce Frequency-Aware Calibra-

tion (FAC). FAC proves that adapting models directly in the raw temporal domain frequently amplifies high-frequency measurement noise into destructive gradient shocks; conversely, isolating adapter updates to orthogonal low- and mid-frequency Fourier sub-bands preserves base model stability while yielding consistent accuracy improvements across non-stationary streaming horizons.

- 4) Living Leaderboard Ecosystems (The TIME Benchmark): Addressing the rapid obsolescence of static benchmark datasets, Qiao et al. [158] introduce the TIME benchmark (50 fresh datasets, 98 tasks) and the public Hugging Face TIME-Leaderboard (<https://huggingface.co/spaces/Real-TSF/TIME-leaderboard>). By pairing automated weekly evaluation pipelines with pattern-level time series decomposition, TIME evaluates foundation models across multi-scale periodicity, non-stationarity, and noise regimes, establishing a living, contamination-resistant community reference standard for zero-shot time-series intelligence.

TABLE 7
Systematic Synthesis of Test-Time Adaptation (TTA) and Causal Time-Series Foundation Models.

Framework	Key Reference	Core Mechanism & Adaptation / Causal Objective	Base Model	Param Footprint	Inference Overhead	Target Shift / Causal Capabilities
Category A: Test-Time Adaptation (TTA) under Non-Stationary Drift						
TSF-TTA	Kim et al. [74] (AAAI'25)	Online update of normalization statistics and linear projection heads via temporal consistency loss $\mathcal{L}_{TTA} = \ \hat{\mathbf{y}}_t - \mathcal{T}(\hat{\mathbf{y}}_{t-1})\ _2^2$	Frozen PatchTST / DLinear	Small (< 5M)	+12% latency	Abrupt mean/variance non-stationarity without source data
AdaNODEs	Dang et al. [75] (ICASSP'26)	Continuous latent trajectory adaptation via Neural ODEs: $\frac{d\mathbf{z}(t)}{dt} = f_\theta(\mathbf{z}(t), t)$ with self-supervised reconstruction	Any Frozen Forecaster	Minimal (0.15M)	+8% latency	Source-free continuous non-linear temporal drift
RG-TTA	Kumar et al. [76] (arXiv'26)	Regime-guided meta-control dynamically detecting stationary vs. non-stationary regimes, modulating gradient steps $\eta_t = \sigma(\text{RegimeGate}(X_t))$	Pretrained Transformer	Zero extra params	+5% latency	Multi-regime streaming series; prevents catastrophic forgetting
FAC Framework	Wang et al. [169] (arXiv'26)	Frequency-Aware Calibration projecting updates onto orthogonal spectral sub-bands to suppress high-frequency noise explosion	Pretrained TSFM	Lightweight	+10% latency	Leakage-free streaming adaptation; eliminates noisy gradient divergence
Category B: Causal Discovery, Structural Priors, and Counterfactual Foundation Models						
Causal-PT	Stein et al. [78] (arXiv'24)	Direct transformer mapping from multivariate temporal tokens to causal DAG adjacency matrix $\mathbf{A} \in \{0, 1\}^{D \times D}$ pre-trained across 10M SCMs	Transformer Encoder	25M	Single forward pass	Zero-shot discovery of directed causal graphs from observational series
CausalTimePrior	Thumm & Chen [79] (arXiv'26)	Prior-Data Fitted Network (PFN) pre-trained on 500M synthetic interventional TSCMs to approximate exact in-context posterior $P(\mathbf{Y} do(\mathbf{X}), \mathcal{D}_{obs})$	Transformer PFN	85M	Single forward pass	In-context causal effect estimation and zero-shot counterfactual forecasting
CaTSG	Xia et al. [80] (arXiv'25)	Structural causal score-based diffusion model conditioning reverse denoising trajectories on time-varying causal graphs $\mathcal{G}_t$	Causal Diffusion Net	45M	Multi-step ODE/SDE	Controllable trajectory generation under interventions ($do(X_t = \alpha)$)
CausalTime Bench	Cheng et al. [170] (NeurIPS'23)	Deep normalizing flow generator providing realistic multivariate time series paired with verifiable ground-truth causal graphs	Benchmark Generator	N/A	Offline evaluation	Ground-truth benchmark suite for time-series causal discovery algorithms
Jander Audit	Jander et al. [77] (arXiv'26)	Synthetic interventional audit harness probing persistence bias and structural distortion across foundation models	Chronos-2, TimesFM, MOIRAI	N/A	Offline audit	Uncovers persistence bias and regime switch degradation in TSFMs

6.9 Living Streaming Benchmarks, Temporal Generalization, and Agentic Multi-Turn Evaluation

Static benchmark splits (e.g., standard 70/10/20 train-val-test partitioning on historical repositories) have increasingly become liabilities in foundation model evaluation. Beyond the acute risk of pretraining data contamination and lookahead leakage, static benchmarks fail to measure how models perform under continuous open-world temporal evolution [171]. Simultaneously, standard single-step error metrics (MSE, MAE, CRPS) cannot evaluate multi-turn agentic workflows where conclusions emerge from cumulative dialogue turns, tool executions, and state reflections [134].

Table 8 provides a systematic taxonomy and architectural comparison across the emerging generation of agentic time-series reasoning frameworks and living temporal benchmarks.

These living and agentic evaluation frameworks illuminate two fundamental paradigms:

- 1) Continuous Live Evaluation vs. Static Overfitting: Garza et al. [171] instantiate Impermanent on top-400 GitHub activity streams. Because software repositories experience organic shocks (major version releases, viral social adoption, tooling migrations), they exhibit intense non-stationarity without synthetic artifacts. Crucially, Impermanent scores forecasts sequentially over time on live data streams with daily leaderboard updates (<https://impermanent.timecopilot.dev>). Their findings expose that zero-shot foundation models showing near-optimal static

error rates frequently deteriorate when evaluated under sustained streaming evolution, underscoring that static benchmarks reward memorization and lookahead leakage rather than true temporal generalizability.

- 2) Multi-Turn Dialogue and Decision Gaps: Kong et al. [134] construct TimeSage-MT, bridging the gap between single-step predictions and practical decision-making workflows. Across 2,680 dialogue turns in eight domains (finance, energy, transportation, healthcare, meteorology, manufacturing, retail, and server telemetry), they discover that even frontier closed-source LLMs (GPT-4o, Claude 3.5 Sonnet) suffer sharp performance drops exceeding 35% when progressing from exploratory questions to complex multi-step decision tasks. These failures stem primarily from three bottlenecks: loss of intermediate memory over long dialogue horizons, inability to calibrate forecast uncertainty across tool invocations, and domain hallucination when resolving conflicting statistical evidence.

6.10 Data Contamination, Pretraining Familiarity Bias, and Automated Continuous Auditing

The rapid proliferation of time-series foundation models has exposed an acute methodological challenge: benchmark data contamination and temporal leakage. Unlike natural language corpora where test sets can be filtered via string matching, time series consist of continuous, normalized numerical sequences that undergo varying degrees of preprocessing (standardization, differencing, patch discretization). Furthermore,

TABLE 8
Taxonomy and Architectural Profiling of Agentic Time-Series Reasoning Frameworks and Living Benchmarks.

Framework	Key Reference	Core Role & Paradigm	Base Model / Backbone	Toolset & Action Space	State & Memory Mechanism	Evaluated Benchmarks & Capabilities
Category A: Interactive Reasoning and Tool-Augmented Agents						
TimeInteract	Pan et al. [135] (arXiv'26)	Streaming Real-Time Interactive Intelligence	Qwen2-7B Backbone with Dual-View TS Encoder	Streaming perception, silent/response gating, decoupled generation	Dual-view short/long-term rolling perceptual memory	StreamTSI-34K (34.5K episodes); near-zero stall; 2.15× speedup
Cast-R1	Tao et al. [136] (arXiv'26)	Tool-Augmented Sequential Decision Forecaster	Qwen2.5-7B (SFT + Multi-Turn RL)	Seasonal-trend decomposition, stationarity tests, lightweight model calls	Episodic decision memory maintaining historical evidence	Complex non-stationary benchmarks; multi-turn self-reflective forecasting
TimeART	Wu et al. [137] (arXiv'26)	Agentic Data Scientist for Time-Series QA	8B Time Series Reasoning Model (TSRM) on Llama-3	Data science tool suite, statistical tests, plotting, forecasting heads	Trajectory memory with experience replay	TimeToolBench (100k expert trajectories); high-precision TSQA
TS-Reasoner	Ye et al. [138] (arXiv'24)	Domain-Specialized Multi-Step Inference Agent	7B / 14B Language Models	Domain-specific computational routines, symbolic reasoning tools	Iterative error feedback loop with constraint validator	TimeSeriesExam; complex multi-step compositional inference
DCATS	Yeh et al. [139] (arXiv'25)	Data-Centric AutoML Agent for Time Series	GPT-4 / Llama-3-70B Planner	Data cleaning routines, anomaly imputation, pipeline configurators	Pipeline state tracker optimizing data quality	Large-scale traffic volume datasets; 6% average error reduction across models
Category B: Living Temporal Benchmarks and Multi-Turn Diagnostic Suites						
Impermanent	Garza et al. [171] (arXiv'26)	Continuous Live Streaming Generalization Benchmark	TimeCopilot Live Platform (Top 400 GitHub Repos)	Daily automated ingestion of PRs, issues, commits, stargazers	Rolling daily window with sequential scoring	Living leaderboard; evaluates temporal robustness against distribution breaks
TimeSage-MT	Kong et al. [134] (arXiv'26)	Multi-Turn Agentic Reasoning Benchmark Suite	Frontier LLMs + Structured TimeSage Agent	Comprehensive time-series skill library (decomposition, anomaly, stats)	Multi-turn conversational dialogue state tracking	240 tasks, 2,680 turns across 8 domains; audits memory and uncertainty failure

because the pretraining datasets of many prominent models (e.g., UTSD, Chronos-Data, Monash) aggregate open-source archives, classic benchmarks (ETT, Weather, Electricity, Traffic) are frequently ingested during pretraining, rendering reported zero-shot performance suspect.

Recent literature systematically tackles this vulnerability through automated auditing frameworks, leakage-free simulation gyms, and temporal familiarity metrics:

- **TSFMAudit: Automated Temporal Contamination Scanning** (Li et al., 2026) [172]: Li et al. develop TSFMAudit, establishing the first formal auditing suite for detecting pretraining contamination in time-series foundation models. TSFMAudit introduces two complementary statistical tests: (1) Quantile-Bin N-Gram Hashing, which maps continuous floating-point sequences into discrete symbolic tokens to detect verbatim sequence memorization, and (2) Mutual Information Permutation Testing, which quantifies non-linear statistical dependence between candidate pre-training segments and benchmark test sets. Auditing multiple prominent open-weight foundation models, TSFMAudit reveals that up to 18.4% of standard benchmark sequences exhibit statistically significant memorization signatures, leading to artificially deflated zero-shot test errors.
- **Pretraining Familiarity Bias Survives Temporal Hold-Outs** (Moghadasi & Ghaderi, 2026) [173]: Moghadasi and Ghaderi challenge the pervasive research convention that temporal hold-outs (e.g., training on pre-2020 data and testing on post-2020 data) guarantee an uncontaminated evaluation. They show that because physical data-generating processes (e.g., weather cycles, power grid load curves) exhibit stationary dynamical attractors, a foundation model exposed to historical records retains profound pretraining familiarity. When evaluated on temporally separated hold-outs from familiar domains versus genuine cross-domain datasets, foundation models experience performance gaps of up to 42%, proving that reported zero-shot scores fre-

quently conflate cross-domain generalization with domain-specific familiarity.

- **Empirical Stress-Testing on Volatile Infrastructure** (Pan & Ezzat, 2026) [174]: Pan and Ezzat conduct an exhaustive empirical audit of zero-shot foundation models (Chronos-Bolt, TimesFM, Moirai) on electricity price markets (PJM, EPEX, and Australian NEM). Characterized by extreme price spikes and renewable penetration, electricity markets violate Gaussian stationarity. Their audit uncovers that univariate foundation models degrade by 28% compared to localized gradient-boosted trees (LightGBM) due to their inability to ingest critical exogenous covariates (temperature, transmission capacity). Furthermore, contamination risk is heightened because public market data is ubiquitously crawled into pretraining corpora without provenance tracking.
- **Forecast-Dojo: Replayable Environments for LLM Forecasting Agents** (Ye et al., 2026) [175]: Ye et al. introduce Forecast-Dojo, a leak-free replayable simulation gym that benchmarks multi-agent forecasting systems on resolved prediction market questions (from Polymarket and Metaculus). Forecast-Dojo deterministically pairs real-world questions with dated, timestamped global news archives, strictly eliminating future lookahead bias. In this controlled environment, agents must gather historical evidence, write Python code, and generate calibrated probabilistic forecasts, providing a rigorous testbed for evaluating genuine temporal reasoning without data contamination.

Table 9 synthesizes the frontier across multi-agent deliberation systems, data contamination auditors, and extreme low-bit edge quantization foundations.

6.11 Distribution-Free Conformal Prediction and Statistical Coverage Guarantees

A profound deficiency in contemporary time series foundation models is their vulnerability to uncalibrated proba-

TABLE 9
Taxonomy and Empirical Characterization of Multi-Agent Deliberation Systems, Contamination Scanners, and Edge Quantization Foundations.

Framework	Key Reference	Core Category & Paradigm	Architectural Mechanism	Target Hardware / Scope	Key Metric / Empirical Finding	
Category A: Multi-Agent Swarm Deliberation and Self-Evolving Systems						
MC-Debate	Trirat et al. [141] (arXiv’26)	Multi-Agent Swarm Deliberation	Cross-modal debate across Numerical, and Textual specialists	Server (7B/14B bones) GPUs Back-	−14.2% reasoning error; eliminates 76% of hallucinated trend reversals	
TimeEvo	Yang et al. [142] (arXiv’26)	Failure-Driven Self-Evolution	Evolutionary agent autonomously synthesizing Python tools on task failure	Server (8B/70B controllers) Clusters Con-	Dynamic tool synthesis; continuous improvement on TimeQA without human engineering	
Traceable-Agent	Kang et al. [140] (arXiv’26)	Traceable Multi-Agent	Enterprise	Four sub-agents with immutable DAG linking 10-K filings to forecasts	Enterprise Infrastructure Cloud	100% decision traceability; verifiable causal citations for regulatory compliance
MetaCaster	Shen et al. [144] (arXiv’26)	Meta-Harness Forecaster Synthesis	LLM agent architecting compact models (<1M) via dataset fingerprints	Edge / Low-power IoT nodes	Matches 100M+ TSFM zero-shot accuracy with 20× lower inference compute	
Category B: Extreme Low-Bit Quantization and Edge Inference Foundations						
SparseBinary	Gorbett et al. [71] (arXiv’23)	1-Bit Weight Trans-former	Extreme binary weights ({−1, +1}) with addition-only accumulation	Embedded / ASICs	FPGAs	87% parameter reduction; multiplication-free multivariate attention
Q-DEQ	Yang et al. [72] (arXiv’26)	Quantized Deep Equilibrium Model	Implicit fixed-point solving ($\mathbf{z}^* = f_\theta(\mathbf{z}^*, \mathbf{x})$) under INT4/INT8	Bare-metal MCUs (<100KB SRAM)		0.3M params; 75 KB footprint; zero dynamical stability loss
SGA	Hu et al. [73] (arXiv’26)	State-Guided Autoregressive Adapter	Latent state curvature tracking to scale autoregressive rollout variance	TSFM Backbones (15M Adapter)		Calibrated prediction intervals over long horizons without Monte Carlo sampling
Category C: Data Contamination Auditing, Familiarity Bias, and Replayable Gyms						
TSFMAudit	Li et al. [172] (arXiv’26)	Data Contamination Scanner	Quantile n-gram hashing and mutual information permutation testing	Pretraining Corpora (UTSD, Chronos)		Detects up to 18.4% sequence memorization in public open-weight models
Familiarity-Audit	Moghadas et al. [173] (arXiv’26)	Pretraining Familiarity Evaluator	Pretraining familiarity index across pre/post temporal benchmark splits	Benchmark Archives (ETT, Weather)		Proves temporal hold-outs preserve domain attractor bias (up to 42% gap)
Forecast-Dojo	Ye et al. [175] (arXiv’26)	Replayable Agent Simulation Gym	Prediction markets paired with dated news streams in leakage-free gym	Multi-Agent Evaluation		Leakage-free benchmark for multi-agent temporal decision policies

bilistic uncertainty. While foundation models deploy various heads—such as categorical cross-entropy over quantized bins (Chronos), continuous Student’s t / Gaussian parameters (TimesFM, Moirai), or diffusion score matching—their nominal prediction intervals lack rigorous finite-sample coverage guarantees. Under out-of-distribution drift, non-stationary variance scaling, or heavy-tailed shocks, standard foundation model intervals suffer from catastrophic empirical undercoverage (e.g., nominal 90% prediction intervals achieving only 65–75% empirical coverage).

Conformal Prediction (CP) provides an elegant, model-agnostic, distribution-free mathematical framework that converts any point or quantile forecaster into set-valued prediction regions $\mathcal{C}_{1-\alpha}(\mathbf{X}_{n+1})$ guaranteeing exact finite-sample marginal coverage:

$$\mathbb{P}(\mathbf{Y}_{n+1} \in \mathcal{C}_{1-\alpha}(\mathbf{X}_{n+1})) \geq 1 - \alpha, \quad (43)$$

for a user-specified significance level $\alpha \in (0, 1)$, without requiring Gaussianity, linearity, or asymptotic sample sizes.

Recent breakthroughs systematically integrate conformal inference with time series foundation models:

- **Foundation Models for Time Series in Conformal Prediction** (Achour et al., 2025) [176]: Achour et al. uncover a structural synergy between zero-shot foundation models and Split Conformal Prediction (SCP). In classical machine learning, limited data must be bifurcated between parameter fitting $\mathcal{D}_{\text{train}}$ and nonconformity calibration $\mathcal{D}_{\text{calib}}$, severely impairing calibration sample size. Because zero-shot foundation models (Lag-Llama, TimesFM, Chronos) require zero target training data, 100% of available in-domain historical data can be dedicated to calibration. Evaluating across ERCOT, NN5, and M3 benchmarks, Achour et al. show that zero-shot TSFM conformal intervals achieve strictly valid $1 - \alpha$ coverage while producing significantly narrower interval widths than classical GBDT or ARIMA baselines, demonstrating that foundation backbones stabilize post-hoc conformal calibration.
- **Adaptive Conformal Anomaly Detection for Signal Monitoring** (Martinez Gil et al., 2026) [177]: Martinez Gil et al. formulate adaptive conformal anomaly detection for industrial telemetry. Rather than applying arbitrary thresholding heuristics to foundation model reconstruction residuals, the authors utilize weighted quantile conformal

prediction bounds to transform residuals into exact, false-alarm-rate controlled p-values. By adaptively updating nonconformity weights online as temporal distributions drift, this framework maintains a guaranteed false alarm rate ($\leq \alpha$) on streaming IoT sensor workloads without retraining foundation backbones.

- **RareCP: Regime-Aware Retrieval for Efficient Conformal Prediction** (Heurich et al., 2026) [178]: Standard conformal inference assumes exchangeability, an assumption systematically violated by temporal autocorrelation and regime-switching dynamics. Heurich, Granz, and Landgraf introduce RareCP, which decouples smooth distribution drift from distinct macroeconomic/operational regimes. RareCP deploys a Mixture of Experts (MoE) with cosine attention alongside a hypernetwork that retrieves the top- k most relevant historical calibration contexts for any query sequence. By computing asymmetric weighted conformal quantiles over retrieved residuals on the GIFT-Eval benchmark, RareCP preserves nominal coverage guarantees while shrinking interval width by up to 22% compared to Adaptive Conformal Inference (ACI) and raw foundation model quantiles.

Table 10 provides a unified synthesis of the Iteration 9 frontiers, comparing post-training policy optimization architectures, conformal prediction frameworks, physics-informed foundations, and continuous asynchronous event models.

6.12 Differentially Private Generative Twins and Provable Leakage-Free Benchmark Curation

As foundation models scale to ingest petabytes of uncured web and institutional telemetry, benchmark data contamination and sensitive training leakage emerge as primary obstacles to empirical evaluation. Evaluating models on real-world industrial, financial, and medical time series risks not only memorization bias, but also severe privacy violations through membership and attribute inference attacks. Constructing synthetic generative data twins with provable (ϵ, δ) -Differential Privacy (DP) guarantees provides a principled pathway toward guaranteed leak-free benchmark curation:

Formally, a randomized generative or forecasting algorithm $\mathcal{M}$ satisfies (ϵ, δ) -Differential Privacy if, for all neighboring sequential datasets $\mathcal{D}, \mathcal{D}'$ differing by at most one individual’s temporal trajectory, and for all measurable output subsets $\mathcal{S} \subseteq \text{Range}(\mathcal{M})$:

$$\mathbb{P}(\mathcal{M}(\mathcal{D}) \in \mathcal{S}) \leq e^\epsilon \mathbb{P}(\mathcal{M}(\mathcal{D}') \in \mathcal{S}) + \delta. \quad (44)$$

In continuous and autoregressive time series, achieving meaningful privacy guarantees is notoriously difficult due to strong temporal autocorrelation: naive step-wise Gaussian noise injection destroys multi-step dynamical coherence, whereas trajectory-level clipping yields catastrophically loose privacy budgets ϵ . Recent breakthroughs systematically bridge this tension:

- **Privacy Amplification by Structured Subsampling for Time Series** (Schuchardt et al., 2025) [179]: Accepted as an ICML 2025 Spotlight, Schuchardt et al. establish the first mathematically rigorous framework for training deep foundation forecasting models under Differential Privacy with structured temporal subsampling. Standard DP-SGD assumes independent uniform Poisson subsampling,

which completely violates sequential temporal dependencies when dividing long series into sliding training windows. Schuchardt et al. develop a novel Rényi Differential Privacy (RDP) accountant specifically designed for structured overlapping subsampling schemes. By accounting for cross-window temporal correlations, their framework enables deep patch transformers to train with strictly amplified privacy budgets ($\epsilon < 2.0$), maintaining high forecasting utility across electricity, traffic, and clinical EHR datasets while mathematically proving that sensitive temporal history cannot be reconstructed by an adversary.

- **TSGBench: Standardized Benchmarking of Generative Time Series** (Ang et al., 2024) [180]: In VLDB 2024, Ang et al. introduce TSGBench, the first unified empirical benchmark specifically assessing time series generation methods across fidelity, privacy preservation, and domain adaptation. TSGBench establishes the rigorous “Train on Synthetic, Test on Real” (TSTR) protocol alongside “Train on Real, Test on Synthetic” (TRTS), evaluating 15 generative paradigms across 12 diverse real-world datasets using 12 complementary metrics (including Maximum Mean Discrepancy (MMD), discriminative classifier error, predictive autocorrelation fidelity, and empirical distance-to-closest-record membership inference risk). TSGBench reveals that generative foundation models paired with privacy-preserving regularization produce synthetic data twins that match real-world forecasting utility while provably defending against downstream data extraction, paving the way for standardized leak-free benchmark distribution.

6.13 Long-Context Memory Retention, Dynamic Graph Benchmarks, and Relational Conformal Inference

As time-series foundation models scale across context horizons and complex relational topologies, evaluating empirical performance requires investigating the fundamental nature of memory retention and graph-structured uncertainty guarantees. In long-sequence forecasting, capturing multi-scale seasonal patterns and non-local regime transitions depends on how past temporal observations are stored and retrieved:

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_\theta(\mathbf{X}_{t-L+1:t}, \mathcal{M}), \quad (45)$$

where $\mathcal{M}$ represents the model’s memory architecture. In standard self-attention transformers, $\mathcal{M}$ is an uncompressed external KV-cache $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{L \times d}$, providing exact associative recall at the cost of $O(L^2)$ computation and memory. In structured state-space models (Mamba/S4), $\mathcal{M}$ is compressed into a fixed-size internal recurrent state $\mathbf{h}_t \in \mathbb{R}^N$ with linear $O(L)$ training and $O(1)$ generation memory. However, whether internal state compression incurs catastrophic loss of long-term temporal information has remained a disputed empirical question. Furthermore, extending foundation models to dynamic graphs and microservices demands structure-aware benchmark suites and distribution-free relational uncertainty quantification:

- **Memory Retention and the Spectrum of Deep Time-Series Architectures** (Nguyen et al., 2026) [181]: Nguyen et al. formalize a unified taxonomy of memory mechanisms across deep time-series models, organizing architectures along a continuum from internal memory (fixed-size states in RNNs and SSMs) to external memory (transformer KV-caches,

TABLE 10

Taxonomy and Empirical Characterization of Post-Training Policy Optimization (RLHF/RLPF), Distribution-Free Conformal Prediction, Physics Foundations, and Asynchronous Event-Stream Models.

Framework	Key Reference	Core Paradigm	Category &	Policy / Conformal / Physical Mechanism	Target Domain / Hardware	Key Empirical Finding / Theoretical Guarantee
Category A: Reinforcement Learning and Post-Training Policy Optimization (RLHF / RLPF)						
TimeRFT	Li et al. [145] (arXiv’26)	RL Fine-Tuning for TSFMs		Quality-aware temporal rewards + difficulty-aware sample selection	Multi-Domain (TimesFM, Chronos)	Up to -18.6% forecast error under distribution shift; overcomes SFT exposure bias
TimeHF	Qi et al. [146] (arXiv’25)	6B Human-Aligned Foundation		Timeseries Policy Optimization (TPO) with pairwise preference rewards	Enterprise Supply Chain (20k products)	$+33.21\%$ replenishment decision accuracy; aligns asymmetric stockout costs
COUNTS	Parker et al. [147] (NeurIPS’25)	Reasoning LLM for Time Series		RVQ-VAE discrete tokens + Group Relative Policy Optimization (GRPO)	Medical ECG-QA / UCR Archive	Elicits verifiable Chain-of-Thought reasoning; outperforms direct prompting by large margins
TimeMaster	Zhang et al. [148] (arXiv’25)	Multimodal Reasoning via RL		Step-by-step diagnostic rewards over spectrograms and waveforms	Physiological Signals (EMG, ECG, HAR)	Eliminates visual hallucinations; enables structured multimodal diagnostic reasoning
LAST SToP	Gupta et al. [149] (ICML’25)	Asynchronous LLM Adaptation		Language-modeled Asynchronous TS (LASTS) + Stochastic Soft Prompting	Irregular Clinical ICU Telemetry	Outperforms QLoRA on irregular time intervals while preserving pre-trained LLM priors
Category B: Distribution-Free Conformal Prediction and Coverage Guarantees						
TSFM-Conformal	Achour et al. [176] (arXiv’25)	Split Conformal Prediction (SCP)		Zero-shot TSFM base forecasters dedicating 100% target data to calibration	ERCOT, NN5, M3 Benchmarks	Exact $1 - \alpha$ marginal coverage guarantee; produces tighter intervals than GBDTs
Adaptive-CAD	Martinez et al. [177] (ICLR’26)	Conformal Anomaly Detection		Weighted conformal quantile bounds outputting p-values	Industrial IoT Telemetry Streams	Strict false alarm rate control ($\leq \alpha$); stable calibration under non-stationary drift
RareCP	Heurich et al. [178] (arXiv’26)	Regime-Aware Conformal MoE		Cosine-attention retrieval of calibration residuals + hypernetwork drift adaptation	GIFT-Eval Benchmark Suite	Preserves nominal coverage while reducing interval width by up to 22% over ACI
Category C: Physics-Informed Foundations and Continuous Asynchronous Event Streams						
GridSFM	Bhan et al. [83] (arXiv’26)	Physics-Inspired Grid Model		Newton power flow operators with logarithmically penalized slack slacks	Electrical Transmission (10k buses)	2.45% zero-shot generation cost error on unseen grids; $10\times$ solver speedup
Physics-SynHist	Longarini et al. [84] (ICML-W’26)	Cold-Start Pipeline	Physical	Deterministic physical chains (pvlb) generating surrogate histories	Solar PV Facilities (440 sites)	Up to 50% MAE reduction on zero-shot cold start without target fine-tuning
SurF	Rezaei et al. [81] (arXiv’26)	Generative Foundation	Event	Time Rescaling Theorem (TRT) as continuous flow bijection to $\text{Exp}(1)$	Earthquakes, Retweets, commerce	Multi-dataset continuous pre-training; eliminates quadrature integration bottlenecks
TradeFM	Kawawa-Beaudan et al. [82] (arXiv’26)	524M Microstructure	Market	Scale-invariant universal tokenization of limit order books	9,000+ Equities (US, APAC)	Replicates non-linear financial stylized facts; zero-shot international market rollout

retrieval-augmented buffers, and agentic key-value stores). Through extensive empirical stress testing across extended lookback horizons ($L \in [10^2, 10^5]$), the authors uncover a fundamental architectural trade-off: while selective SSMS (Mamba) eliminate transformer memory explosion and maintain sub-linear latency scaling, their fixed-dimensional recurrent state exhibits gradual information decay on high-entropy multi-frequency signals. Conversely, full-attention transformers preserve distant dependencies without decay, but experience quadratic memory bottlenecks and latency degradation beyond 8K tokens. This systematic audit demonstrates that hybrid architectures interleaving selective state-space layers with sparse retrieval stores represent the optimal frontier for million-step context windows.

- ChronoGraph: Real-World Graph-Structured Microser-

vice Telemetry (Lutu et al., 2025) [182]: Presented at the NeurIPS 2025 Workshop on Foundation Models, Lutu et al. introduce ChronoGraph, a novel benchmark suite constructed from real-world production cloud microservices. Each node in ChronoGraph represents a distributed microservice emitting multivariate telemetry streams (CPU, memory, and network throughput), while directed graph edges encode real-time call dependencies. Uniquely, ChronoGraph provides expert-annotated incident outage windows, enabling rigorous evaluation of foundation model robustness during cascading system failures. Empirical evaluation demonstrates that standard flat multivariate foundation models fail to localize anomaly cascades, whereas graph-aware foundation architectures incorporating topological message passing reduce incident detection

TABLE 11
Architectural, privacy, and symbolic characteristics of representative Earth Observation, Differentially Private, and Neuro-Symbolic temporal foundation models (2023–2026).

Model / Framework	Key Reference	Core Category & Paradigm	Architectural & Theoretical Mechanism	Modalities / Constraints / Privacy Guarantee	Key Empirical Finding / Formal Result
Category A: Earth Observation, Satellite Radar, and Planetary Remote Sensing Foundations					
Prithvi W/C	Schmidt et al. [85] (arXiv'24)	Planetary Atmosphere	2.3B Scaled Encoder-Decoder on spherical harmonic grids	MIRRA-2, ERA5 (160 atmospheric variables)	1000× faster than numerical weather prediction; SOTA hurricane tracking
EarthPT	Smith et al. [86] (NeurIPS-W'23)	Pixel-Level Remote Sensing	700M Autoregressive Decoder over Sentinel-2 pixel series	Optical & Near-Infrared surface reflectance (> 10B tokens)	Unsupervised cloud inpainting; zero-shot vegetation biomass forecasting
Prithvi EO-2.0	Sewarman et al. [87] (arXiv'24)	Multi-Temporal Geospatial	300M Spatio-Temporal ViT with 3D attention factorization	Harmonized Landsat & Sentinel-2 (HLS; 42M global scenes)	SOTA transfer in flood inundation, crop mapping, and planetary burn scar delineation
AgriFM	Li et al. [88] (arXiv'25)	Radar-Optical Agricultural	80M Hierarchical Cross-Modal Multimodal Transformer	Sentinel-1 SAR (VV/VH backscatter) + Sentinel-2 optical	Bypasses cloud occlusions; accurate zero-shot agricultural crop phenology tracking
SpectralGPT	Hong et al. [89] (IEEE TPAMI'24)	3D Spatial-Spectral-Temporal	900M 3D Masked Autoencoder tensor reconstruction	Hyperspectral & Multispectral image cubes (> 1M cubes)	SOTA 3D spectral-temporal change detection; models non-linear spectral signatures
Change2	Zeng et al. [90] (arXiv'24)	Generative Remote Sensing	120M Continuous Flow / Diffusion Generative Transformer	Multi-temporal satellite pairs (> 100k bi-temporal scenes)	Unsupervised change detection; synthesizes realistic counterfactual terrain evolutions
Presto	Teng et al. [91] (NeurIPS'23)	Compact Sensor-Agnostic	4.8M Channel-Time Masked Pixel Transformer	Sentinel-1, Sentinel-2, Landsat, ERA5 weather series	Matches models 100× larger on fuel moisture and crop type classification
Category B: Differentially Private Generative Twins and Leakage-Free Benchmarks					
DP-StructuresSub	Schuchardt et al. [179] (ICML'25)	Provable Privacy Forecasting	Rényi DP accountant for sliding-window structured subsampling	Provable (ϵ, δ)-DP bounds ($\epsilon < 2.0$) on sequential data	Prevents temporal training leakage without catastrophic downstream utility degradation
TSGBench	Ang et al. [180] (VLDB'24)	Synthetic Benchmark Suite	Train-on-Synthetic Test-on-Real (TS/TR) standardized evaluation	15 Generative architectures across 12 multi-domain datasets	Rigorously benchmarks fidelity, domain adaptation, and membership inference risks
Category C: Neuro-Symbolic Temporal Logic and Differentiable Verifiable Specifications					
LogicDecoder	Candusso et al. [156] (ECML-PKDD'25)	Logic Inversion Transformer	Autoregressive transformer decoding continuous embeddings to STL syntax	Signal Temporal Logic grammar tokens over continuous latent vectors	Guarantees syntactically valid STL formulas; enables latent gradient optimization
STARs	Ferfoglia et al. [151] (arXiv'25)	Concept Bottleneck Model	Neural representations constrained to STL robustness degrees $\rho(\phi, \mathbf{x}, t)$	Naval ship telemetry, ICU patient streams, automotive sensors	Matches deep baseline accuracy while providing mathematically verifiable concept explanations
Confidence-STL	Mao et al. [152] (arXiv'26)	LLM Reasoning Monitor	Signal Temporal Logic trajectory mining over token-level logit confidence	Dynamic confidence traces across 7B to 70B LLMs (Llama, Qwen)	Detects reasoning hallucinations up to 40% earlier than static logit thresholds
ReasonSTL	Ye et al. [153] (arXiv'26)	Tool-Augmented Agent	8B LLM with Process-Supervised RL (PRM) & SMT / dReal solvers	Natural Language requirements to executable STL specifications	91.4% semantic correctness rate; eliminates syntax hallucinations in formal verification

delay by up to 42%.

- **STOIC: Relational and Sequential Conformal Prediction over Graphs** (Niresi et al., 2026) [183]: Niresi et al. address the open challenge of providing distribution-free uncertainty quantification on complex graph-structured energy time series. In interconnected electricity grids and district heating networks, point forecasts generated by spatial-temporal graph neural networks (STGNNs) lack safety guarantees. Standard split conformal prediction fails because spatial graph correlations violate exchangeability assumptions. Niresi et al. propose STOIC (Spatial-Temporal Graph Conformal Prediction with In-Context Learning), which reformulates STGNN prediction residuals into tabular representations and leverages the zero-shot calibration power of tabular foundation models. Without task-specific retraining, STOIC calibrates prediction intervals that simultaneously account for temporal autocorrelation and graph relational dependencies, delivering strictly valid $1 - \alpha$ marginal coverage with narrower interval widths than traditional conformal baselines across five real-world utility networks.

6.14 Transferability Estimation, Vintage Data Revisions, and Heterogeneous Batch Scheduling

As the foundation paradigm matures, critical systemic questions shift from “can foundation models predict time series?” to: (1) How can practitioners predict whether a foundation model will transfer positively or negatively to a new target domain without incurring the prohibitive compute cost of full fine-tuning? (2) How do real-world revisions and publication lags contaminate historical benchmarks? and (3) How should foundation models be scheduled across heterogeneous compute clusters serving multi-tenant closed-loop workloads?

- **TimeTic: In-Context Transferability Estimation** (Yao et al., 2025) [184]: While zero-shot performance is widely used as a heuristic proxy for selecting foundation models, Yao et al. demonstrate that zero-shot error correlates poorly with post-fine-tuning adaptation quality. Evaluating 10 foundation models across 10 diverse datasets, they introduce TimeTic. TimeTic reformulates transferability estimation as an in-context learning problem solved by tabular foundation models. By extracting statistical dataset meta-features paired with layer-wise entropy evolution vectors (capturing how representation uncertainty shifts across transformer depths), TimeTic predicts the downstream ranking of foundation models before fine-tuning. TimeTic achieves a

mean Spearman rank correlation of ≈ 0.60 with ground-truth fine-tuned accuracy—representing a 30% improvement over zero-shot ranking proxies—effectively preventing costly fine-tuning on ill-suited architectures.

- **Diversified Scaling Inference: Expanding Generative Support** (Hua et al., 2026) [185]: Hua et al. investigate whether computational scaling laws can be unlocked strictly at inference time without modifying model parameters. In standard autoregressive generation, foundation models frequently collapse into conservative mode seeking, underexploring low-probability but high-impact future branches. Hua et al. propose Diversified Scaling Inference, injecting tailored signal perturbations during decoding to broaden generative support. They theoretically establish a critical sample threshold beyond which diversified sampling provably outperforms standard temperature sampling. Furthermore, they define the RobustMSE metric to quantify the latent headroom performance of frozen TSFMs under fixed inference budgets, proving that controlled diversity scaling yields up to 14.8% error reductions across Chronos, TimesFM, and MOIRAI.
- **VINTAGE-TS: Auditing Hindsight Bias in Revised Time Series** (Ahmad, 2026) [186]: A pervasive but largely unaddressed contamination vector in macroeconomic, climate, and epidemiological forecasting is data revision. Statistical agencies routinely revise historical time series months or years after the fact as additional censuses arrive. Foundation models pretrained on contemporary retrospective downloads inadvertently train on “mature” data that was fundamentally unavailable to human decision-makers at prediction time, creating artificial performance inflations. Ahmad formulates VINTAGE-TS, building upon a frozen Chronos-2 backbone. VINTAGE-TS introduces dual temporal indexing: distinguishing observation time t from information-availability time $\tau \geq t$. Evaluated on the ALFRED real-time vintage database, VINTAGE-TS isolates the severe performance degradation (18%–35% higher RMSE) that occurs when models forecast unrevised initial releases versus revised mature series, providing a reproducible benchmarking harness for revision-consistent foundation modeling.
- **Armory: Lookahead Batch Scheduling for Robot Policy Serving** (Bansal et al., 2026) [187]: Transitioning foundation models to physical embodiment, Bansal et al. address the challenge of serving large policy models across heterogeneous multi-robot fleets. In closed-loop robotic control, foundation models generate “action chunks” (sequences

TABLE 12

Architectural, relational, and memory characteristics of representative Selective State Space Models (Mamba/S4), Spatio-Temporal Graph Foundations, and In-Context Latent PFNs (2023–2026).

Model / Framework	Key Reference	Core Paradigm & Topology	Algorithmic & Mathematical Mechanism	Memory Footprint / Complexity	Key Empirical Finding / Formal Result
Category A: Selective State-Space Models (Mamba/S4) and Linear-Time Sequence Modeling					
S-Mamba	Wang et al. [96] (arXiv:24)	Native TSFM (Multivariate)	Bidirectional selective state space across temporal patches & channels	Linear $O(L)$ compute; constant $O(1)$ recurrent step	Outperforms quadratic transformers with $3 \times -5 \times$ inference speedup on long series
TimeMachine	Ahamed & Cheng [97] (ICAT24)	Native TSFM (Dual-Axis)	Quadraplet 4-Mamba architecture routing 2D forward-backward & channel axes	Linear memory $O(L)$; avoids GPU out-of-memory	Resolves long-term dependency degradation on lookback horizons $> 10,000$ steps
Bi-Mamba+	Liang et al. [98] (arXiv:24)	Native TSFM (Bidirectional)	Interleaved forward-backward selective state space with state fusion	Linear $O(L)$ complexity; eliminates causal scan bias	Captures future contextual dependencies in offline LTSM without quadratic self-attention cost
QuantFlow	Baile et al. [99] (arXiv:26)	Polarized Native TSFM	Quantized selective state-space blocks replacing self-attention	70% lower memory footprint; zero row data leakage	Enables collaborative foundation pretraining across resource-constrained edge nodes
Category B: Hierarchical Spatio-Temporal Graph Foundations and Geometric Mesh Manifolds					
GPT-ST	Li et al. [92] (NeurIPS:23)	Spatio-Temporal Graph TSFM	Spatio-temporal masked autoencoder with hierarchical spatial clustering	Hierarchical region graph attention; easy-to-hard mask	Generative pretraining consistently boosts downstream traffic flow prediction across PoMS networks
AmazonSWE	Cartuyvels et al. [93] (arXiv:26)	Continental Hydrology Graph	Bidirectional selective SSM over flattened DAG river topology with SWAT altimetry	Scalable to 19K river reaches; handles $< 1\%$ daily observations	Cuts gauge elevation RMSE by 18%–39% over physical-statistical hydrodynamic baselines
DePSource-FM	Shu et al. [94] (arXiv:26)	DeFi Financial Graph TSFM	Graph-tabular encoder pretrained on 43.7M entries across 4,300+ DeFi protocols	Joint protocol-level flow and dynamic inter-protocol link prediction	SOTA macroprudential contagion monitoring and systemic credit exposure stress testing
EarthGeometry	Ranjan [95] (arXiv:26)	Planetary Manifold Foundations	Physically typed coordinate-invariant representations on spherical geodesic meshes	Hodge/Helmholtz vector-tensor decomposition; SO(3) invariance	Proves physical field typing yields superior OOD generalization in planetary weather forecasting
Category C: Long-Context Memory Retention, Dynamic Benchmarks and In-Context Latent PFNs					
Nguyen Memory	Nguyen et al. [181] (arXiv:26)	Memory Taxonomy & Audit	Unified taxonomy profiling internal state vs external KV-cache memory retention	Empirical stress testing across horizons $L \in [10^4, 10^7]$	Demonstrates SSMs avoid KV explosion while sparse retrieval buffers prevent state decay
ChronoGraph	Lutu et al. [182] (NeurIPS:W:25)	Microservice Dynamic Graph	Multivariate node streams (CPU, memory, net) + directed dependency graphs	Real production incident anomaly labels; structure-aware scoring	Proves topological message passing reduces incident detection delay by 42% in cascading outages
STOIC	Niemi et al. [183] (arXiv:26)	Relational Conformal Inference	Graph STCNN residuals reformulated as tabular contexts for foundation calibration	Finite-sample valid $1 - \alpha$ marginal coverage guarantees	Delivers tighter valid prediction intervals over energy and district heating network graphs
Lat-PFN	Veldhuis et al. [53] (arXiv:24)	In-Context Latent PFN	Prior-data Fitted Network integrated with Joint Embedding Predictive Architecture	Normalized abstract time axis; zero-gradient Bayesian inference	Spontaneously induces discrete multi-step patch tokens in latent space without patch supervision

of future motor actions) executed over receding horizons. Because different robots consume actions at varying speeds and experience disparate network latencies, standard naive batching induces either action queue starvation (causing robot stalling) or excessive latency. Bansal et al. propose Armory, formulating action chunk scheduling as a Markov Decision Process (MDP). Armory deploys a Lookahead scheduler that simulates candidate batch execution over a server-side mirror of each robot’s queue state, maximizing latency-normalized execution throughput by up to 18% over standard FIFO and priority schedulers.

6.15 Anytime-Valid Game-Theoretic Auditing and the Category Error Critique

As foundation models transition into automated industrial control, financial risk assessment, and clinical monitoring, the fundamental limits of offline empirical evaluation have become apparent. Static test-set benchmarks provide no finite-sample safety certificates once models encounter non-stationary streaming distributions, while the overarching philosophy of building a single “universal” foundation model for all temporal data has drawn rigorous foundational critique:

- **Bet on Features: Anytime-Valid Auditing of Quantile Forecasters** (Antonov et al., 2026) [188]: In risk-sensitive decision pipelines (such as inventory replenishment and grid dispatch under asymmetric underage/overage costs), forecasters output conditional quantiles $\hat{q}_\alpha(\mathbf{x})$. Standard conformal prediction and batch quantile tests require fixed sample sizes and exchangeability, failing to detect localized drifts in streaming deployments. Antonov et al. formulate a distribution-free, game-theoretic auditing framework. An auditor bets against the forecaster’s conditional validity using a sequential betting martingale:

$$M_t = \prod_{i=1}^t \left(1 + \boldsymbol{\lambda}_i(\mathbf{x}_i)^\top \mathbf{s}_i \right), \quad \mathbf{s}_i = \mathbb{I}(y_i \leq \hat{q}_\alpha(\mathbf{x}_i)) - \alpha, \quad (46)$$

where betting vectors $\boldsymbol{\lambda}_i(\mathbf{x}_i)$ are optimized online via Follow-the-Regularized-Leader (FTRL). Under the null hypothesis of perfect conditional calibration, M_t is a non-negative test martingale with $\mathbb{E}[M_t] = 1$. By Ville’s inequality, $\mathbb{P}(\exists t \geq 1 : M_t \geq 1/\delta) \leq \delta$, guaranteeing finite-sample Type-I error control at any arbitrary stopping time. Crucially, by parameterizing bets as functions of contextual features $\mathbf{x}_i$, Antonov et al. demonstrate that their auditor detects localized subgroup miscalibrations (e.g., severe under-coverage in specific supply chain tiers) up to $4.2\times$ faster than standard unconditional scalar tests.

- **Position: Universal TSFMs Rest on a Category Error** (Dai et al., 2026) [189]: In a seminal position paper, Dai et al. challenge the prevailing consensus that time series analysis should pursue a monolithic, multi-billion-parameter “Universal Foundation Model” analogous to GPT-4 in natural language processing. They argue that this pursuit conflates a structural container (sequentially ordered real numbers $(t, \mathbf{x}_t)$) with a semantic modality (a shared linguistic or perceptual space governed by common generative rules). While text corpora across medicine, law, and literature share common semantic syntax and world knowledge, the physical data-generating processes of distinct time series domains—such as Navier-Stokes turbulence in fluid dynamics, chaotic Navier-Stokes weather flows, high-frequency limit order book market microstructure, and cardiac electrophysiology—are governed by fundamentally incompatible, non-transferable differential laws. Forcing disparate physical domains into a unified transformer backbone creates an ill-posed optimization task, causing models to collapse into high-cost “generic low-pass filters”. Dai et al. formalize this limitation via the Autoregressive Blindness Bound:

$$\inf_{\hat{y}} \mathbb{E} [|y_{t+1} - \hat{y}(\mathbf{x}_{1:t})|] \geq \Delta_{\text{intervention}} \cdot \mathbb{P}(\text{do}(I_t)), \quad (47)$$

which proves that any history-only foundation model is fundamentally blind to causal interventions and regime shifts ($\text{do}(I_t)$) that lack historical precedent. As an alternative to monolithic universality, they advocate the Causal Control Agent paradigm: an intelligent meta-orchestrator that dynamically leverages domain context to coordinate specialized solvers, lightweight Just-in-Time (JIT) adapters, and physics-informed models, replacing stationary zero-shot MSE benchmarks with Time-to-Recovery (TTR) adaptation metrics.

7 Open Challenges and Future Horizons

Despite rapid progress in foundation models and LLM adaptations, temporal machine learning remains in a formative phase. In this section, we delineate six critical open research horizons that must be addressed to achieve truly universal temporal intelligence.

7.1 Irregular Sampling and Continuous-Time Dynamics

Existing foundation models (e.g., Chronos, TimesFM, MOIRAI) predominantly assume regularly sampled, synchronous discrete grids. However, real-world physical

TABLE 13
Architectural mechanisms, transferability estimation, active information acquisition, and heterogeneous scheduling across Iteration 12 foundation models and frameworks (2025–2026).

Model / Framework	Key Reference	Core Paradigm & Topology	Mathematical & Algorithmic Formulation	Hardware / Latency Envelope	Key Empirical Finding / Formal Result
Category A: Heterogeneous Multivariate Decoupling, Covariate Homogenization & Memory Distillation					
Falcon-X	Liu et al. [100] (arXiv'26)	Native TSFM (Multivariate)	Unified Prototype Diff-Attention (UPDA) + Variate Reassembly Router	230M params; parallel prototype projection	Decouples competing physical channels; cuts multivariate zero-shot error by 17.4%
UniCA	Han et al. [101] (arXiv'25)	Native TSFM (Covariates)	Covariate Homogenization mapping multimodal features to dense continuous tokens	120M params; modular Pre/Post-Fusion	Enables zero-shot conditioning on categorical/text covariates; +22.1% GIFT-Eval gain
TS-Memory	Lynn et al. [102] (KDD'26)	Memory Adapter (Native TSFM)	Plug-and-play adapter with confidence-gated distillation from INN teacher	0.5M params; constant $O(1)$ inference latency	Prevents negative transfer under domain shift; zero catastrophic forgetting without DB backup
JanTime	Saha & Shmakov [103] (arXiv'26)	Native TSFM (In-Context)	Hierarchical Multi-Scope Transformer + Task-Conditioned Patch Decoder	350M params; zero parameter updates	In-context demonstration adaptation across 120B+ observations; multi-task zero-shot transfer
ST-Prune	Chen et al. [104] (arXiv'26)	Native TSFM (Spatio-Temporal)	Real-time loss-gradient and complexity-aware dynamic sample pruning	Accelerated pretraining; 46% wall-clock speedup	Eliminates redundant spatial slices without sacrificing downstream generalization
Category B: Active Information Acquisition, Online Deferral & Temporal Causal PPNs					
STAP	Kim et al. [105] (ICML'25)	Active Learning (PDE TSFM)	Selective time-step acquisition optimizing expected global variance reduction	45M params; non-uniform solver querying	Explores 3.4x wider nonlinear physical regimes under fixed trajectory budget
LED-SLDS	Montreuil et al. [106] (arXiv'26)	Streaming Online Deferral	Factorized Switching Linear-Gaussian State-Space + Information Query Score	153M params; hard real-time streaming	Balances cost vs latent information gain; maintains < 2% deferral under concept drift
TCPTN	Talasila & Shama [107] (arXiv'26)	Causal PPN (Native TSFM)	Synthetic SCM pretraining with explicit Causal Judgment Head	85M params; sub-second zero-shot inference	Outputs calibrated reliability signals & confounding strength across thousands of industrial series
Category C: Transferability Estimation, Vintage Auditing, Modality Routing & Batch Scheduling					
TimeTic	Yao et al. [184] (arXiv'25)	Transferability Benchmark	In-context transferability predictor via layer-wise entropic evolution	Tabular FM meta-learner; zero-fine-tuning cost	Achieves 0.60 rank correlation with fine-tuned accuracy (+30% over zero-shot ranking)
Diversified Scaling	Hua et al. [185] (arXiv'26)	Inference Compute Scaling	Tailored temporal signal perturbations expanding generative support	Evaluated on Chronos, TimesFM, MOIRAI	Proves critical sample threshold; RobustMSE headroom metric reduces error by up to 14.8%
VINTAGE-TS	Ahmad [186] (arXiv'26)	Contamination Benchmark	Dual temporal indexing (observation time t vs availability time τ)	120M Chronos-2 joint adapter	Identifies 18%–35% error inflation from retrospective data revision contamination
Armory	Bansal et al. [187] (arXiv'26)	Heterogeneous Scheduling	Lookahead MDP scheduler over temporal action chunk queues for VLA serving	Multi-tenant GPU serving for heterogeneous robots	Improves closed-loop robot execution throughput by up to 18% over FIFO heuristics
TSRouter	Yu et al. [154] (COLM'26)	Multimodal Routing (LLMTS)	Candidate scoring on heterogeneous bipartite task-query-modality model graph	8B parameter routing envelope	+16%–46% relative reasoning accuracy gain while reducing inference dollars by > 60%
ZARA	Li et al. [155] (ACL'26)	Agentic Reasoning (LLMTS)	Placement-specific vector stores + verifiable natural language evidence traces	7B LLM + frozen MOMENT backbone	Eliminates zero-shot sensor hallucinations across body placement domain shifts

TABLE 14
Architectural mechanisms, continual adaptation, cross-sensory telemetry, and formal verification across Iteration 15 foundation models and frameworks (2024–2026).

Model / Framework	Key Reference	Core Paradigm & Topology	Mathematical & Algorithmic Formulation	Hardware / Scale Envelope	Key Empirical Finding / Formal Result
Category A: Online Continual Learning, Temporal Plasticity & Black-Box Error Adaptation					
ELF	Lee et al. [108] (ICML'25)	Native TSFM (Online Adapter)	Dual-module fast frequency forecaster + softmax gating	<1M adapter; frozen FM backbone	12%–28% error reduction over static FMs; zero full retraining
ORCA	Dai et al. [109] (NeurIPS'26)	Native TSFM (Black-Box)	Context of errors $P(\mathbf{e}_t \mathbf{x}_{1:t}, \mathbf{y}_t)$ + Boltzmann router	<2M adapter; API serving mode	Cuts black-box streaming error by up to 34.2% with zero gradient backprop
NatSR	Uretitni et al. [110] (ICLR'26)	Native TSFM (Score-Driven)	Natural gradient descent with Student- t score-driven replay	Standard transformer footprint	Robust tracking under heavy-tailed spikes; prevents catastrophic forgetting
Temporal Plasticity	Liu et al. [111] (IJCNN'25)	Evaluation & Benchmark	Continual incremental fine-tuning plasticity evaluation suite	Evaluated on Time-MoE 2.4B, Chronos	Discovers emerging scalar resilience: billion-scale TSFMs retain plasticity longer
Category B: Cross-Sensory Telemetry, Industrial Multimodality & Tactile Generalist Policies					
FISHER	Fan et al. [112] (IEEE TII'25)	Native TSFM (Industrial)	Sub-band spectral tokenization mapping variable f_c to sub-bands	28M params; RMIS benchmark	Solves M5 heterogeneity; 16x more parameter-efficient than VLMs
PowerPM	Tu et al. [113] (NeurIPS'24)	Native TSFM (Power Systems)	Relational GCN over grid hierarchy + Temporal Transformer	115M params; city-district-user grid	SOTA transfer on electricity forecasting, stability, and customer profiling
FTP-1	Yuan et al. [114] (arXiv'26)	Native TSFM (Tactile Policy)	Morphology-aware unified latent tokens across 21 sensors	180M params; 3,000h tactile corpus	+17.2% seen sensor gain; +31% zero-shot success gain on unseen sensors
TouchWorld	Zhou et al. [115] (arXiv'26)	Native TSFM (Tactile World)	Decoupled predictive contact world model + > 100Hz reactive policy	350M params; Ego/Touch dataset	Resolves millisecond slip and tactile hysteresis in bimanual manipulation
TS-JEPA	Ennadir et al. [116] (NeurIPS'24W)	Native TSFM (JEPA)	Latent-space energy minimization discarding observation reconstruction	22M params; UCR/UEA, Monash	Filters high-frequency sensor noise; superior linear probing and zero-shot transfer
RATFM	Maru & Sato [117] (arXiv'25)	Native TSFM (Retrieval)	Test-time nearest-neighbor retrieval of in-domain normal prototypes	110M params; UCR Anomaly Archive	Matches in-domain fine-tuning accuracy without updating a single weight
Category C: Decoupled Formal Safety, Anytime Auditing & Foundational Critique					
PEACE	Uysal et al. [156] (ICRA'26W)	LLMTS (Formal Agent)	Decoupled Planner above the loop + Deterministic Kinematic Barrier	7B LLM planner + real-time filter	Eliminates UAV flight hallucinations; guaranteed geofencing; $O(1)$ real-time execution
Bet on Features	Antonov et al. [188] (arXiv'26)	Evaluation & Benchmark	Game-theoretic betting martingales with ϵ -values and FTFL	Model-agnostic sequential auditor	Anytime-valid finite-sample Type-I error control; detects localized drift 4.2x faster
Category Error	Dai et al. [189] (arXiv'26)	Survey & Foundations	Autoregressive Blindness Bound: structural container vs semantic modality	Theoretical proof & benchmark harness	Proves history-only TSFMs blind to interventions; advocates Causal Control Agents

telemetry—such as ICU medical records, IoT sensor dropouts, and financial trade ticks—arrives at asynchronous, irregular time intervals. Emerging approaches such as FlowState [41] leverage continuous flow matching and continuous-time neural state-space models to achieve sampling-rate equivariance. Future foundation architectures must natively encode irregular time intervals (Δt) into latent manifolds rather than relying on crude imputation or resampling heuristics.

7.2 Prior-Data Fitted Bayesian Networks

Standard foundation models require thousands of gradient optimization steps during fine-tuning. A compelling alternative is demonstrated by TabPFN-TS [51], which adapts Prior-Data Fitted Networks to time series. By training a transformer exclusively on synthetic prior stochastic processes (e.g., Gaussian processes, Ornstein-Uhlenbeck, and Markov switching models), the network learns to perform Bayesian posterior inference in a single forward pass without parameter updates. Expanding synthetic prior spaces to encompass complex non-linear dynamics represents a fertile research direction.

7.3 Extreme Long-Horizon Context and Serial Scaling

In natural language, modern models handle context windows exceeding 1M tokens. In temporal analysis, historical series often span millions of high-frequency timestamps. Capturing decadal climate shifts, annual retail cycles, and second-level power grid vibrations within a single context window demands extreme context scaling. Architectural innovations such as long-context state-space models (Mamba), hierarchical linear attention, and serial scaling (as explored in Timer-XL [16] and Timer-S1 [18]) will be essential to eliminate context truncation.

7.4 Unified Multimodal Perception

Real-world decision-making rarely relies on isolated scalar numbers. An industrial operator monitors pressure gauges alongside diagnostic maintenance logs; a macroeconomist reads financial reports while analyzing stock tickers; an environmental scientist tracks satellite imagery across seasonal vegetation cycles [63]. Future foundation models must advance beyond ad-hoc text reprogramming to train native multi-modal architectures from scratch—treating numerical time series, textual reports, spatial optical imagery, and spectral vibrations as co-equal perceptual modalities, as pioneered by Chronicle [132], VLT [131], and ChronoSteer [133].

7.5 Calibrated Uncertainty under Distribution Shift

In safety-critical applications (such as healthcare and grid management), an uncalibrated point forecast is hazardous. As demonstrated by recent reliability audits [26], current zero-shot models frequently underestimate predictive variance when confronted with unprecedented macroeconomic shocks or extreme weather events. Developing conformal prediction techniques, robust Bayesian heads, and flow-based generative sampling that maintain rigorous coverage guarantees under out-of-distribution shifts is imperative.

7.6 Edge Deployment, Hardware Quantization, and Green AI

While research models scale to billions of parameters, industrial IoT environments require deployment on low-power microcontrollers, edge gateways, and automotive Electronic Control Units (ECUs). Systematic cross-platform benchmarking (HoliBench [166]), dynamical systems trajectory sensitivity planning (TQS-PTQ [68]), percentile-based 4-bit activation calibration [69], on-device fully quantized microcontroller

training (MCU-FQT [70]), integer mixed-precision quantization (INT8/INT4 on embedded FPGAs [167]), ultra-compact patch encoders [64], [66], and carbon-aware dynamic dispatch (FM-CAC [168]) will be essential to transform temporal foundation research into sustainable, energy-efficient cyber-physical systems.

7.7 Autonomous Living Foundations and Continuous Discovery

With dozens of new preprints posted weekly across machine learning repositories, academic surveys quickly become obsolete. Future research must embrace autonomous, living benchmark harnesses (e.g., LiveHouse-TS [163], Impermanent [171]), continuous automated snowballing pipelines that monitor arXiv and Semantic Scholar streams, vintage-consistent simulation pipelines eliminating lookahead bias (MACROCAST [67]), and automated multi-granularity leakage audits (It’s TIME [158]) to maintain living, contamination-free community standards.

7.8 Interactive Multi-Turn Agents and Strategic Tool-Use

Finally, the paradigm of treating time-series intelligence as passive, single-pass point prediction is rapidly yielding to autonomous, tool-augmented agentic ecosystems [134]. Future temporal systems will combine streaming asynchronous perception (TimeInteract [135]), reinforcement-learned sequential decision policies (Cast-R1 [136]), expert trajectory reasoning models (TimeART [137]), domain-specialized constraint verifiers (TS-Reasoner [138]), and automated data cleaning planners (DCATS [139]). Developing formal verification methods for multi-turn agentic actions, tool-use error boundaries, and streaming conversational state retention represents the definitive next frontier for temporal artificial intelligence.

8 Conclusion

In this survey, conducted under the PRISMA 2020 systematic review protocol, we have synthesized the foundational developments at the intersection of Large Language Models and Time Series Foundation Models from 2021 through 2026. We formalized mathematical definitions of temporal sequences, subseries patching, and quantization; introduced a unified five-dimensional taxonomy; and compared 41 core benchmark and foundation models across architecture, tokenization, pre-training corpora, and uncertainty formulations.

Our critical synthesis highlights a decisive architectural demarcation:

- Native TSFMs (such as Chronos, TimesFM, MOIRAI, Timer, and TTM) demonstrate superior parameter efficiency, faster inference latency, and stronger zero-shot numeric forecasting accuracy by training directly on temporal dynamics without linguistic mismatch.
- Repurposed LLMs (such as Time-LLM and multimodal reasoning frameworks) excel not at raw numeric extrapolation, but as cognitive orchestrators—grounding numerical time series within textual narratives, metadata, and decision-support systems.

Looking ahead, the frontier of temporal intelligence lies in resolving benchmark contamination, achieving continuous-time sampling equivariance, calibrating uncertainty under

non-stationary shifts, and delivering ultra-efficient edge deployment. We hope this survey provides a rigorous compass for researchers and practitioners building the next generation of universal temporal models.

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